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SUPPLEMENT TO "INTELLIGENT SAMPLING AND INFERENCE FOR MULTIPLE CHANGE POINTS IN EXTREMELY LONG DATA SEQUENCES"

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8. Supplement Part A (Single Change Point Problem).

8.1. *Problem Setup.* Instead of proving Theorems 1 and 2 directly, we shall consider a more general nonparametric result from which the two theorems will follow as a special case. As before suppose the time series data is $(x_1, Y_1), ..., (x_N, Y_N)$, where $x_i = i/N$ and $Y_i = f(x_i) + \varepsilon_i$ for $i = 1, ..., N$. We will make the weaker assumptions that

- f is a right continuous function in $[0, 1]$ with a single left discontinuity at some point $\tau_0 \in (0, 1)$, with jump size $f(\tau_0+) - f(\tau_0-) = \Delta$
- there exists a $\beta_f > 0$ where $|f(x) - f(y)| \leq \beta_f |x - y|$ whenever $(x - \tau_0)(y - \tau_0) > 0$
- the errors ε_i's are iid $N(0, \sigma^2)$ error terms

The main difference between this model and the model presented in section 2.1 is the looser restriction on the signal f: here f could be any Lipschitz continuous function with a single discontinuity and not be constrained to the family of piecewise constant functions.

We will first remark on some background regarding this this model before moving on to proving some results. Estimation procedures for such a dataset can be found in [4], where one-sided polynomial fitting was used to obtain an estimate $\hat{\tau}_N$ for $\tau_N := \lfloor N\tau_0 \rfloor / N$. In summary, fix a sequence of bandwidth $h = h_N$, a non-negative integer p, and a kernel function K with support on $[-1, 1]$. Next, for all $x_m \in (h, 1 - h)$, consider the signal estimates

$$\hat{f}_-(x_m) := \pi_1 \left(\underset{(a_0,...,a_p)\in\mathbb{R}^{p+1}}{\arg\min} \left(\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) \left(Y_{m-j-1} - a_0 - a_1 j - ... - a_p j^p\right)^2 \right) \right)$$

$$\hat{f}_+(x_m) := \pi_1 \left(\underset{(a_0,...,a_p)\in\mathbb{R}^{p+1}}{\arg\min} \left(\sum_{j=0}^{Nh} K\left(\frac{j}{nh}\right) \left(Y_{m+j} - a_0 - a_1 j - ... - a_p j^p\right)^2 \right) \right), \tag{36}$$

where π_1 is the projection functions such that $\pi_1(a_0, \ldots, a_p) = a_0$. The change point estimate is

$$\hat{\tau}_N := \underset{x_i\in(h,1-h)}{\arg\max} |\hat{f}_+(x_i) - \hat{f}_-(x_i)|. \tag{37}$$

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This estimator is consistent under a few regularity conditions on the kernel K and conditions on how fast h converges to 0. For the sake of brevity we will not mention all those conditions here, but we will note that under said conditions, $\mathbb{P}[N(\hat{\tau}_N - \tau_N) = k] \to \mathbb{P}[L(\Delta/\sigma) = k]$ for all $k \in \mathbb{Z}$, which is the exactly asymptotic result for $\hat{\tau}_N$ obtained by least squares in a stump model setting. Finally, for our purposes we propose estimators $\hat{\alpha}$ and $\hat{\beta}$ for $f(\tau_0-)$ and $f(\tau_0+)$, respectively, defining them as

$$\hat{\alpha} := \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) Y_{N\hat{\tau}_N - j - 1}}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)}$$

$$\hat{\beta} := \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) Y_{N\hat{\tau}_N + j}}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)}. \tag{38}$$

These two estimators are consistent:

LEMMA 3. *$|\hat{\alpha} - f(\tau_0+)|$ and $|\hat{\beta} - f(\tau_0-)|$ are $O_p(h \vee (Nh)^{-1/2})$.*

It is possible to perform intelligent sampling to this nonparametric setting as in steps (ISS1)-(ISS4), though with a slight adjustment. Instead of fitting a stump function at step (ISS2), use one-sided local polynomial fitting with bandwidth h on the first stage subsample to obtain estimates $(\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N)$ for the parameters $(f(\tau_0-), f(\tau_0+), \tau_N)$. These first stage estimators satisfy the following consistency result:

$$\mathbb{P}\left[|\hat{\tau}_N^{(1)} - \tau_N| \le w(N); \quad |\hat{\alpha}^{(1)} - f(\tau_0-)| \vee |\hat{\beta}^{(1)} - f(\tau_0+)| \le \rho_N\right] \to 1 \tag{39}$$

for the sequence $w(N) = CN^{1-\gamma+\delta}$ where δ and C can be any positive constants, and some sequence $\rho_N \to 0$ (an explicit sequence can be derived by Lemma 3). The consistency condition in (39) is sufficient for a generalized versions of Theorems 1 and 2:

THEOREM 6.

$$\hat{\tau}^{(2)} - \tau_N = O_p(N^{-1}) \tag{40}$$

THEOREM 7. *Suppose the conditions of Theorem 7 are satisfied, then for all integers $k \in \mathbb{Z}$ we have*

$$\mathbb{P}\left[\lambda_2\left(\tau_N, \hat{\tau}_N^{(2)}\right) = k\right] \to \mathbb{P}\left[L_{\Delta/\sigma} = k\right] \tag{41}$$

These results can hold under a general nonparametric setting, but they still hold for the stump model from Section 2. Since consistency condition (39) is if f is a stump function and least square fitting was used at step (ISS2) as it was written in Section 2.2, Theorems 6 and 7 do imply Theorems 1 and 2. The proof of Theorem 6 will be covered in Section 8.2.1, while the proof of Theorem 7 will be covered in Appendix B at Section 8.2.2.

REMARK 9. *We note that not only do the consistency results of intelligent sampling for stump models generalize to this nonparametric setting, the computational time aspects translates also does not change. Local polynomial fitting on n data points takes $O(N)$ computational time, see e.g. [5]. Therefore the computational time analysis in Section 2.1 still holds for this nonparametric case.*

REMARK 10. *It is not possible derive asymptotic distribution results for $N(\hat{\tau}_N^{(2)} - \tau_N)$. Consider the case where $\tau = 0.5$, $N_1 = \sqrt{N}$ (or $\gamma = 0.5$), and the two subsequences $N = 2^{2j}$ or $N = 3^{2j}$ for some large integer j. In such cases the first stage subsample would choose points that have integer multiples of $1/2^j$ or $1/3^j$ as their x-coordinate.*

- *If $N = 2^{2j}$, $\tau_N = \frac{\lfloor 2^{2j} \cdot 0.5 \rfloor}{2^{2j}} = 0.5$ is an integer multiple of $1/2^j$, and hence τ_N is an x-coordinate used in the first stage.*
- *If $N = 3^{2j}$, then $\tau_N = \frac{\lfloor 3^{2j} \cdot 0.5 \rfloor}{3^{2j}}$, and it can be checked that $\lfloor 3^{2j} \cdot 0.5 \rfloor$ is an even integer not divisible by 3. Since the x-coordinate of every first stage data point takes the form $\frac{k}{3^j} = \frac{3^j k}{3^{2j}}$ for some integer k, this means τ_N is not used in the first stage.*

Hence, in the former case we cannot ever have $\hat{\tau}_N^{(2)} = \tau_N$, while in the latter case, we have $\tau_N^{(2)} = \tau_N$ and Theorem 2 tells us that $\mathbb{P}[\hat{\tau}_N^{(2)} = \tau_N]$ converges to the nonzero $\mathbb{P}[L(\Delta/\sigma) = 0]$ as j increases. Clearly, we have two subsequences for which $\mathbb{P}[\hat{\tau}_N^{(2)} = \tau_N]$ converges to different values.

To further validate the extension to this nonparametric setting, we also ran a set of simulations for when $Y_i = 2\sin(4\pi x_i) + 2 \cdot 1(x_i > 0.5) + \varepsilon_i$, where ε_i are iid $N(0, 1)$. We took 15 values of N between 2500 and 10^6, chosen evenly on the log scale, and applied intelligent sampling on 1000 replicates. For each of these values of N. First stage used roughly $N_1 = \sqrt{N}$ points, which were subjected to one sided local polynomial fitting with a parabolic kernel and bandwidth $h = N_1^{-0.3}$, while the second stage interval had half-width $8/\sqrt{N}$. Figures 17 and 18 show results consistent with Theorems 6 and 7.

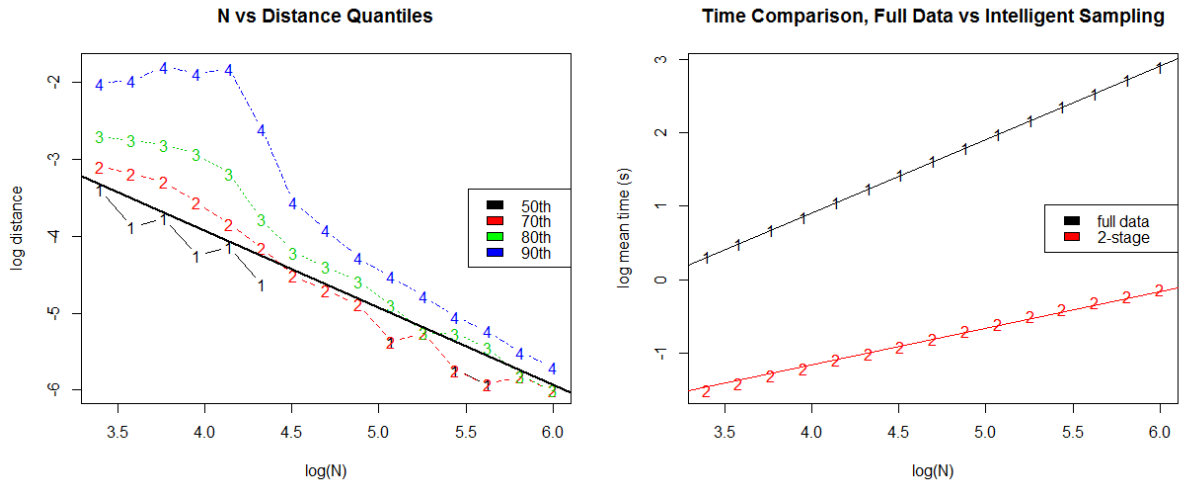


Fig 17: left graph shows log-log plot of the quantiles of $|\hat{\tau}_N^{(2)} - \tau_N|$ versus N, with the solid black line having a slope of exactly -1. Some datapoints for the quantiles of the 50th quantiles do not appear since for some N, the median of $|\hat{\tau}_N^{(2)} - \tau_N|$ was 0. Right graph is a log-log plot of the mean computational time of using all datapoints (black) and intelligent sampling (red), with the solid black line having a slope of exactly 1 and the solid red a slope of exactly 0.5.

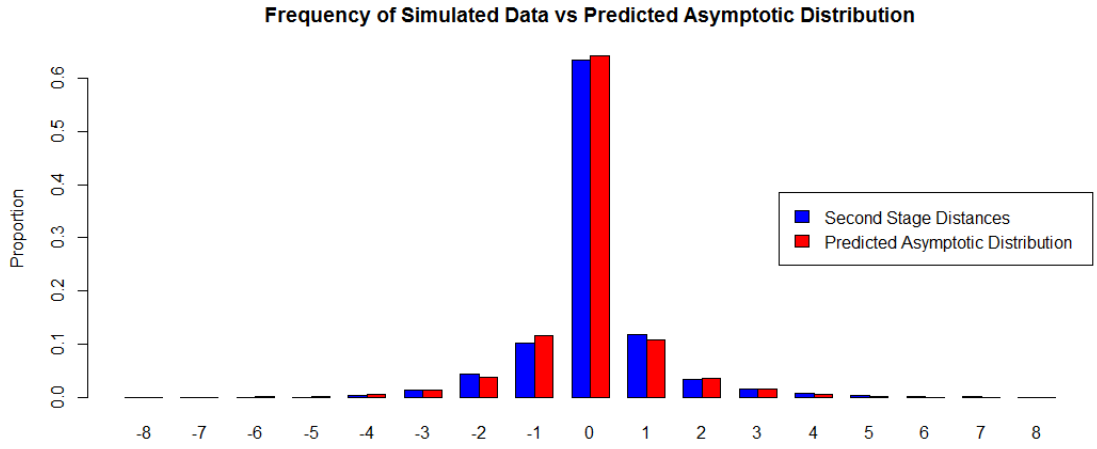


Fig 18: distribution of $\lambda_2\left(\tau_N, \hat{\tau}^{(2)}\right)$ values (blue) compared with the distribution of L from Theorem 2.

8.2. *Proof of Corollary 3.*

PROOF. We will show that for any $\epsilon > 0$,

$$\mathbb{P}\left[|\hat{\beta} - f(\tau+)| > C_0(h \vee (Nh)^{-1/2})\right] \leq \epsilon. \tag{42}$$

We start off by utilizing the rate of convergence of the change point estimator: there is a constant $C_1 > 0$ such that

$$\mathbb{P}\left[|\hat{\tau}_N - \tau| > \frac{C_1}{N}\right] < \frac{\epsilon}{2}$$

for all sufficiently large N. Hence, for any $C > 0$ we have,

$$\begin{aligned}
&\mathbb{P}\Big[|\hat{\beta}_N) - f(\tau+)| > C\Big] \leq \\
&\mathbb{P}\left[|\hat{\beta}_N - f(\tau+)| > C \text{ and } |\hat{\tau}_N - \tau| \leq \frac{C_1}{N}\right] + \mathbb{P}\left[|\hat{\tau}_N - \tau| > \frac{C_1}{N}\right] \leq \\
&\mathbb{P}\left[|\hat{\beta}_N(t) - f(\tau+)| > C \text{ for some } |t - \tau| \leq \frac{C_1}{N}\right] + \frac{\epsilon}{2} \qquad (43)
\end{aligned}$$

where

$$\hat{\beta}_N(t) := \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) Y_{Nt+j}}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)} \tag{44}$$

Next, we bound the first term above, so consider only the case where $|t - \tau| \leq \frac{C_1}{N}$. By expanding we have

$$\begin{aligned}
|\hat{\beta}(t) - f(\tau+)| &= \left|\frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)[f(t + j/N) + \varepsilon_{Nt+j}]}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)} - f(\tau+)\right| \\
&\leq \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)|f(t + j/N) - f(\tau+)|}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)} + \left|\frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\varepsilon_{Nt+j}}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)}\right| \\
&:= A(t) + |B(t)| \qquad (45)
\end{aligned}$$

First, we derive a bound for $A(t)$. If $t \geq \tau$ then we have

$$\begin{aligned} A(t) &= \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) |f(t+j/N) - f(\tau+)|}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &\leq \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) \beta_f |t+j/N-\tau|}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &\leq \frac{\beta_f \sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) \left(\frac{C_1+j}{N}\right)}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &= \beta_f h \frac{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\left(\frac{j}{Nh}\right)}{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} + \frac{C_1}{N} \end{aligned} \tag{46}$$

Note that since $Nh \to \infty$ as $N \to \infty$, we have $\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\frac{j}{Nh} \to \int_0^1 xK(x)\,dx$ (which exists) and $\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right) \to \int_0^1 K(x)\,dx = 1$, as $N \to \infty$, hence we can find a constant $M > 0$ such that

$$A(t) \leq \beta_f M h + \frac{C_1}{N} \tag{47}$$

for all sufficiently large N. On the other hand, suppose $t < \tau$. For sufficiently large N we would have $N(\tau - t) \leq C_1 < Nh$ and so

$$\begin{aligned} A(t) &= \frac{\sum_{j=0}^{N(\tau-t)-1} K\left(\frac{j}{Nh}\right) |f(t+j/N) - f(\tau+)|}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} + \frac{\sum_{j=N(\tau-t)}^{Nh} K\left(\frac{j}{Nh}\right) |f(t+j/N) - f(\tau+)|}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &\leq \frac{\sum_{j=0}^{N(\tau-t)-1} K\left(\frac{j}{Nh}\right) (\Delta + \beta_f(\tau - t - j/N))}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} + \frac{\sum_{j=N(\tau-t)}^{Nh} K\left(\frac{j}{Nh}\right) \beta_f (t+j/N-\tau)}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &\leq \frac{K^{\uparrow} N(\tau-t)\Delta}{\sum_{j=1}^{Nh} K\left(\frac{j}{Nh}\right)} + \frac{\beta_f \sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\left(\frac{C_1+j}{N}\right)}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \\ &\quad (K^{\uparrow} \text{ is any constants that uniformly bounds the function } K \text{ from above on} [0,1]) \\ &\leq (K^{\uparrow}\Delta) \frac{\frac{1}{Nh} C_1}{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} + \beta_f h \frac{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\left(\frac{j}{Nh}\right)}{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} + \frac{C_1}{N} \end{aligned} \tag{48}$$

which, for sufficiently large N, can be bounded by $\frac{M_1}{Nh} + \beta_f M h + \frac{C_1}{N}$ for some constants $M, M_1 > 0$. Hence, this shows that $A(t)$ itself is $O(h \vee (Nh)^{-1})$ for all t where $|t - \tau| \leq \frac{C_1}{N}$.

Next, we consider the random term

$$B(t) = \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right) \varepsilon_{Nt+j}}{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)} \tag{49}$$

which satisfies

$$\begin{aligned}
\mathbb{E}[B(t)] &= 0 \\
\operatorname{var}(B(t)) &= \frac{\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)^2}{\left(\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\right)^2} \\
&= (Nh)^{-1} \frac{\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)^2}{\left(\frac{1}{Nh}\sum_{j=0}^{Nh} K\left(\frac{j}{Nh}\right)\right)^2} \\
&\leq (Nh)^{-1} 2 \int_0^1 K(x)^2\, dx, \qquad \text{for all sufficiently large } N. \qquad (50)
\end{aligned}$$

Thus $B(t) = O_p((Nh)^{-1/2})$ by Chebychev's inequality.

Combining these results on $A(t)$ and $B(t)$ derived above, one can find constants $C_2, C_3 > 0$ such that for all $N > N_2$ for some integer N_2 we have

$$\begin{aligned}
A(t) &\leq C_2[h \vee (Nh)^{-1}] \\
\mathbb{P}[|B(t)| > C_3(h \vee (Nh)^{-1/2})] &\leq \frac{\epsilon}{2(2C_1+3)} \qquad (51)
\end{aligned}$$

for all $|t-\tau| \leq \frac{C_1}{N}$, to get from (43):

$$\begin{aligned}
&\mathbb{P}\Big[|\hat{\beta}_N - f(\tau+)| > (C_1+C_2)(h \vee (Nh)^{-1/2})\Big] \leq \\
&\mathbb{P}\left[|\hat{\beta}_N(t) - f(\tau+)| > (C_1+C_2)(h \vee (Nh)^{-1/2}) \text{ for some } |t-\tau| \leq \frac{C_1}{N}\right] + \frac{\epsilon}{2} \leq \\
&\mathbb{P}\left[A(t) + |B(t)| > (C_1+C_2)(h \vee (Nh)^{-1/2}) \text{ for some } |t-\tau| \leq \frac{C_1}{N}\right] + \frac{\epsilon}{2} \leq \\
&\sum_{t:|t-\tau|\leq C_1/N} \mathbb{P}\left[A(t) + |B(t)| > (C_1+C_2)(h \vee (Nh)^{-1/2})\right] + \frac{\epsilon}{2} \leq \\
&\sum_{t:|t-\tau|\leq C_1/n} \mathbb{P}\left[|B(t)| > C_2(h \vee (Nh)^{-1/2})\right] + \frac{\epsilon}{2} \leq \\
&\sum_{t:|t-\tau|\leq C_1/N} \frac{\epsilon}{2(2C_1+3)} + \frac{\epsilon}{2} \leq \epsilon \qquad (52)
\end{aligned}$$

for all $N \geq N_1 \vee N_2$. This establishes that $|\hat{\beta}_N - f(\tau+)|$ is $O_p(h \vee (Nh)^{-1/2})$, and the proof for $|\hat{\alpha}_N - f(\tau-)|$ proceeds similarly. □

8.2.1. *Proof of Theorem 6.* The structure of this proof will be similar to the rate of convergence proof found in Lan et al (2007). We will initially set some notations: let $\tau_N := \lfloor N\tau \rfloor / N$, and define

$$\tau_N^{(2)} := \begin{cases} \tau_N & \text{if } \tau_N \text{ is not in first subsample} \\ \tau_N - 1/N & \text{if } \tau_N \text{ is in first subsample} \end{cases} \tag{53}$$

We will show that $\left(\hat{\tau}_N^{(2)} - \tau_N^{(2)}\right)$ is $O_p(1/N)$, which will also demonstrate the same rate of convergence for $\left(\hat{\tau}_N^{(2)} - \tau_N\right)$. An additional property of $\tau_N^{(2)}$, used later on, is the fact that $\lambda_2\left(\tau_N^{(2)}, \hat{\tau}_N^{(2)}\right) = \lambda_2\left(\tau_N, \hat{\tau}_N^{(2)}\right)$. This will be utilized in the proof of Theorems 2 and 4.

Proof. Denote G_N as the joint distribution of $(\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)})$. Given any constant $\epsilon > 0$, there is a positive constant C_ϵ such that for all sufficiently large N we have

$$\begin{aligned} (\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)}) &\in [f(\tau-) - \rho_N, f(\tau-) + \rho_N] \\ &\times [f(\tau+) - \rho_N, f(\tau+) + \rho_N] \times [\tau - C_\epsilon/N^\gamma, \tau + C_\epsilon/N^\gamma] \end{aligned} \tag{54}$$

with probability at least $1 - \epsilon$. Denote this event as R_N. It follows that for any sequence $\{a_N\}$,

$$\begin{aligned} &\mathbb{P}\Big[N|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| > a_N\Big] \leq \\ &\int_{R_N} \mathbb{P}\Big[N|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| > a_N \Big| (\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)}) = (\alpha, \beta, t)\Big]\, dG_N(\alpha, \beta, t) + \epsilon \leq \\ &\sup_{(\alpha,\beta,t)\in R_N} \mathbb{P}\Big[N|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| > a_N \Big| (\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)}) = (\alpha, \beta, t)\Big] + \epsilon \end{aligned} \tag{55}$$

Next, we show that this first term is smaller than any $\epsilon > 0$ for a sequence $a_N = O(1/N)$ and all sufficiently large N, by bounding the probability that

$$\mathbb{P}_{\alpha,\beta,t}\left[N|\hat{\tau}_N^{(2)} - \tau| > a_N\right] := \mathbb{P}\left[N|\hat{\tau}_N^{(2)} - \tau| > a_N \Big| (\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)}) = (\alpha, \beta, t)\right]$$

for any given $(\alpha, \beta, t) \in R_N$.

Conditional on the first stage estimates equaling (α, β, t), we can rewrite $\hat{\tau}_N^{(2)}$ and τ as

maximizers of:

$$\begin{aligned}
\hat{\tau}_N^{(2)} &= \underset{d\in S^{(2)}}{\arg\min}\left(\frac{1}{\lambda_2(S^{(2)}(t))}\sum_{i:i/N\in S^{(2)}(t)}\left(Y_i-\frac{\alpha+\beta}{2}\right)(1(i/N\le d)-1(i/N\le\tau))\right)\\
&:= \underset{d\in S^{(2)}}{\arg\min}\,\mathbb{M}_n(d)\\
\tau_N^{(2)} &= \underset{d\in S^{(2)}}{\arg\min}\left(\frac{1}{\lambda_2(S^{(2)}(t))}\sum_{i:i/N\in S^{(2)}(t)}\left(Y_i-\frac{\alpha+\beta}{2}\right)(1(i/N\le d)-1(i/N\le\tau))\right)\\
&:= \underset{d\in S^{(2)}}{\arg\min}\,M_n(d) \qquad (56)
\end{aligned}$$

Since $\tau_N^{(2)}-C_\epsilon/N^\gamma\le t\le\tau_N^{(2)}+C_\epsilon/N^\gamma+2/N$, for N large enough so that $KN^\delta/2>(C_\epsilon+2)$, we have $t-KN^{-\gamma+\delta}<\tau_N^{(2)}-KN^{-\gamma+\delta}/2<\tau_N^{(2)}+KN^{-\gamma+\delta}/2<t+KN^{-\gamma+\delta}$. This enables us to define the function $A(r)$ in the domain where $8N^{-1+\gamma-\delta}<r<K/2$, such that

$$\begin{aligned}
a(r) &:= \min\{M_n(d):|d-\tau_N^{(2)}|\ge rN^{-\gamma+\delta}\}\\
&= \min_{|d-\tau^{(2)}|\ge rN^{-\gamma+\delta}}\frac{\sum_{i:i/N\in S^{(2)}}\left(f(i/N)-\frac{\alpha+\beta}{2}\right)(1(i/N\le d)-1(i/N\le\tau))}{\lambda_2[t-KN^{-\gamma+\delta},t+KN^{-\gamma+\delta}]} \qquad (57)
\end{aligned}$$

To make $a(r)$ simpler to work with, we show that for sufficiently large N, there exists a constant $A>0$ such that $a(r)\ge Ar$. First, because

$$\begin{aligned}
S^{(2)}(t) &:= \left\{i/N:i\quad\in\mathbb{N},\quad i/N\in[t\pm KN^{-\gamma+\delta}],\quad \lfloor N/N_1\rfloor \text{ does not evenly divide } i\right\}\\
&\subset [t-KN^{-\gamma+\delta},t+KN^{-\gamma+\delta}]\\
&\subset [\tau^{(2)}-2KN^{-\gamma+\delta},\tau^{(2)}+2KN^{-\gamma+\delta}] \qquad (58)
\end{aligned}$$

this implies

$$\begin{aligned}
|f(i/N)-f(\tau+)|&\le 2\beta_f KN^{-\gamma+\delta} &&\text{for all } i/N\in S^{(2)}, i/n>\tau\\
|f(i/n)-f(\tau-)|&\le 2\beta_f KN^{-\gamma+\delta} &&\text{for all } i/n\in S^{(2)}, i/n\le\tau
\end{aligned}$$

Combine this with the fact that $|\alpha-f(\tau-)|$ and $|\beta-f(\tau+)|$ are $o(1)$, which implies for sufficiently large N, and for any $i/n\in S^{(2)}(t)$,

$$\begin{aligned}
f(i/n)-\frac{\alpha+\beta}{2}&>\frac{\Delta}{4} &&\text{if } i/n>\tau\\
f(i/n)-\frac{\alpha+\beta}{2}&<-\frac{\Delta}{4} &&\text{if } i/n\le\tau
\end{aligned}$$

The preceding fact implies that every term in the summand of (57) is positive, and therefore the

minimizing d for (57) would be either $\tau^{(2)} \pm rN^{-\gamma+\delta}$:

$$\begin{aligned} a(r) &= \left(\frac{\sum_{i:i/n \in S^{(2)}(t)} \left(f(i/n) - \frac{\alpha+\beta}{2} \right) (1(i/n \le \tau^{(2)} + rN^{-\gamma+\delta}) - 1(i/n \le \tau))}{\lambda_2[t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}]} \right) \wedge \\ &\quad \left(\frac{\sum_{i:i/n \in S^{(2)}(t)} \left(f(i/n) - \frac{\alpha+\beta}{2} \right) (1(i/n \le \tau^{(2)} - rN^{-\gamma+\delta}) - 1(i/n \le \tau))}{\lambda_2[t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}]} \right) \\ &\ge \frac{\Delta}{4} \cdot \frac{\lambda_2(\tau^{(2)}, \tau^{(2)} + rN^{-\gamma+\delta}] \wedge \lambda_2(\tau^{(2)} - rN^{-\gamma+\delta}, \tau^{(2)}]}{\lambda_2[t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}]} \end{aligned} \tag{59}$$

It can also be shown that for N large enough (specifically $\lfloor N^{1-\gamma} \rfloor \ge 2$) and any $d_1, d_2 \in [t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}]$ such that $d_2 - d_1 \ge 8/N$, we have

$$\lambda_2(d_1, d_2] \ge [N(d_2 - d_1) - 2] - \left[\frac{N(d_2 - d_1)}{\lfloor N^{1-\gamma} \rfloor} + 1 \right] \ge \frac{N(d_2 - d_1)}{8}$$

In a slightly similar fashion, it can be argued that for all large N, $\lambda_2[t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}] \le 3KN^{1-\gamma+\delta}$. Since we restricted r to be greater than $8N^{-1+\gamma-\delta}$, this means

$$\begin{aligned} a(r) &\ge \frac{\Delta}{4} \cdot \frac{NrN^{-\gamma+\delta}}{8 \cdot 3KN^{1-\gamma+\delta}} \\ &\ge \frac{\Delta}{96K} r \end{aligned} \tag{60}$$

Hence, this shows that $a(r)$ is greater than some linear function with 0 intercept.

Now define $b(r) = (a(r) - M_n(\tau_N^{(2)}))/3 = a(r)/3$, then we have the following relation:

$$\sup_{d \in S^{(2)}} |\mathbb{M}_n(d) - M_n(d)| \le b(r) \quad \Rightarrow \quad |\hat{\tau}_N^{(2)} - \tau_N^{(2)}| \le rN^{-\gamma+\delta} \tag{61}$$

To show the above is true, suppose $d \in [t - KN^{-\gamma+\delta}, t + KN^{-\gamma+\delta}]$ and $|d - \tau_N^{(2)}| > rN^{-\gamma+\delta}$. If, in addition, the left expression above holds, then

$$\begin{aligned} \mathbb{M}_n(d) \ge M_n(d) - b(r) \ge a(r) - b(r) \quad &\Rightarrow \\ \mathbb{M}_n(d) - \mathbb{M}_n(\tau^{(2)}) \ge a(r) - b(r) - M_n(\tau^{(2)}) - b(r) = b(r) > 0 \end{aligned} \tag{62}$$

Since $\mathbb{M}_n(d) > \mathbb{M}_n(\tau_N^{(2)})$ and $\hat{\tau}_N^{(2)}$ minimizes $\mathbb{M}_n$ among all points in $S^{(2)}(t)$, this implies d could not equal $\tau_N^{(2)}$, showing that $|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| \le rN^{-\gamma}$.

Next, we bound $\mathbb{P}_{\alpha,\beta,t}\left[|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| \le rN^{-\gamma+\delta} \right]$. First, we split it into the two parts:

$$\begin{aligned} &\mathbb{P}_{\alpha,\beta,t}\left[|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| > rN^{-\gamma+\delta} \right] \le \\ &\mathbb{P}_{\alpha,\beta,t}\left[rN^{-\gamma+\delta} < |\hat{\tau}_N^{(2)} - \tau_N^{(2)}| \le \eta N^{-\gamma+\delta} \right] + \mathbb{P}_{\alpha,\beta,t}\left[|\hat{\tau}_N^{(2)} - \tau_N^{(2)}| > \eta N^{-\gamma+\delta} \right] := \\ &P_N(\alpha, \beta, t) + Q_N(\alpha, \beta, t) \end{aligned} \tag{63}$$

where $\eta = K/3$. We first consider the term $P_n(\alpha, \beta, t)$. Because

$$\begin{aligned} rN^{-\gamma+\delta} < |\hat{\tau}_N^{(2)} - \tau_N^{(2)}| \le \eta N^{-\gamma+\delta} \quad &\Rightarrow \quad \inf_{\tau_N^{(2)}+rN^{-\gamma+\delta}<d\le\tau_N^{(2)}+\eta N^{-\gamma+\delta}} \mathbb{M}_n(d) \le \mathbb{M}_n(\tau) \\ &\text{or} \quad \inf_{\tau_N^{(2)}-\eta N^{-\gamma+\delta}<d\le\tau_N^{(2)}-rN^{-\gamma+\delta}} \mathbb{M}_n(d) \le \mathbb{M}_n(\tau), \end{aligned} \tag{64}$$

we can first split $P_n(\alpha, \beta, t)$ into the two terms

$$\begin{aligned} P_N(\alpha,\beta,t) &\le P_{N,1}(\alpha,\beta,t) + P_{N,2}(\alpha,\beta,t) \\ &=: \mathbb{P}_{\alpha,\beta,t}\left[\sup_{\tau_N^{(2)}+rN^{-\gamma+\delta}<d\le\tau_N^{(2)}+\eta N^{-\gamma+\delta}} (\mathbb{M}_n(\tau) - \mathbb{M}_n(d)) \ge 0\right] + \\ &\quad \mathbb{P}_{\alpha,\beta,t}\left[\sup_{\tau_N^{(2)}-\eta N^{-\gamma+\delta}<d\le\tau_N^{(2)}-rN^{-\gamma+\delta}} (\mathbb{M}_n(\tau) - \mathbb{M}_n(d)) \ge 0\right] \end{aligned} \tag{65}$$

We first form an upper bound for $P_{n,1}(\alpha, \beta, t)$ for all $(\alpha, \beta, t) \in R_n$. Note that

$$\begin{aligned} &\mathbb{M}_n(\tau_N^{(2)}) - \mathbb{M}_n(d) \\ = \; &-(\mathbb{M}_n(d) - M_n(d)) - M_n(d) \\ = \; &-\frac{\sum_{i:\, i/N\in S^{(2)}(t)} \left[\left(Y_i - \frac{\alpha+\beta}{2}\right) - \left(f(i/N) - \frac{\alpha+\beta}{2}\right)\right] (1(i/N \le d) - 1(i/N \le \tau))}{\lambda_2[t - KN^{-\gamma+\delta}, t + kN^{-\gamma+\delta}]} - M_n(d) \\ = \; &-\frac{\sum_{i:\, i/N\in S^{(2)}(t)\cap(\tau^{(2)},d]} \varepsilon_i}{\lambda_2[t - KN^{-\gamma+\delta}, t + kN^{-\gamma+\delta}]} - \frac{\sum_{i:\, i/N\in S^{(2)}(t)\cap(\tau^{(2)},d]} \left(f(i/N) - \frac{\alpha+\beta}{2}\right)}{\lambda_2[t - KN^{-\gamma+\delta}, t + kN^{-\gamma+\delta}]} \end{aligned} \tag{66}$$

As previously explained, the $\left(f(i/N) - \frac{\alpha+\beta}{2}\right)$ term in the second summand can be bounded below by $\Delta/4$ for all sufficiently large N, and hence this leads to:

$$\begin{aligned} \mathbb{M}_n(\tau) - \mathbb{M}_n(d) \ge 0 \quad &\Rightarrow \\ -\sum_{i:\, i/N\in S^{(2)}\cap(\tau^{(2)},d]} \varepsilon_i \ge \frac{\Delta}{4} &\lambda_2(\tau^{(2)}, d] \end{aligned} \tag{67}$$

It thus follows that

$$P_{N,1}(\alpha,\beta,t) \le \mathbb{P}_{\alpha,\beta,t}\left[\sup_{\substack{\tau_N^{(2)}+rN^{-\gamma+\delta}<d \\ \le\tau_N^{(2)}+\eta N^{-\gamma+\delta}}} \left(\frac{1}{\lambda_2(\tau^{(2)}, d]}\right) \left|\sum_{i:\, i/N\in S^{(2)}(t)\cap(\tau^{(2)},d]} \varepsilon_i\right| \ge \frac{\Delta}{4}\right] \tag{68}$$

and by the Hajek-Renyi inequality, we get

$$\begin{aligned}
&\mathbb{P}_{\alpha,\beta,t}\left[\sup_{\substack{\tau_N^{(2)}+rN^{-\gamma+\delta}<d\\ \leq\tau_N^{(2)}+\eta N^{-\gamma+\delta}}}\left(\frac{1}{\lambda_2(\tau,d]}\right)\left|\sum_{i:\, i/N\in S^{(2)}(t)\cap(\tau^{(2)},d]}\varepsilon_i\right|\geq\frac{\Delta}{4}\right]\\
\leq\;&\frac{16}{\Delta^2}\left(\frac{1}{\lambda_2(\tau^{(2)},\tau^{(2)}+rN^{-\gamma+\delta}]}+\sum_{j=\lambda_2(\tau^{(2)},\tau^{(2)}+rN^{-\gamma+\delta}]}^{\lambda_2(\tau^{(2)},\tau^{(2)}+\eta N^{-\gamma+\delta}]}\frac{1}{j^2}\right)\\
\leq\;&\frac{32}{\Delta^2}\cdot\frac{1}{\lambda_2(\tau^{(2)},\tau^{(2)}+rN^{-\gamma+\delta}]} \qquad (69)
\end{aligned}$$

We argued earlier that $\lambda_2(\tau^{(2)},\tau^{(2)}+rN^{-\gamma+\delta}]\geq rN^{1-\gamma+\delta}/8$ for N sufficiently large enough, thus

$$P_{N,1}(\alpha,\beta,t)\leq\frac{8B}{rN^{1-\gamma+\delta}} \qquad (70)$$

where $B=32/\Delta^2$. From this expression we arrive at $P_{N,1}(\alpha,\beta,t)\leq\epsilon$ (for any $\epsilon>0$) eventually, by setting $r=CN^{-1+\gamma-\delta}$ where C is any constant satisfying $C>8$ and $8B/C\leq\epsilon$.

To bound $Q_N(\alpha,\beta,t)$, from (60) we've argued that $a(r)$ is eventually greater than a multiple of r when $r>8N^{-1+\gamma-\delta}$. Since we've defined $b(r)=a(r)/3$, we can find some positive constant B' where $b(r)\geq B'r$ when $r>8N^{-1+\gamma-\delta}$ (and for all large N). Since $\eta=K/3>8N^{-1+\gamma-\delta}$, eventually, this leads to

$$\begin{aligned}
&\mathbb{P}_{\alpha,\beta,t}\left[|\hat{\tau}^{(2)}-d|>\eta N^{-\gamma+\delta}\right]\\
\leq\;&\mathbb{P}_{\alpha,\beta,t}\left[\sup_{d\in S^{(2)}(t)}|\mathbb{M}_n(d)-M_n(d)|>b(\eta)\right]\\
\leq\;&\mathbb{P}_{\alpha,\beta,t}\left[\sup_{d\in S^{(2)}(t)}|\mathbb{M}_n(d)-M_n(d)|>B'\eta\right]\\
=\;&\mathbb{P}_{\alpha,\beta,t}\left[\sup_{d\in S^{(2)}(t)}\frac{\left|\sum_{i:\, i/N\in S^{(2)}(t)}\epsilon_i(\mathbb{1}(i/N\leq d)-\mathbb{1}(i/N\leq\tau))\right|}{\lambda_2(S^{(2)}(t))}>B'\eta\right] \qquad (71)
\end{aligned}$$

Using Corollary 8.8 from [2], the latter expression is bounded by $C_1\exp(-C_2\eta^2\lambda_2(S^{(2)}(t)))$ for some positive constants C_1,C_2, which converges to 0. □

8.2.2. *Proof of Theorem 7.*

PROOF. Let $\{x_1^{(2)},x_2^{(2)},...\}$ be the x-coordinates of the data, not used in the first stage, with corresponding response variable $(Y_1^{(2)},Y_2^{(2)},...)$ and error terms $(\varepsilon_1^{(2)},\varepsilon_2^{(2)},...)$. As a set,

$\{x_1^{(2)}, x_2^{(2)}, ...\}$ equals $\{x_1, ..., x_N\} - \left\{\frac{\lfloor N/N_1 \rfloor}{N}, \frac{2\lfloor N/N_1 \rfloor}{N}, ...\right\}$. Note that we do not have $x_j^{(2)} = j/N$ for every integer j, and additionally we can write $\tau^{(2)} = x_m^{(2)}$ for some integer m. Since our estimate will also be one of the $x_i^{(2)}$'s, we can then denote $\hat{m}$ be the integer such that $\hat{\tau}^{(2)} = x_{\hat{m}}^{(2)}$. Note that we have the following relation between $\hat{m} - m$ and the λ_2 function on intervals:

$$\hat{m} - m = \begin{cases} \lambda_2(\tau^{(2)}, \hat{\tau}^{(2)}] & \text{when } \hat{\tau}^{(2)} > \tau \\ -\lambda_2(\hat{\tau}^{(2)}, \tau^{(2)}] & \text{when } \hat{\tau}^{(2)} \leq \tau \end{cases} \tag{72}$$

Hence we can write results on $\lambda_2(\tau^{(2)}, \hat{\tau}^{(2)})$ in terms of $\hat{m} - m$.

After taking a subset $S^{(2)}$ of $\{x_1^{(2)}, x_2^{(2)}, ..., x_{N-N_1}^{(2)}\}$ (specifically $S^{(2)}$ are those within $KN^{-\gamma+\delta}$ of the pilot estimate $\hat{\tau}^{(1)}$), we minimize

$$\begin{aligned} \hat{\Delta}^{(2)}(t) &:= \sum_{i: x_i \in S^{(2)}} \left(Y_i - \frac{\hat{\alpha}_N^{(1)} + \hat{\beta}_N^{(1)}}{2} \right) (1(x_i \leq t) - 1(x_i \leq \tau)) \\ &= \sum_{i: x_i^{(2)} \in S^{(2)}} \left(Y_i^{(2)} - \frac{\hat{\alpha}_N^{(1)} + \hat{\beta}_N^{(1)}}{2} \right) (1(x_i^{(2)} \leq t) - 1(x_i^{(2)} \leq \tau^{(2)})) \end{aligned} \tag{73}$$

over all points $t \in S^{(2)}$ to obtain the estimate for the change point. Equivalently the domain of $\hat{\Delta}^{(2)}(t)$ can be extended to all $t \in \{x_1^{(2)}, x_2^{(2)}, ...\}$, letting

$$\hat{\Delta}^{(2)}(t) = \max\left\{\hat{\Delta}^{(2)}(r) : r \in S^{(2)}\right\} + 1 \qquad \text{for } t \notin S^{(2)}$$

The argmin of this extension is the argmin of the function restricted to $S^{(2)}$. This extended definition will be used for the next result:

LEMMA 4. *For any fixed positive integer $j_0 > 0$,*

$$\begin{aligned} \hat{\Delta}^{(2)}\left(x_{m+j}^{(2)}\right) &= \frac{j\Delta}{2} + \epsilon_{m+1}^{(2)} + ... + \epsilon_{m+j}^{(2)} + o_p(1) \qquad \textit{for } 1 \leq j \leq j_0 \\ \hat{\Delta}^{(2)}\left(x_m^{(2)}\right) &= 0 + o_p(1) \\ \hat{\Delta}^{(2)}\left(x_{m-j}^{(2)}\right) &= \frac{j\Delta}{2} - \epsilon_m^{(2)} - ... - \epsilon_{m-j+1}^{(2)} + o_p(1) \qquad \textit{for } 1 \leq j \leq j_0 \end{aligned} \tag{74}$$

From this lemma it is straightforward to show the asymptotic distribution of $\lambda_2\left(\tau_N, \hat{\tau}_N^{(2)}\right)$ is the distribution of $L_{\Delta/\sigma}$, the argmax of the random process

$$X_{\Delta/\sigma}(j) = \begin{cases} \frac{|j|\Delta}{2} - \varepsilon_{-1}^* - ... - \varepsilon_j^* & \text{, for } j < 0 \\ 0 & \text{, for } j = 0 \\ \frac{j\Delta}{2} + \varepsilon_1^* + ... + \varepsilon_j^* & \text{, for } j > 0 \end{cases} \tag{75}$$

where the $\{\varepsilon_j\}_{j\in\mathbb{Z}}$ are iid $N(0,\sigma^2)$ random variables.

For any fixed $\epsilon > 0$ and integer j, we will show that $\left|\mathbb{P}\left[\hat{m} - m = j\right] - \mathbb{P}\left[L_{\Delta/\sigma} = j\right]\right| \leq \epsilon$ for all sufficiently large N. To do this we will first establish 3 probability bounds.

First Bound: First we will show that with high probability we can approximate the stochastic process $L_{\Delta/\sigma}$, which has support $\mathbb{Z}$, with a stochastic process $L^{(k)}_{\Delta/\sigma}$, which has a finite support $\mathbb{Z} \cap [-k, k]$.

We note that there exists an integer $j_1 > |j|$, such that $|L_{\Delta/\sigma}| > j_1$ with probability less than $\epsilon/3$. For any integer k with $k \geq j_1$, define $L^{(k)}_{\Delta/\sigma} := \underset{|i|\leq k}{\arg\min}\{X_{\Delta/\sigma}(i)\}$. In the case that $|L_{\Delta/\sigma}| \leq k$, we have $L^{(k)}_{\Delta/\sigma} = L_{\Delta/\sigma}$, and using this we can show that $\mathbb{P}[L_{\Delta/\sigma} = j]$ is within $\epsilon/3$ of $\mathbb{P}[L^{(k)}_{\Delta/\sigma} = j]$:

$$\begin{aligned}
& \left|\mathbb{P}[L_{\Delta/\sigma} = j] - \mathbb{P}[L^{(k)}_{\Delta/\sigma} = j]\right| \\
= & \left|\mathbb{P}\left[L_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right] - \mathbb{P}\left[L^{(k)}_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right] - \mathbb{P}\left[L^{(k)}_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| > k\right]\right| \\
\leq & \left|\mathbb{P}\left[L_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right] - \mathbb{P}\left[L^{(k)}_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right]\right| + \mathbb{P}[|L_{\Delta/\sigma}| > k] \\
= & \left|\mathbb{P}\left[L^{(k)}_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right] - \mathbb{P}\left[L^{(k)}_{\Delta/\sigma} = j, |L_{\Delta/\sigma}| \leq k\right]\right| + \mathbb{P}[|L_{\Delta/\sigma}| > k] \\
\leq & \ 0 + \frac{\epsilon}{3} \qquad (76)
\end{aligned}$$

Second Bound: We will show that there exists an integer $j_0 > j_1$ such that $|\hat{m} - m| \leq j_0$ with probability greater than $1 - \frac{\epsilon}{3}$. From our theorem on the rate of convergence, we can find some integer $j_0 > j_1$ such that for all sufficiently large N,

$$\mathbb{P}\left[|\hat{\tau}^{(2)} - \tau| \leq \frac{j_0 - 2}{N}\right] > 1 - \frac{\epsilon}{3}. \qquad (77)$$

When $|\hat{\tau}^{(2)} - \tau| \leq \frac{j_0-2}{N}$, we have $|\hat{m} - m| \leq j_0$; first we can show

$$\begin{aligned}
\left|x^{(2)}_{\hat{m}} - x^{(2)}_{m}\right| & \leq \left|\hat{\tau}^{(2)} - \tau\right| + \left|\tau - \tau^{(2)}\right| \\
& \leq \frac{j_0 - 2}{N} + \frac{2}{N} \\
& = \frac{j_0}{N}, \qquad (78)
\end{aligned}$$

and second, because the $\left\{x^{(2)}_1, x^{(2)}_2, ...\right\}$ grid is just the equally spaced $\{1/N, 2/N, ..., N/N\}$ with

some points taken out, the result of (78) implies $|\hat{m} - m| \leq j_0$. Hence

$$\begin{aligned} \mathbb{P}\left[|\hat{m} - m| \leq j_0\right] &\geq \mathbb{P}\left[|\hat{\tau}^{(2)} - \tau| \leq \frac{j_0 - 2}{N}\right] \\ &> 1 - \frac{\epsilon}{3} \end{aligned} \tag{79}$$

Third Inequality: Define $\hat{\tau}^{(2)}_{j_0}$ to be the minimizer of $\hat{\Delta}^{(2)}(\cdot)$ on the set $\left\{x^{(2)}_{m-j_0}, x^{(2)}_{m-j_0+1}, ..., x^{(2)}_{m+j_0}\right\}$, and let $\hat{m}_{j_0}$ be its corresponding index such that $\hat{\tau}^{(2)}_{j_0} = x^{(2)}_{\hat{m}_{j_0}}$. In the case when $|\hat{m} - m| \leq j_0$, then $\hat{\tau}^{(2)}_{j_0}$ would be equal to $\hat{\tau}^{(2)}$, and $\hat{m} = \hat{m}_{j_0}$. Using this notation we can obtain the following bound:

$$\begin{aligned} &|\mathbb{P}\left[\hat{m} - m = j\right] - \mathbb{P}\left[\hat{m}_{j_0} - m = j\right]| \\ &= \left|\mathbb{P}\left[\hat{m} - m = j, |\hat{m} - m| \leq j_0\right] - \mathbb{P}\left[\hat{m}_{j_0} - m = j, |\hat{m} - m| \leq j_0\right] - \mathbb{P}\left[\hat{m}_{j_0} - m = j, |\hat{m} - m| > j_0\right]\right| \\ &= \left|\mathbb{P}\left[\hat{m}_{j_0} - m = j, |\hat{m} - m| \leq j_0\right] - \mathbb{P}\left[\hat{m}_{j_0} - m = j, |\hat{m} - m| \leq j_0\right] - \mathbb{P}\left[\hat{m}_{j_0} - m = j, |\hat{m} - m| > j_0\right]\right| \\ &\leq \mathbb{P}\left[|\hat{m} - m| > j_0\right] \\ &\leq \epsilon/3 \end{aligned} \tag{80}$$

Consider the stochastic process $\hat{\Delta}^{(2)}(x^{(2)}_{m+i})$ for $i \in \{-j_0, ..., 0, ..., j_0\}$. The previous lemma showed that, as a random variable in $\mathbb{R}^{2j_0+1}$, $\left(\hat{\Delta}^{(2)}(x^{(2)}_{m-j_0}), ..., \hat{\Delta}^{(2)}(x^{(2)}_{m+j_0})\right)$ converges in distribution to

$$(X_{\Delta/\sigma}(-j_0), ..., X_{\Delta/\sigma}(j_0)).$$

Also consider the function $\text{Ind}_{min} : \mathbb{R}^{2j_0+1} \to \mathbb{Z}$, defined as

$$\text{Ind}_{min}(a_1, ..., a_{2j_0+1}) = \left(\underset{i=1,...,2j_0+1}{\arg\min}\ (a_i)\right) - (j_0 + 1). \tag{81}$$

It can be easily checked that Ind_{min} is a continuous function, and by definition, we also have

$$\begin{aligned} L^{j_0}_{\Delta/\sigma} &= \text{Ind}_{min}(X_{\Delta/\sigma}(-j_0), ..., X_{\Delta/\sigma}(j_0)) \\ \hat{m}_{j_0} - m &= \text{Ind}_{min}\left(\hat{\Delta}^{(2)}(x^{(2)}_{m-j_0}), ..., \hat{\Delta}^{(2)}(x^{(2)}_{m+j_0})\right). \end{aligned} \tag{82}$$

Hence, by the continuous mapping theorem we have $\hat{m}_{j_0} - m$ converging to $L^{j_0}_{\Delta/\sigma}$ in distribution. For sufficiently large N, the absolute difference between $\mathbb{P}[L^{j_0}_{\Delta/\sigma} = j]$ and $\mathbb{P}[\hat{m}_{j_0} - m = j]$ will be

less than $\epsilon/3$.

Combining what we have just shown, for sufficiently large N we have

$$\begin{aligned} & \left|\mathbb{P}\left[\hat{m}-m=j\right]-\mathbb{P}\left[L_{\Delta/\sigma}=j\right]\right| \\ \leq & \left|\mathbb{P}\left[\hat{m}-m=j\right]-\mathbb{P}\left[\hat{m}_{j_0}-m=j\right]\right|+\left|\mathbb{P}\left[\hat{m}_{j_0}-m=j\right]-\mathbb{P}\left[L_{\Delta/\sigma}^{j_0}=j\right]\right| \\ & +\left|\mathbb{P}\left[L_{\Delta/\sigma}^{j_0}=j\right]-\mathbb{P}[L_{\Delta/\sigma}=j]\right| \\ \leq & \frac{\epsilon}{3}+\frac{\epsilon}{3}+\frac{\epsilon}{3} \end{aligned} \tag{83}$$

□

Proof of Lemma 4

PROOF. First note that with probability increasing to 1, $x_{m-j_0}^{(2)}, x_{m-j_0+1}^{(2)}, ..., x_{m+j_0}^{(2)}$ are all contained inside $S^{(2)}$, and this fact will be shown first. Since $\hat{\tau}^{(1)}-\tau=O_p(N^{-\gamma})$, for any $\epsilon>0$ it is possible to find a constant $C>0$ such that

$$\mathbb{P}\left[\hat{\tau}^{(1)}-CN^{-\gamma}\leq\tau\leq\hat{\tau}^{(1)}+CN^{-\gamma}\right]>1-\epsilon \tag{84}$$

for all sufficiently large N. Additionally, for all sufficiently large N we have $\frac{4+j_0}{N}\leq(KN^{\delta}-C)N^{-\gamma}$, and which means that if $|\tau^{(1)}-\tau|\leq CN^{-\gamma}$ then

$$\begin{aligned} \hat{\tau}^{(1)}-KN^{-\gamma+\delta} &= \hat{\tau}^{(1)}-(KN^{-\gamma}-C)N^{-\gamma}-CN^{-\gamma} \\ &\leq \tau-\frac{4+2j_0}{N} \\ \hat{\tau}^{(1)}+KN^{-\gamma+\delta} &= \hat{\tau}^{(1)}+(KN^{-\gamma}-C)N^{-\gamma}+CN^{-\gamma} \\ &\geq \tau+\frac{4+2j_0}{N} \end{aligned} \tag{85}$$

Finally, for all sufficiently large N, we have $\lfloor N^{1-\gamma}\rfloor>2$, i.e. the first stage subsample chooses points which are spaced more than $2/N$ points apart. Hence,

$$\begin{aligned} x_{m-j_0}^{(2)} &\geq x_m^{(2)}-2\left(\frac{j_0+2}{N}\right)=\tau^{(2)}-2\left(\frac{j_0+2}{N}\right) \\ &\geq \tau-\frac{2j_0+4}{N} \\ x_{m+j_0}^{(2)} &\leq x_m^{(2)}+2\left(\frac{j_0+2}{N}\right) \\ &\leq \tau+\frac{2j_0+4}{N}, \end{aligned} \tag{86}$$

which leads to the conclusion that for all N large enough, we have

$$\begin{aligned}
1-\epsilon &< \mathbb{P}\left[\hat{\tau}^{(1)} - CN^{-\gamma} \le \tau \le \hat{\tau}^{(1)} + CN^{-\gamma}\right] \\
&\le \mathbb{P}\left[\hat{\tau}^{(1)} - KN^{-\gamma+\delta} \le \tau - \frac{2j_0+4}{N} < \tau + \frac{2j_0+4}{N} \le \hat{\tau}^{(1)} + KN^{-\gamma+\delta}\right] \\
&\le \mathbb{P}\left[x^{(2)}_{m-j_0} \ge \hat{\tau}^{(1)} - KN^{-\gamma+\delta} \text{ and } x^{(2)}_{m+j_0} \le \hat{\tau}^{(1)} + KN^{-\gamma+\delta}\right] \\
&= \mathbb{P}\left[x^{(2)}_{m-j_0} \text{ and } x^{(2)}_{m+j_0} \text{ are in } S^{(2)}\right] \qquad (87)
\end{aligned}$$

Therefore, consider the case for which $x^{(2)}_{m-j_0}$ through $x^{(2)}_{m+j_0}$ are contained in $S^{(2)}$. Under this condition we have $\hat{\Delta}^{(2)}\left(x^{(2)}_m\right) = 0$ by simple calculation, and for any $0 < j \le j_0$,

$$\begin{aligned}
&\left|\hat{\Delta}^{(2)}(x^{(2)}_{m+j}) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j}\epsilon^{(2)}_{m+i}\right)\right| \\
= &\left|\sum_{i:x^{(2)}_i \in S^{(2)} \cap \left(x^{(2)}_m, x^{(2)}_{m+j}\right]} \left(Y^{(2)}_i - \frac{\hat{\alpha}_N + \hat{\beta}_N}{2}\right) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j}\epsilon^{(2)}_{m+i}\right)\right| \\
= &\left|\sum_{i=1}^{j}\left(f(x^{(2)}_{m+i}) - \frac{\hat{\alpha}^{(1)}_N + \hat{\beta}^{(1)}_N}{2}\right) - \frac{j\Delta}{2}\right| \\
= &\left|\sum_{i=1}^{j}\left[\left(f(x^{(2)}_{m+i}) - f(\tau+)\right) + \left(\frac{f(\tau+)+f(\tau-)}{2} - \frac{\hat{\alpha}^{(1)}_N + \hat{\beta}^{(1)}_N}{2}\right) + \right.\right. \\
&\left.\left.\left(\frac{f(\tau+)-f(\tau-)}{2} - \frac{\Delta}{2}\right)\right]\right| \\
\le &\left|\sum_{i=1}^{j}\left(f(x^{(2)}_{m+i}) - f(\tau+)\right)\right| + \frac{j_0}{2}\left(\left|\hat{\alpha}^{(1)}_N - f(\tau-)\right| + \left|\hat{\beta}^{(1)}_N - f(\tau+)\right|\right) \qquad (88)
\end{aligned}$$

For the first term above, using an earlier argument we make the case that for sufficiently large N we have $\lfloor N/N_1 \rfloor > 2$ and $x^{(2)}_{m+i} \le \tau + \frac{2i+4}{N}$, hence

$$\begin{aligned}
\left|\sum_{i=1}^{j}\left(f(x^{(2)}_{m+i}) - f(\tau+)\right)\right| &\le \sum_{i=1}^{j}\beta_f\left|x^{(2)}_{m+i} - \tau\right| \\
&\le \beta_f \sum_{i=1}^{j}\frac{2i+4}{N} \\
&\le \frac{\beta_f}{N}(2j_0^2 + 4j_0) \qquad (89)
\end{aligned}$$

For the second term, it was shown earlier that both $\left|\hat{\alpha}_N^{(1)} - f(\tau-)\right|$ and $\left|\hat{\beta}_N^{(1)} - f(\tau+)\right|$ are $O_p\left(h \vee \sqrt{\frac{1}{N_1 h}}\right)$, and hence, so is their sum. Overall, this shows that for sufficiently large N, $\left|\hat{\Delta}^{(2)}(x_{m+j}^{(2)}) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j} \epsilon_{m+i}^{(2)}\right)\right|$ is (uniformly for all $0 \le j \le j_0$) bounded above by the random variable

$$\frac{\beta_f}{N}(2j_0^2 + 4j_0) + \frac{j_0}{2}\left(\left|\hat{\alpha}_N^{(1)} - f(\tau-)\right| + \left|\hat{\beta}_N^{(1)} - f(\tau+)\right|\right), \tag{90}$$

which is $o_p(1)$. Similarly, again for any $0 < j \le j_0$,

$$\begin{aligned} & \left|\hat{\Delta}^{(2)}(x_{m-j}^{(2)}) - \left(\frac{j\Delta}{2} - \sum_{i=1}^{j} \epsilon_{m-i+1}^{(2)}\right)\right| \\ \le\ & \left|\sum_{i=0}^{j-1}\left(f(x_{m-i}^{(2)}) - f(\tau)\right)\right| + \frac{j_0}{2}\left(\left|\hat{\alpha}_N^{(1)} - f(\tau-)\right| + \left|\hat{\beta}_N^{(1)} - f(\tau+)\right|\right) \end{aligned} \tag{91}$$

which is again uniformly bounded by the expression in (90).

Therefore, given any $\epsilon > 0$, we have

$$\begin{aligned} & \mathbb{P}\left[\left|\hat{\Delta}^{(2)}(x_{m+j}^{(2)}) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j} \epsilon_{m+i}^{(2)}\right)\right| \ge \epsilon\right] \\ \le\ & \mathbb{P}[x_{m-j_0}^{(2)} \notin S^{(2)} \text{ and/or } x_{m+j_0}^{(2)} \notin S^{(2)}] + \\ & \mathbb{P}\left[x_{m-j_0}^{(2)}, x_{m+j_0}^{(2)} \in S^{(2)}, \text{ and } \left|\hat{\Delta}^{(2)}(x_{m+j}^{(2)}) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j} \epsilon_{m+i}^{(2)}\right)\right| \ge \epsilon\right] \\ \le\ & \mathbb{P}[x_{m-j_0}^{(2)} \notin S^{(2)} \text{ and/or } x_{m+j_0}^{(2)} \notin S^{(2)}] + \\ & \mathbb{P}\left[\frac{\beta_f}{N}(2j_0^2 + 4j_0) + \frac{j_0}{2}\left(\left|\hat{\alpha}_N^{(1)} - f(\tau-)\right| + \left|\hat{\beta}_N^{(1)} - f(\tau+)\right|\right) \ge \epsilon\right] \\ \to\ & 0 + 0 \qquad \text{for all } 0 < j \le j_0 \end{aligned} \tag{92}$$

and similarly,

$$\begin{aligned} & \mathbb{P}\left[\left|\hat{\Delta}^{(2)}(x_m^{(2)})\right| \ge \epsilon\right] \to 0 \\ & \mathbb{P}\left[\left|\hat{\Delta}^{(2)}(x_{m-j}^{(2)}) - \left(\frac{j\Delta}{2} + \sum_{i=1}^{j} \epsilon_{m-i+1}^{(2)}\right)\right| \ge \epsilon\right] \to 0 \end{aligned} \tag{93}$$

□

9. Supplement Part B (Multiple Change Point Problem). Here we will provide analysis of topics and verification of results found in Section 3 of the main paper. In the first subsection, we will introduce and describe a method that is very practicable for calibrating the width of the intervals used at step (ISM5). In the following subsection, we have a subsection on the analysis used for Section 4.2. This is followed by two discussion sections on the numerical savings of intelligent sampling in practice and a discussion on dependent error terms. All subsections that follow will be proofs of the results found in the main paper and in the earlier subsections here.

9.1. *Calibration of intervals used in Stage 2 of Intelligent Sampling.* Constructing confidence intervals based on Lemma 1 would require working with either the $C_2 \log(N^*)$ expression, where an explicit expression for C_2 is unavailable; or with a much wider interval, say of half-width $\beta_{N^*} \log N^*$ with a series $\beta_{N^*} \to \infty$ ($\beta_{N^*} = \log N^*$ being a possibility). The latter method also presents issues; there are no guiding principles to choosing the right β_{N^*} sequence to work well in practice, and an arbitrarily chosen β_{N^*} could have sub-par performance on a fixed data set despite being sound in the asymptotic sense. To address this issue, we introduce a calibration method which allows the construction of confidence intervals with explicitly calculable width around the first stage estimates $\hat{\tau}_j^{(1)}$'s: the idea is to fit stump models on data with indices $[\hat{\tau}_{j-1}^{(1)} + 1, \hat{\tau}_{j+1}^{(1)}]$, as each of these stretches forms a stump model with probability going to 1.

Consider starting right after step (ISM4) (e.g., Figure 19), where we have rough estimates $\hat{\tau}_i^*$'s of the change points (with respect to the $\{Z_i\}$ sequence) and $\hat{\nu}_i^{(1)}$'s of the signals, obtained from the N^* sized subsample $\{Z_i\}$.

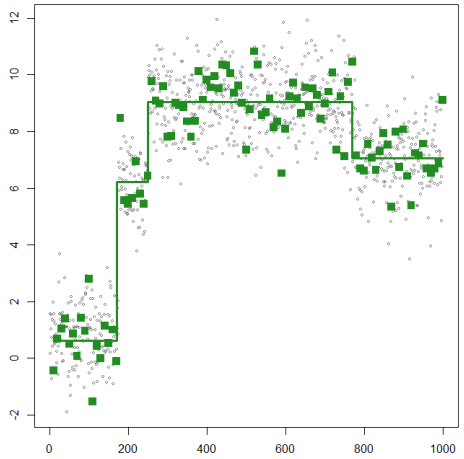


Fig 19: Green points are Z_i's, solid green line is the BinSeg estimate.

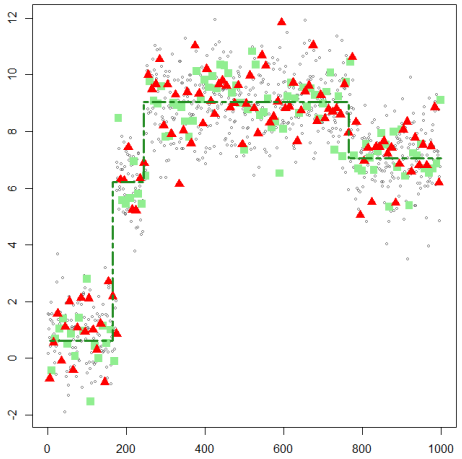


Fig 20: Z_i's are light green points, BinSeg estimates as dashed green line, V_i's as red points.

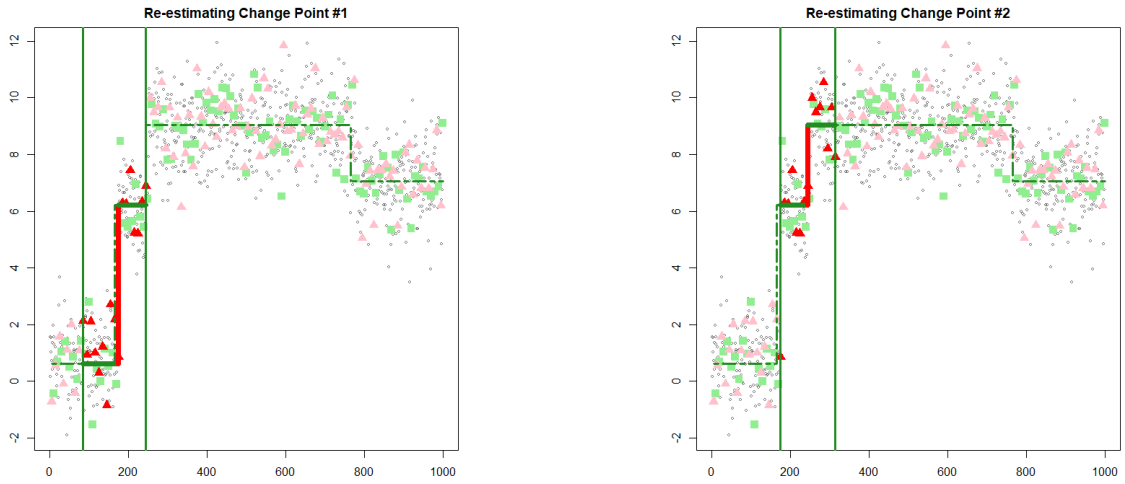


Fig 21: Re-estimation of the first (**Left Panel**) and second (**Right Panel**) detected change points (similar procedure for third estimated change point not shown). Solid green and solid red lines denote stump estimates using V_i's from $\{V_k : \hat{\tau}_j^* - \hat{d}_j \leq k \leq \hat{\tau}_j^* + \hat{d}_j\}$ intervals.

We then pick a different subsample $\{V_i\}$ of equal size to the $\{Z_i\}$ subsample and consider the $\hat{\tau}_i^*$'s and $\hat{\nu}_i^{(1)}$'s as estimates for the parameters of this data sequence (e.g. Figure 20). For each j, fit a one-parameter stump model $f_j^{(d)}(k) = \hat{\nu}_{j-1}^{(1)} 1(k \leq d) + \hat{\nu}_j^{(1)} 1(k > d)$ (here d is the discontinuity parameter) to the subset of $\{V_k : \hat{\tau}_{j-1}^* + 1 \leq k \leq \hat{\tau}_{j+1}^*\}$ given by $\{V_k : \hat{\tau}_j^* - \hat{d}_j \leq k \leq \hat{\tau}_j^* + \hat{d}_j\}$ where $\hat{d}_j = \min\{(\hat{\tau}_j^* - \hat{\tau}_{j-1}^*), (\hat{\tau}_{j+1}^* - \hat{\tau}_j^*)\}$, to get an updated least squares estimate of τ_j (e.g. Figure 21)[1].

Formally, the calibration steps are:

(ISM4-1): pick a positive integer k_N less than $\lfloor N/N_1 \rfloor$ [2], take a subsample $\{V_i\}$ from the dataset of $\{Y_i\}$ which is the same size as the $\{Z_i\}$ subsample, by letting $V_i = Y_{i\lfloor N/N_1 \rfloor - k_N}$ for all i.

The $\{V_i\}$ subsample also conforms to the model given in (11), with change points $\tau_i^{**} = \max\{j \in \mathbb{N} : j\lfloor N/N_1 \rfloor - k_N \leq \tau_i\}$ and minimum spacing $\delta_{N^*}^{**} := \min_k(\tau_{k+1}^{**} - \tau_k^{**})$ which satisfies $|\delta_{N^*}^{**} - \delta_{N^*}^{*}| \leq 1$.

(ISM4-2): For each $i = 1, ..., J$, consider the estimates $\hat{\tau}_i^*$ (obtained from the $\{Z_j\}$ subsample at step (ISM3)) as estimators for τ_i^{**}. From (13) it is possible to derive that

$$\mathbb{P}\left[\hat{J} = J, \max_{i=1,...,J} |\hat{\tau}_i^* - \tau_i^{**}| \leq w^*(N^*) + 1, \max_{i=0,...,J} |\hat{\nu}_i^{(1)} - \nu_i| \leq \rho_N\right] \to 1 \tag{94}$$

where $w^*(N) + 1 \to \infty$ and $(w^*(N^*) + 1)/\delta_{N^*}^{**} \to 0$.

[1] This subset is used instead of the full interval to avoid situations where $\frac{\tau_j - \tau_{j-1}}{\tau_{j+1} - \tau_j} \to 0$ or ∞, which makes matters easier for theoretical derivations.

[2] A good pick is $k_N = \left\lfloor \frac{\lfloor N/N_1 \rfloor}{2} \right\rfloor$

(ISM4-3): For each $i = 1, ..., \hat{J}$, define $\hat{d}_i = \min\{(\hat{\tau}^*_{i+1} - \hat{\tau}^*_i), (\hat{\tau}^*_i - \hat{\tau}^*_{i-1})\}$ (with $\hat{\tau}^*_0 = 0$ and $\hat{\tau}^*_{\hat{J}+1} = \lfloor N/\lfloor N/N_1 \rfloor \rfloor$), and re-estimate the change points by letting

$$\hat{\tau}^{re}_i := \underset{t:|t-\hat{\tau}^*_i|<\hat{d}_i}{\arg\min} \left[\sum_{j:|j-\hat{\tau}^*_i|<\hat{d}_i} \left(V_i - \hat{\nu}^{(1)}_{i-1} 1(j \le t) - \hat{\nu}^{(1)}_i 1(j > t) \right)^2 \right] \tag{95}$$

for $i = 1, ..., \hat{J}$.

(ISM4-4): To translate the $\hat{\tau}^{re}_i$'s (change point estimates for the subsample for the $\{V_j\}$'s) into estimates for $\tau_1, \ldots, \tau_J$ (change points for the full data set), set the first stage change point estimators as $\hat{\tau}^{(1)}_i := \hat{\tau}^{re}_i \lfloor N/N_1 \rfloor - k_N$.

REMARK 11. *Note that any change point detection procedure used at stage (ISM3) can be supplemented by these procedures, so long as it satisfies certain consistency conditions. More broadly, these steps can be used outside of the intelligent sampling framework. Given a consistent change point estimation scheme* for which a method to construct explicit confidence intervals is not known*, one could split the data in two subsamples, the odd points (first, third, fifth, etc data points) and the even points. The aforementioned estimation scheme could be applied to the odd points, and the subsequent steps (ISM4-1) to (ISM4-4) could be applied to the even points. A result similar to that of Theorem 8, presented below, could then be used to construct confidence intervals.*

Using these steps, we can express the deviations of the $\hat{\tau}^{re}$ estimators using the quantiles $Q_{\Delta,\sigma_1,\sigma_2}$ that were defined in the paragraph before (20).

THEOREM 8. *Suppose conditions (M1) to (M3), and (M4 (SBS)) are satisfied, and the estimation method used in step (ISM3) satisfies (13), and the pertinent ρ_N appearing in (13) also satisfies $J\rho_N \to 0$. For any sequence*[3] *α_N between 0 and 1 such that $\alpha_N \ge CN^{-\eta}$ for some positive constants C and η, we have*

$$\mathbb{P}\left[\hat{J} = J, |\hat{\tau}^{re}_j - \tau^{**}_j| \le Q_{\Delta_j,\sigma,\sigma}\left(1 - \frac{\alpha_N}{J}\right) \text{ for } j = 1, \ldots, J \right] \ge 1 - \alpha_N + o(\alpha_N) \tag{96}$$

PROOF. See Section 9.9 of Supplement Part B. □

The practical implication of this result for implementing intelligent sampling is that we can obtain explicitly calculable simultaneous confidence intervals for the change-points. The intervals $[\hat{\tau}^{re}_j \pm Q_{\Delta_j,\sigma,\sigma}\left(1 - \frac{\alpha_N}{J}\right)]$ for $j = 1, \cdots, J$ capture the sparse scale change points τ^{**}_j for $j = 1, \cdots, J$ with probability approaching 1 if we choose some $\alpha_N \to 0$. Converting back to the original scale, the intervals $[\hat{\tau}^{(1)}_j \pm (Q_{\Delta_j,\sigma,\sigma}\left(1 - \frac{\alpha_N}{J}\right) + 1)\lfloor N/N_1 \rfloor]$ for $j = 1, \cdots, \hat{J}$ have the properties that they

[3]The sequence could be a sequence that converges to 0 or not converge to 0, it only needs to stay between 0 and 1, and satisfy the bound $\alpha \ge cN^{-\eta}$.

simultaneously capture $\tau_1, \cdots, \tau_J$ with probability approaching 1.[4] The second stage samples are then picked as points within each of these intervals that are not the form $i\lfloor N/N_1 \rfloor$ or $i\lfloor N/N_1 \rfloor - k$ for some integer i (since the first stage points cannot be used at steps (ISM5) and (ISM6)).

Similar to Theorem 5, the condition $J\rho_N \to 0$ is required here. This is because the proofs of both results are similar in structure. For intelligent sampling with SBS used in stage 1 (i.e. at step (ISM3)), conditions (M1) to (M3) and (M4 (SBS)) are already requirements for the consistency results of the procedure (Lemmas 1 and 2). but the condition of $J\rho_N \to 0$ is not automatically clear from these conditions alone. To satisfy this additional point, we place the requirement of

(M7 (SBS)) The Ξ of condition (M3) is restricted to $\Xi \in [0, 1/3)$, and the γ chosen at step (ISM2) satisfies $\gamma > 3\Xi$.

With this additional condition, we can arrive at the following result which will permit us to use steps (ISM4-1) to (ISM4-4):

LEMMA 5. *Suppose conditions (M1), (M2), (M3), (M4 (SBS)), and (M7 (SBS)) are satisfied. There exist constants C_1 and C_2 such that if SBS is performed at step ISM3 on the subsample $Z_1, \ldots, Z_{N^*}$ with $\zeta_{N^*} = C_1\sqrt{\log(N^*)}$, and signal estimates are obtained as in (31), then there exists a sequence $\rho^*_{N^*} \to 0$ such that*

$$\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\ldots,J} |\hat{\tau}^*_i - \tau^*_i| \leq C_2 \log(N^*); \quad \max_{i=0,\ldots,J} |\hat{\nu}^*_i - \nu^*_i| \leq \rho^*_{N^*}\right] \to 1 \tag{97}$$

and $J\rho^*_{N^*} \to 0$ as $N^* \to \infty$.

The verification of this result can be found in Section 9.10.

9.2. *Detailed Computational Time Analysis.* In this section we will give a more detailed analysis on the computational time that was briefly touched upon in Section 4.2 of the main paper. We will make the same assumption that $\delta_N/N^{1-\Xi} \to C_1$ and $J(N)/N^{\Lambda} \to C_2$ for some $\Lambda \in [0, \Xi]$ and some positive constants C_1, C_2.

The SBS procedure, when applied to a data sequence of n points, has $O(n \log n)$ computational complexity (see [3]). Since first stage of intelligent sampling (steps up to (ISM3)) involves applying SBS to $O(N^\gamma)$ points, it therefore takes $O(N^\gamma \log(N))$ time to obtain the first stage estimators. After the SBS estimates are obtained, we have the option to use the method described in Section 9.1, in order to upgrade them to estimates whose deviation quantiles are known. Even if we choose to use this subsequent step, it only involves least squares fitting upon $O(N^\gamma)$ points and therefore requires only $O(N^\gamma)$ computational time, and the total computational order is still $O(N^\gamma \log(N))$ up to this point.

[4] In practice, the Δ_j/σ's are estimated from data.

Whether one implements the Section 9.1 procedures, denote $\left\{\hat{\tau}_1^{(1)}, \ldots \hat{\tau}_{\hat{J}}^{(1)}\right\}$ as the estimates for $\{\tau_1, \ldots, \tau_J\}$ obtained up to right before implementing step (ISM5). At this point, we discuss possibilities in terms of how to construct the second stage intervals:

(Int-1): We can use expression (33) to construct intervals $\left[\hat{\tau}_j^{(1)} \pm \beta_{N^*} \log(N) \lfloor \frac{N}{N_1} \rfloor\right]$ that we will use to obtain the final estimates. Setting $\beta_{N^*} = \log(N)$ would mean the total amount of points used at stage (ISM5) would be $O\left(\hat{J} N^{1-\gamma} \log^2 N\right)$.

(Int-2): The more methodologically sound method would be to use Theorem 8 and construct confidence intervals $\left[\hat{\tau}_j^{(1)} \pm \left(Q_{\Delta_j,\sigma,\sigma}(\sqrt[\hat{J}]{1-\alpha}) + 1\right) \lfloor \frac{N}{N_1} \rfloor\right]$ for $j = 1, \ldots, \hat{J}$. Lemma 11 (see Section 10.2, Supplement Part C) tells us that $Q_{\Delta_j,\sigma,\sigma}(\sqrt[\hat{J}]{1-\alpha})$ can be bounded by a multiple of $\log(\hat{J})$, which in turn is no greater than $\log(N)$. This would result in step (ISM5) using a total of $O\left(\hat{J} N^{1-\gamma} \log N\right)$ points.

Depending on whether (Int-1) or (Int-2) was performed before step (ISM5), there will be least squares fitting on $O\left(\hat{J} N^{1-\gamma} \log^a N\right)$ points, taking $O\left(\hat{J} N^{1-\gamma} \log^a N\right)$ time to compute, where $a = 2$ or 1, respectively. Although the distribution of the $\hat{J}$ obtained from SBS is not known, a consequence of Lemma 1 is that $\mathbb{P}\left[\hat{J} = J\right] \geq 1 - CN^{-1}$ for some constant C, and therefore

$$\mathbb{E}\left[\hat{J}\right] \leq J + C\frac{N}{N} = O(J). \tag{98}$$

This leads to the conclusion that the second stage has a computational time that is on average $O\left(JN^{1-\gamma} \log^a(N)\right) = O\left(N^{1-\gamma+\Lambda} \log^a N\right)$, $a = 2$ or 1 depending on whether (Int-1) or (Int-2) was performed, respectively.

Using this result we could choose the γ for an optimal computational time. This can be done by setting the order of the first stage to equal the order of time for the second stage. Ignoring the log factors, the two orders are equal when $\gamma = \frac{1+\Lambda}{2}$, with the restriction that $\gamma > \Xi$ for (Int-1) and $\gamma > 3\Xi$ for (Int-2) (due to condition (M7 (SBS))). Hence the minimizing γ would be $\gamma_{min} = \frac{1+\Lambda}{2} \vee (\Xi + \eta)$ or $\gamma_{min} = \frac{1+\Lambda}{2} \vee (3\Xi + \eta)$ (η any tiny positive value), resulting in order $N^{1-\gamma_{min}+\Lambda} \log^a(N)$ computational time for (ISM5), with $a = 2$ or 1 depending on whether (Int-1) or (Int-2) was used, respectively. The computational time for (Int-1) is as follows

- For $\Xi \in [0, 1/2)$, we have $\frac{1+\Lambda}{2} < \Xi$ and hence $\gamma_{min} = \frac{1+\Lambda}{2}$ and the computational time is order $N^{(1+\Lambda)/2} \log^2(N)$.
- For $\Xi \in [1/2, 1)$, we either have $\frac{1+\Lambda}{2} > \Xi$ or $\frac{1+\Lambda}{2} \leq \Xi$:
  - In the case that $\frac{1+\Lambda}{2} > \Xi$, we again have $\gamma_{min} = \frac{1+\Lambda}{2}$ and the computational time is order $N^{(1+\Lambda)/2} \log^2(N)$, due to the second stage having a higher order than the first stage.

– Otherwise, $\gamma_{min} = \Xi + \eta$ where η is any small positive value less than $1 - \Xi$. The computational time order will be determined by the first stage procedure, which is of order $N^{\Xi+\eta} \log N$.

TABLE 9

Table of γ_{min} and computational times for various values of Ξ, for SBS with (Int-1) performed at the first stage. Also shown are their values for extreme value of Λ ($\Lambda = 0$ and $\Lambda = \Xi$).

Ξ	$[0, 1/2)$	$[1/2, 1)$
γ_{min}	$\frac{1+\Lambda}{2}$	$\max\left\{\frac{1+\Lambda}{2}, \Xi + \eta\right\}$
Order of Time	$N^{(1+\Lambda)/2} \log^2 N$	$\max\{N^{(1+\Lambda)/2} \log^2 N, N^{\Xi+\eta} \log N\}$
γ_{min} $(\Lambda = 0)$	$\frac{1}{2}$	$\Xi + \eta$
Time $(\Lambda = 0)$	$N^{1/2} \log^2 N$	$N^{\Xi+\eta} \log N$
γ_{min} $(\Lambda = \Xi)$	$\frac{1+\Xi}{2}$	$\frac{1+\Xi}{2}$
Time $(\Lambda = \Xi)$	$N^{(1+\Xi)/2} \log^2 N$	$N^{(1+\Xi)/2} \log^2 N$

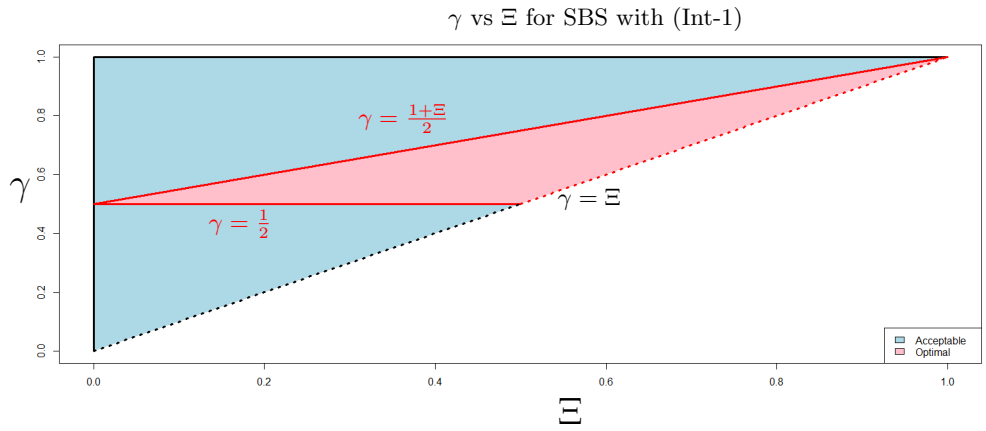


Fig 22: Blue triangle encompasses all valid values of γ vs Ξ. Pink region, solid red lines, and dotted red lines denotes γ_{min} for each Ξ (γ_{min} can vary for different values of Λ even when Ξ is fixed, hence the red region).

For (Int-2), the minimal γ and order of computational time behaves differently:

- For $\Xi \in [0, 1/6)$, we have $\frac{1+\Lambda}{2} < 2\Xi$ and hence $\gamma_{min} = \frac{1+\Lambda}{2}$ and the computational time is order $N^{(1+\Lambda)/2} \log^2(N)$.
- For $\Xi \in [1/6, 1/5)$, we either have $\frac{1+\Lambda}{2} > 3\Xi$ or $\frac{1+\Lambda}{2} \leq 3\Xi$. This will lead to, respectively, $\gamma_{min} = \frac{1+\Lambda}{2}$ or $\gamma_{min} = 3\Xi + \eta$ (η being any small value less than $1 - 3\Xi$), with computational time orders $N^{\gamma_{min}} \log N$.
- For $\Xi \in [1/5, 1/3)$, we have $\frac{1+\Xi}{2} > 3\Xi$, hence $\gamma_{min} = 3\Xi + \eta$, with computational time order $N^{3\Xi+\eta} \log N$.

- For $\Xi \in [1/3, 1]$, condition (M7 (SBS)) is not satisfied and we do not have the theoretical guarantee that (Int-2) will produce intervals of satisfactory coverage.

TABLE 10

Table of γ_{min} and computational times for various values of Ξ, for SBS with (Int-2) performed at the first stage. Also shown are their values for extreme value of Λ ($\Lambda = 0$ and $\Lambda = \Xi$).

Ξ	$[0, 1/6)$	$[1/6, 1/5)$	[1/5,1/3)	[1/3,1]
γ_{min}	$\frac{1+\Lambda}{2}$	$\max\left\{\frac{1+\Lambda}{2}, 3\Xi + \eta\right\}$	$3\Xi + \eta$	N/A
Order of Time	$N^{(1+\Lambda)/2}\log(N)$	$\max\{N^{(1+\Lambda)/2}, N^{3\Xi+\eta}\} \cdot \log(N)$	$N^{3\Xi+\eta}\log(N)$	N/A
γ_{min} ($\Lambda = 0$)	$\frac{1}{2}$	$3\Xi + \eta$	$3\Xi + \eta$	N/A
Time ($\Lambda = 0$)	$N^{1/2}\log(N)$	$N^{3\Xi+\eta}\log(N)$	$N^{3\Xi+\eta}\log(N)$	N/A
γ_{min} ($\Lambda = \Xi$)	$\frac{1+\Xi}{2}$	$\frac{1+\Xi}{2}$	$3\Xi + \eta$	N/A
Time ($\Lambda = \Xi$)	$N^{(1+\Xi)/2}\log(N)$	$N^{(1+\Xi)/2}\log(N)$	$N^{3\Xi+\eta}\log(N)$	N/A

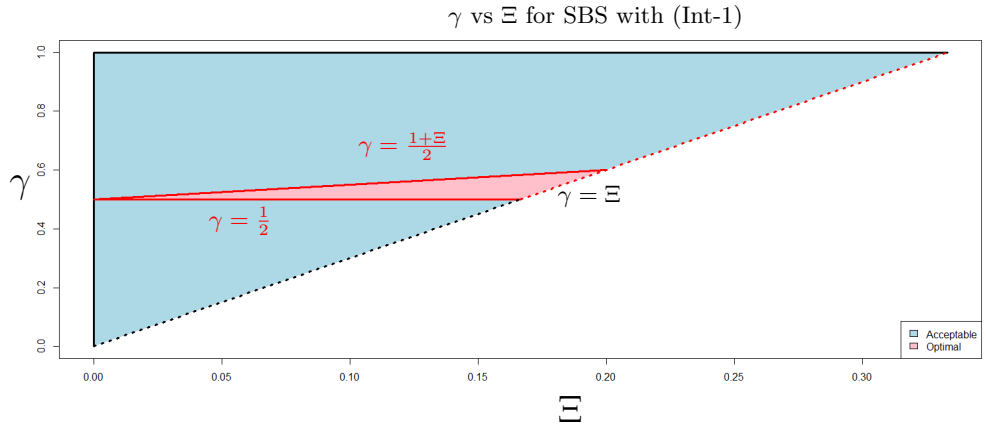


Fig 23: Blue triangle encompasses all valid values of γ vs Ξ as set by (M7 (SBS)). Pink region, solid red lines, and dotted red lines denotes γ_{min} for each Ξ (γ_{min} can vary for different values of Λ even when Ξ is fixed, hence the red region).

It can be seen that the biggest decrease in order of average computational time is for small values of Ξ and Λ, and in fact for $\Lambda = 0$ the computational complexity can reach $O(\sqrt{N}\log(N))$, which is marginally slower than intelligent sampling on a single change point. For larger values of Ξ, there is less than a square root drop in $N\log(N)$ (order of using SBS on the whole data) to $N^{\gamma_{min}}\log(N)$ (intelligent sampling), to the point where both procedures take near the same order of time for high values of Ξ.

REMARK 12. *Note that when implementing the intelligent sampling strategy knowledge of Ξ is desirable, but in practice, its value is unknown. If one is willing to impose an upper bound on Ξ, intelligent sampling can be implemented with this (conservative) upper-bound.*

9.3. *Sample Size Considerations from a Methodological Angle.* In the asymptotic setting of Section 9.2, we were concerned about minimizing the *order* of computational time required for locating the change points through intelligent sampling, assuming that certain important quantities were known. The focus in this section is on obtaining explicit expressions for the minimum sample size that the procedure requires to correctly identify the underlying change points. Obviously, the minimum sample size is the key driver in the computational time formlas provided, albeit not the single one, and also addresses computer memory usage issues. In order to develop explicit expressions for the total computational time, one would need to know exactly how fast the first stage procedure used at (ISM3) (e.g., SBS, binary segmentation) runs versus data size and model parameters[5], and such expressions are unavailable as an exact expression. Therefore, we look at minimizing the subsample utilized as a proxy, with the added benefit of deriving the least amount of data that must be held in memory at a single time.

We have already investigated the optimal order of the first stage subsample, denoted N_1, and demonstrated in Section 9.2 that in the best cases the size of both the first and second stage subsamples scales as $\sqrt{N}\log(N)$. Although valid, these previous analyses only apply to an abstract asymptotic setting. In practice, given a data set with fixed (large) N, a different approach is needed to determine the optimal number of points to use at each different stage.

Given the number of change points and their associated SNR's, we show below how to optimally allocate samples in order to minimize the total number used, for intelligent sampling. For simplicity we assume the error terms satisfy condition (M4 (SBS)). We start with the two-stage intelligent sampling procedure and assume that in stage 1, roughly N_1 points are used and another N_1 points for the calibration steps described in Section 9.1. At stage 2, we work with $\hat{J}$ (which is $\approx J$) intervals. Using Theorem 8, setting the width of the second stage intervals to be $\left(Q_{\Delta,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\left\lfloor\frac{N}{N_1}\right\rfloor$ for a small α will ensure that they cover the true change points with high probability (close to $1-\alpha$ if not greater). Assuming N_1 is large enough so that the first stage is accurate (ie $\hat{J}=J$ and $\max_j|\hat{\tau}_j^{(1)}-\tau_j|$ is small with high probability) , the number of points used in the two stages, combined, is approximately

$$2N_1+\frac{2\left(\sum_{j=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\right)N}{N_1}. \tag{99}$$

This presents a trade-off, e.g. if we decrease N_1 by a factor of 2, the second term in (99) increases by a factor of 2. To use a minimal number of points in both stages, we need to set $N_1=\sqrt{N\sum_{j=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)}$. In turn this yields a minimum of $4\sqrt{N\sum_{j=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)}$ when plugged into (99). For any given values of N, J, and SNR, this provides a lower bound on the minimum number of points that intelligent sampling must utilize, and Tables 24 and 25 depict some of these lower bounds for a select number of these parameters.

[5]For example, binary segmentation will generally terminate in fewer steps on a dataset with fewer change points than on another dataset of the same length but more change points.

Fig 24: For $N = 1.5 \times 10^7$, the minimal percentage of data that must be used for various values of J and SNR, assuming all jumps have equal SNR and $\alpha = 0.01$.

Δ/σ \ J	40	60	80	100
1	4.33	5.43	6.4	7.3
1.25	3.46	4.38	5.14	5.84
1.5	2.92	3.67	4.33	4.84
1.75	2.53	3.2	3.7	4.26
2	2.26	2.77	3.33	3.72
2.25	1.96	2.53	2.92	3.43
2.5	1.85	2.26	2.61	3.1
2.75	1.73	2.12	2.44	2.73
3	1.6	1.96	2.26	2.53

Fig 25: For $N = 1.5 \times 10^{10}$, the minimal percentage of data that must be used for various values of J and SNR, assuming all jumps have equal SNR and $\alpha = 0.01$.

Δ/σ \ J	100	250	500	1000
1	0.231	0.386	0.57	0.839
1.25	0.185	0.31	0.456	0.669
1.5	0.157	0.258	0.386	0.566
1.75	0.135	0.225	0.327	0.484
2	0.118	0.2	0.292	0.426
2.25	0.108	0.179	0.263	0.386
2.5	0.098	0.163	0.231	0.343
2.75	0.086	0.146	0.219	0.31
3	0.08	0.137	0.207	0.292

Note that while the fraction of points used for the larger N above is smaller, in absolute terms, this still translates to very large subsamples: even just 0.57% of 1.5×10^{10} (for SNR 1 and $J = 500$ on Table 25) is a very large dataset of 8.6×10^7, which almost requires server type computer capabilities. The situation becomes more tenuous for larger values of N. This suggests that *a larger number of stages* is in order for sample sizes of N exceeding 10^{10}.

For a three-stage implementation, suppose $\approx N_1$ points are utilized at stage 1, letting us form simultaneous confidence intervals that are (approximately) of the form

$$\left[\hat{\tau}_j^{(1)} - \sum_{j=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\frac{N}{N_1}, \hat{\tau}_j^{(1)} + \sum_{j=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\frac{N}{N_1}\right] \qquad \text{for } j = 1,\ldots,J$$

(assuming $\hat{J} = J$ for simplification). At stage 2, suppose at the j'th confidence interval we subsample roughly $N_2^{(j)}$ points, giving us a subsample which skips approximately every $2Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)\frac{N}{N_1 N_2^{(j)}}$ points. Hence, at stage 3 we work with confidence intervals that are (approximately) of the form

$$\left[\hat{\tau}_j^{(2)} \pm \left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\left(2\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\frac{N}{N_1 N_2^{(j)}}\right)\right] \tag{100}$$

for $j = 1,\ldots,J$. In total all three stages use around a total of

$$2N_1 + \sum_{j=1}^{J} N_2^{(j)} + \frac{4N}{N_1}\left(\sum_{j=1}^{J}\frac{\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)^2}{N_2^{(j)}}\right) \tag{101}$$

points. This expression is minimized by setting

$$N_1 = N^{1/3}\left(\sum_{k=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\right)^{2/3} \qquad \text{and}$$

$$N_2^{(j)} = 2N^{1/3}\frac{Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1}{\left(\sum_{k=1}^{J}\left(Q_{\Delta_k,\sigma,\sigma}(\alpha,J)+1\right)\right)^{1/3}} \tag{102}$$

for $j = 1,\ldots,J$, which in turn gives a minimum of $6N^{1/3}\left(\sum_1^J\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\right)^{2/3}$ for (101). A similar analysis on a four-stage procedure would have the optimal subsample allocation as

$$N_1 = N^{1/4}\left(\sum_{k=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\right)^{3/4} \qquad \text{and}$$

$$N_2^{(j)} = N_3^{(j)} = 2\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)N^{1/4}\left(\sum_{k=1}^{J}\left(Q_{\Delta_k,\sigma,\sigma}(\alpha,J)+1\right)\right)^{-1/4} \qquad \text{for } j=1,\ldots,J \tag{103}$$

which yields a total of $8N^{1/4}\left(\sum_{k=1}^{J}\left(Q_{\Delta_j,\sigma,\sigma}\left(1-\frac{\alpha}{J}\right)+1\right)\right)^{3/4}$ points utilized.

Fig 26: For $N = 1.5\times10^{10}$, minimal percentage of the data that must be used for a three stage procedure, assuming all jumps have equal SNR and $\alpha = 0.01$.

Δ/σ \ J	100	250	500	1000
1	0.03	0.061	0.101	0.168
1.25	0.023	0.045	0.075	0.127
1.5	0.018	0.036	0.06	0.099
1.75	0.015	0.029	0.049	0.082
2	0.012	0.025	0.041	0.068
2.25	0.011	0.022	0.036	0.06
2.5	0.009	0.018	0.031	0.052
2.75	0.009	0.017	0.027	0.046
3	0.008	0.016	0.025	0.043

Fig 27: For $N = 1.5\times10^{12}$, minimal percentage of the data that must be used for a four stage procedure, assuming all jumps have equal SNR and $\alpha = 0.01$.

Δ/σ \ J	250	500	1000	2000
1	8.3e-04	1.5e-03	2.6e-03	4.6e-03
1.25	6.0e-04	1.1e-03	1.9e-03	3.3e-03
1.5	4.6e-04	8.2e-04	1.4e-03	2.5e-03
1.75	3.6e-04	6.6e-04	1.1e-03	2.0e-03
2	3.1e-04	5.5e-04	9.6e-04	1.7e-03
2.25	2.5e-04	4.5e-04	8.0e-04	1.4e-03
2.5	2.2e-04	4.0e-04	6.8e-04	1.2e-03
2.75	1.9e-04	3.5e-04	5.9e-04	1.1e-03
3	1.8e-04	3.0e-04	5.5e-04	9.2e-04

Comparing Figures 25 and 26, we focus on the case of 1000 change points with SNR 1.5: using three stages allows us to decrease the minimal required points by a factor of around five. The ease on computations is greater when looking at the largest amount of data the computer must handle at a time:[1] this is $N_1 \approx 2.1\times10^7$ for two stages and $N_1 \approx 2.5\times10^6$ for three stages, a decrease by

[1]For intelligent sampling the largest data subset the computer has to work with and hold in memory at any moment, under these optimal allocations and when all change points have equal SNR, is the roughly N_1 sized data set used at the initial step for BinSeg. All subsequent steps can work with sub-intervals of data less than N_1 in size.

a factor of 9. Meanwhile for a dataset of size 1.5 trillion, using four stages allows us to work with subsamples of size at most $N_1 \approx 4.7 \times 10^6$ for the more demanding scenario of SNR 1.5 and 2000 change points, a very manageable dataset for most computers.

We note here that these optimal allocations are valid assuming that the first stage procedure is able to pin down $\hat{J}$ and the change points with the initial subsample. In general, this will be the case provided the SNR is reasonable, and the initial subsample is large enough so that the change-points are adequately spaced apart. For example, in the context of the above tables, one can ask whether the correct number of change points can be estimated for a 2.4 million length dataset with 1000 evenly spaced change points, or 2000 change points on a 4.7 million length data with 2000 evenly spaced change points, under a constant SNR of 1.5 (which is of modest intensity). To this end, we ran a set of simulations and concluded that if there are over 1000 data points between consecutive change points of SNR 1.5, based on these two settings and for appropriate tuning parameters, binary segmentation estimators satisfy $\hat{J} = J$ and $\max |\hat{\tau}_j - \tau_j| \leq 150$ with probability over 99%.

Observe also that the formulas provided depend on the values of the SNRs at the change points and the actual number of change points (J). In practice, neither will be known, and the the practitioner will not be able to determine the derived allocations exactly. In such situations, conservative lower bounds on the SNRs and a conservative higher bound on J, can yield valid (but conservative) sampling allocations when plugged in to the expressions derived through this section. Such bounds can be obtained if background information about the problem and the data are available, or via rough pilot estimates on an appropriately sparse subsample.

It is also worth pointing out that the intelligent sampling procedure is readily adaptable to a distributed computing environment, which can come into play, especially with data sets of length exceeding 10^{12} that are stored sequentially across several storage disks. In such cases, the two sparse subsamples at the first stage, which are of much smaller order, can be transferred over to a central server (a much easier exercise than transferring all the data on to one server), where the first is analyzed via binary segmentation to determine the initial change-points, and the other used for the re-estimation procedure and associated confidence intervals as described in Section 9.1. As the number of disks on which the data are stored is of a much smaller order than the length of the data, each re-estimated change-point and its associated confidence interval will typically belong to a stretch of data completely contained within one storage disk, and the subsequent resampling and estimation steps can be performed on the local processor, after the information on the confidence interval has been transferred back from the central server. An occasional communication between two machines may be necessary.

9.4. *Dependent Errors.* The proposed intelligent sampling procedure for multiple change point problems has so far been presented in the setting of i.i.d. data for a signal-plus-noise model. However, many data sequences (such as time series) usually exhibit temporal correlation. Hence, it is of interest to examine the properties of the procedure under a non-i.i.d. data generating mechanism.

While we believe that results akin to Theorem 3 (growing number of change-points in the i.i.d. error regime) should go through under various forms of dependence among errors, a theoretical treatment of this would require a full investigation of the tail properties of random walks under

dependent increments and is outside the scope of this paper. An asymptotic distributional result, analogous to Theorem 4, under finitely many change points in the dependent regime is also expected to hold. We present below a proposition for the finite J case under a set of high-level assumptions.

Suppose that the data sequence is in the form (11) and satisfies conditions (M1) to (M3). Upon the error terms, we impose the assumption that they have an autocorrelation structure which dies out at a polynomial rate or faster, and locally around the change points assume that the joint distributions of the errors are fixed [i.e. invariant to N]:

(M4-alt1): ε_j's are each marginally $N(0, \sigma_j^2)$, and there exist positive constants σ_{max}, B and α, independent of N, such that $\sigma_j \leq \sigma_{max}$ and $\text{cor}(\varepsilon_j, \varepsilon_{j+k}) \leq Bk^{-\alpha}$ for any j and $j+k$ from 1 to N.

(M4-alt2): there exists a sequence $w_e(N) \to \infty$ and Gaussian sequences $\{\epsilon_{i,j}\}_{i\in\mathbb{Z}}$ (not required to be stationary) for $j = 1, \ldots, J$, such that for all $j = 1, \ldots, J$ and all sufficiently large N, $\{\varepsilon_{\tau_j - w_e(N)}, \ldots, \varepsilon_{\tau_j + w_e(N)}\}$ has the same joint distribution as $\{\epsilon_{-w_e(N),j}, \ldots, \epsilon_{w_e(N),j}\}$.

On a set of data where (M4-alt1) and (M4-alt2) hold (along with assumptions (M1) to (M3),- we want steps (ISM1) and (ISM4) to go through with some procedure that ensures

$$\mathbb{P}\left[\hat{J} = J,\ \max_{i=1,\ldots,J} |\hat{\tau}_i^{(1)} - \tau_i| \leq w(N),\ \max_{i=0,\ldots,J} |\hat{\nu}_i^{(1)} - \nu_i| \leq \rho_N\right] \to 1. \tag{104}$$

for some sequence $w(N) \to \infty$ and $w(N) = o(\delta_N)$.

Next, we desire for the final estimators $\hat{\tau}^{(2)}$ to be $O_p(1)$ consistent and have the property that for each $\epsilon > 0$ there exists a constant C such that

$$\mathbb{P}\left[\hat{J} = J;\quad \max_{i=1,\ldots,J} \left|\hat{\tau}_i^{(2)} - \tau_i\right| \leq C\right] \geq 1 - \epsilon\,. \tag{105}$$

for all sufficiently large N.

Proposition 3. *Suppose conditions (M1) to (M3), (M4-alt1), and (M4-alt2) are satisfied. Next, suppose the first stage estimators satisfy (104) and the second stage estimators, constructed as in the i.i.d. setting but with a minor modification*[2] *satisfy (105). Define the random walks*

$$Z_{i,j} = \begin{cases} \Delta_j(\epsilon_{1,j} + \cdots + \epsilon_{i,j}) - i\Delta_j^2/2, & i > 0 \\ 0, & i = 0 \\ \Delta_j(\epsilon_{i+1,j} + \cdots + \epsilon_{0,j}) - i\Delta_j^2/2, & i < 0\,, \end{cases} \tag{106}$$

with the $\epsilon_{i,j}$'s from condition (M4-alt2), for $j = 1, 2, \ldots, J$, and denote $\tilde{L}_j := \arg\min_{i\in\mathbb{Z}} Z_{i,j}$. Then $|\hat{\tau}_j^{(2)} - \tau_j|$'s for $j = 1, \ldots, J$ jointly converge to the distribution of $(\tilde{L}_1, ..., \tilde{L}_J)$: for any integers $k_1, \cdot, k_J$,

$$\mathbb{P}\left[\hat{J} = J,\ |\hat{\tau}_j^{(2)} - \tau_j| = k_j \text{ for } 1 \leq j \leq J\right] \to \prod_{j=1}^{J} \mathbb{P}[\tilde{L}_j = k_j] \tag{107}$$

[2]See the remark right after the proposition.

REMARK 13. *As in the i.i.d. case, the intervals* $[\hat{\tau}_j^{(1)} - Kw(N), \hat{\tau}_j^{(1)} + Kw(N)]$ *for* $j = 1, \ldots, J$ *[obtained at step (ISM5)] would each contain only one change point with probability approaching one. We are therefore still justified in fitting stump models on each interval, although with a slight modification. Unlike the i.i.d. error terms scenario, the joint distribution of the error terms at the second stage does change when we condition on the estimators* $\hat{\tau}_j^{(1)}$*'s, regardless if we leave out the* $\{Z_j\}$ *subsample at the second stage. We thus make the following modification to (ISM5) in 3.1 (and assume this altered procedure is used from here on in this section):*

(ISM5-alt): take $S^{(2)}\left(\hat{\tau}_i^{(1)}\right)$ *as all integers in* $[\hat{\tau}_i^{(1)} - Kw(N), \hat{\tau}_i^{(1)} + Kw(N)]$ *without any points omitted.*

Step (ISM-6) then proceeds as before.

REMARK 14. *We note that the asymptotic distribution given above and in Theorem 4 have the same form, since as before, conditional on (104) being true, intelligent sampling simplifies the problem into multiple single change point problems. Using Proposition 3 to construct confidence intervals in a practical setting requires an idea of the joint distribution of the* $\{\epsilon_{ij}\}$*'s. In practice, one would have to impose some structural conditions, e.g. assuming an ARMA or ARIMA structure on long stretches of the errors to the left and right of the change points.*

REMARK 15. *The hard work lies in the verification of the high-level conditions (104) and (105) in different dependent error settings, and as mentioned previously, is not dealt with in this paper but should constitute interesting agenda for follow-up research on this problem. We will, however, use Proposition 3 in our simulation and data analysis sections to construct confidence intervals for various dependent error scenarios.*

9.5. *Proof of Theorem 3.* As a reminder, this theorem only required the error terms to be independent and zero-mean subgaussian, with subguassian parameters σ_i for $1 \leq i \leq N$, which means

$$\begin{aligned} \mathbb{E}[\exp(s\varepsilon_i)] &\leq \exp\left(\frac{\sigma_i^2 s^2}{2}\right) \\ \mathbb{P}[|\varepsilon_i| \geq s] &\leq 2\exp\left(-\frac{s^2}{2\sigma_i^2}\right) \end{aligned} \tag{108}$$

for all $s \in \mathbb{R}$. Since condition (M4) also states that the subgaussian parameters (σ_i's) are bounded above by a constant $\sigma_{\max}$ not dependent on N, this means that

$$\begin{aligned} \mathbb{E}[\exp(s\varepsilon_i)] &\leq \exp\left(\frac{\sigma_{\max}^2 s^2}{2}\right) \\ \mathbb{P}[|\varepsilon_i| \geq s] &\leq 2\exp\left(-\frac{s^2}{2\sigma_{\max}^2}\right) \end{aligned} \tag{109}$$

for all $i = 1, \ldots, N$ and $s \in \mathbb{R}$.

Much of the theory used in this proof and subsequent proof will extensively deal with the points used at the second stage, where points used at stage 1 will be skipped. To make these arguments, we will denote some notation to clearly , define the points $\tau_j^{(2)}$'s as

$$\tau_j^{(2)} := \begin{cases} \tau_j - 1 & \text{if } \tau_j \text{ was a first stage subsmaple point} \\ \tau_j & \text{otherwise} \end{cases} \tag{110}$$

for $j = 1, \ldots, J$ define

$$S^{(2)}(t) := \{i \in \mathbb{N} : \ |i - t| \leq Kw(N), \quad Y_i \text{ not used in 1st stage subsample}\}. \tag{111}$$

PROOF. Define the event

$$\mathcal{R}_N := \left\{ \hat{J} = J; \quad \max_{i=1,\ldots,J} \left| \hat{\tau}_i^{(1)} - \tau_i \right| \leq w(N); \quad \max_{i=0,\ldots,J} |\hat{\nu}_i^{(1)} - \nu_i| \leq \rho_N \right\}, \tag{112}$$

Denote G_N as the joint distribution of $J, \hat{\tau}_1^{(1)}, ..., \hat{\tau}_J^{(1)}, \hat{\nu}_0^{(1)}, ..., \hat{\nu}_J^{(1)}$; the domain of G_N would be $\bigcup_{k=0}^{N-1} \mathbb{N}^{k+1} \times \mathbb{R}^{k+1}$.

Then, for any sequence $\{a_N\}$, we can bound $\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\ldots,J} \left| \hat{\tau}_i^{(2)} - \tau_i^{(2)} \right| \leq a_N \right]$ from below

by:

$$
\begin{aligned}
& \mathbb{P}\left[\hat{J}=J;\quad \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\right] \\
\ge & \sum_{k=0}^{N-1}\int_{\substack{0<t_1<t_2<\ldots.<t_k<N \\ v_1,\ldots,v_k\in\mathbb{R}}} \mathbb{P}\left[\hat{J}=J;\ \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\middle|\hat{J}=k,\hat{\tau}_j^{(1)}=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for } j\le k\right] \\
& dG_N(k,t_1,...,t_k,v_0,...,v_k) \\
\ge & \int_{\substack{|t_i-\tau_i|\le Kw(N)\\ |v_i-\nu_i|\le\rho_N \\ \text{for all } i}} \mathbb{P}\left[\hat{J}=J;\ \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\middle|\hat{J}=J,\hat{\tau}_j^{(1)}=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for } j\le J\right] \\
& dG_N(J,t_1,...,t_J,v_0,...,v_J) \\
\ge & \left(\inf_{\substack{|t_i-\tau_i|\le Kw(N)\\ |v_i-\nu_i|\le\rho_N \\ \text{for all } i}} \mathbb{P}\left[\hat{J}=J;\ \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\middle|\hat{J}=J,\hat{\tau}_j^{(1)}=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for } j\le J\right]\right)\cdot \\
& \mathbb{P}\left[\hat{J}=J;\quad \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(1)}-\tau_i\right|\le Kw(N);\quad \max_{i=0,\ldots,J}|\hat{\nu}_i^{(1)}-\nu_i|\le\rho_N\right] \\
\ge & \left(\inf_{\substack{|t_i-\tau_i|\le Kw(N)\\ |v_i-\nu_i|\le\rho_N \\ \text{for all } i}} \mathbb{P}\left[\max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\middle|\hat{J}=J,\hat{\tau}_j^{(1)}=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for } j\le J\right]\right)- \\
& \mathbb{P}[\mathcal{R}_N \text{ is false}] \qquad (113)
\end{aligned}
$$

We wish to show that for all $\epsilon>0$, there exists a sequence $a_N=O(\log(J(N)))$ such that

$$\mathbb{P}\left[\hat{J}=J;\quad \max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\right]>1-\epsilon$$

for all large N. It is sufficient to show this is satisfied by the second to last line of (113), as $\mathcal{R}_N$ is true with probability increasing to 1. Henceforth, we will work with the probability

$$\mathbb{P}\left[\max_{i=1,\ldots,J}\left|\hat{\tau}_i^{(2)}-\tau_i^{(2)}\right|\le a_N\middle|\hat{J}=J,\hat{\tau}_j^{(1)}=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for } j\le J\right]$$

and, in the domain $|t_i-\tau_i|\le Kw(N)$ and $|v_i-\nu_i|\le\rho_N$ for all i, we show that it is greater than $1-\epsilon$ for all sufficiently large N and $a_N=C_1\log J+C_2$ for some $C_1,C_2>0$. In the remainder of the proof, assume that all t_i's and v_i's fall within this domain.

For sufficiently large N, we have $Kw(N)\le\delta_N/4$, and therefore no two of the second stage intervals $([t_i-Kw(N),\,t_i+Kw(N)]$ for $i=1,...,J)$ intersect. Because each $\hat{\tau}_j^{(2)}$ is a function

of all Y_i's in the disjoint index sets $S^{(2)}(t_j) \subset [t_j - Kw(N),\ t_j + Kw(N)]$ and the two level estimates $\hat{\nu}^{(1)}_{j-1}$ and $\hat{\nu}^{(1)}_j$, conditional independence holds:

$$\begin{aligned} & \mathbb{P}\left[\max_{i=1,\ldots,J} \left|\hat{\tau}^{(2)}_i - \tau^{(2)}_i\right| \le a_N \middle| \hat{J} = J, \hat{\tau}^{(1)}_j = t_j, \hat{\nu}^{(1)}_j = v_j \text{ for } j \le J \right] \\ = & \prod_{i=1}^{J} \mathbb{P}\left[\left|\hat{\tau}^{(2)}_i - \tau^{(2)}_i\right| \le a_N \middle| \hat{J} = J, \hat{\tau}^{(1)}_j = t_j, \hat{\nu}^{(1)}_j = v_j \text{ for all } j \right] \end{aligned}$$

To show the above product is eventually greater than some $1-\varepsilon$, it would suffice to show that, for all $1 \le k \le J$ and sufficiently large N,

$$\begin{aligned} & \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\hat{\tau}^{(2)}_k - \tau^{(2)}_k\right| > a_N \right] \\ := & \mathbb{P}\left[\left|\hat{\tau}^{(2)}_k - \tau^{(2)}_k\right| > a_N \middle| \hat{J} = J, \hat{\tau}^{(1)}_j = t_j, \hat{\nu}^{(1)}_j = v_j \text{ for all } j \right] \\ \le & C_\epsilon / J \end{aligned} \tag{114}$$

for some $C_\epsilon < -\log(1-\varepsilon)$.

For any k between 1 to J inclusive, we can write explicit expressions for $\tau^{(2)}_k$ and $\hat{\tau}^{(2)}_k$.

$$\begin{aligned} \hat{\tau}^{(2)}_k &= \operatorname*{arg\,min}_{d \in S^{(2)}(t_k)} \left(\operatorname{sgn}(v_k - v_{k-1}) \sum_{i \in S^{(2)}(t_k)} \left(Y_i - \frac{v_{k-1}+v_k}{2} \right) \left[1(i \le d) - 1\left(i \le \tau^{(2)}_k \right) \right] \right) \\ &:= \operatorname*{arg\,min}_{d \in S^{(2)}(t_k)} \mathbb{M}_k(d) \end{aligned} \tag{115}$$

Next, since, $t_k \in [\tau_k - w(N), \tau_k + w(N)]$, for N large enough so that $\frac{K-1}{2} w(N) > 1$ we have

$$\begin{aligned} & t_k - Kw(N) \le \tau_k - (K-1)w(N) < \tau^{(2)}_k - \frac{K-1}{2} w(N) < \\ & \tau^{(2)}_k + \frac{K-1}{2} w(N) < \tau_k + (K-1)w(N) \le t_k + Kw(N) \end{aligned} \tag{116}$$

That is to say, the set $S^{(2)}(t_k)$ includes the interval $\left[\tau^{(2)}_k - \frac{K-1}{2} w(N), \tau^{(2)}_k + \frac{K-1}{2} \right]$, minus the first stage subsample points, regardless of which t_k was used among those permissible in R_N. Therefore, for any even integer a_N such that $1 < a_N < \frac{K+1}{2} w(N)$ (which would be satisfied for all large N if $a_N = O(\log(N)) = o(w(N))$),

$$\begin{aligned} & \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\hat{\tau}^{(2)}_k - \tau^{(2)}_k\right| > a_N \right] \\ \le & \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left| \lambda_2\left(\tau^{(2)}_k, \hat{\tau}^{(2)}_k \right) \right| > \frac{a_N}{2} \right] \end{aligned} \tag{117}$$

for all sufficiently large N such that the first stage subsample samples more sparsely than taking every other point. In order for $\left|\lambda_2\left(\hat{\tau}_k^{(2)}, \tau_k^{(2)}\right)\right| > a_N/2$, there must exists a d such that

$$\left|\lambda_2(\tau_k^{(2)}, d)\right| > a_N/2 \qquad \text{and} \qquad \mathbb{M}_k(d) \leq \min_{\substack{\ell \in S^{(2)}(t_k) \\ |\lambda(\tau_k^{(2)}, \ell)| \leq \frac{a_N}{2}}} \mathbb{M}_k(\ell) < \mathbb{M}_k(\tau_k^{(2)}) = 0. \tag{118}$$

Therefore,

$$\begin{aligned}
& \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\hat{\tau}_k^{(2)} - \tau_k^{(2)}\right| > a_N\right] \\
\leq\ & \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right)\right| > \frac{a_N}{2}\right] \\
\leq\ & \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\exists d \in S^{(2)}(t_k) \text{ where } \left|\lambda_2(\tau^{(2)}, d)\right| > a_N/2 \text{ and } \mathbb{M}_k(d) < 0\right] \\
\leq\ & \sum_{\substack{d \in S^{(2)}(t_k) \\ \left|\lambda_2(\tau^{(2)}, d)\right| > a_N/2}} \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) \leq 0\right] \qquad (119)
\end{aligned}$$

Each $\mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) \leq 0\right]$ can be bounded as follows by recognizing that each $\mathbb{M}_k(d)$ has subGaussian distribution: for every $d \in S^{(2)}(t_k)$ and sufficiently large N so that $\rho_N \leq |\underline{\Delta}|/2$ (and hence $\operatorname{sgn}(v_{k-v_{k-1}}) = \operatorname{sgn}(\nu_k - \nu_{k-1})$ for all k)

$$\begin{aligned}
& \mathbb{M}_k(d) \\
=\ & \operatorname{sgn}(v_k - v_{k-1}) \sum_{i \in S^{(2)}(t_k)} \left(Y_i - \frac{v_{k-1} + v_k}{2}\right)\left[1(i \leq d) - 1\left(i \leq \tau_k^{(2)}\right)\right] \\
=\ & \begin{cases}
\operatorname{sgn}(\nu_k - \nu_{k-1})\left(\lambda_2(\tau_k^{(2)}, d)\left(\nu_k - \frac{v_{k-1}+v_k}{2}\right) + \sum\limits_{\substack{\tau_k^{(2)} < \ell \leq d \\ \ell \in S^{(2)}(t_k)}} \varepsilon_\ell\right) & \text{for } d > \tau_k^{(2)} \\
0 & \text{for } d = \tau_k^{(2)} \\
\operatorname{sgn}(\nu_k - \nu_{k-1})\left(\lambda_2(\tau_k^{(2)}, d)\left(\nu_{k-1} - \frac{v_{k-1}+v_k}{2}\right) - \sum\limits_{\substack{d < \ell \leq \tau_k^{(2)} \\ \ell \in S^{(2)}(t_k)}} \varepsilon_\ell\right) & \text{for } d < \tau_k^{(2)}
\end{cases} \\
=\ & \begin{cases}
\left|\lambda_2(\tau_k^{(2)}, d)\right|\left(\left|\frac{\nu_{k+1}-\nu_k}{2}\right| + \operatorname{sgn}(\nu_k - \nu_{k-1})\hat{D}_k\right) + \operatorname{sgn}(\nu_k - \nu_{k-1}) \sum\limits_{\substack{\tau_k^{(2)} < \ell \leq d \\ \ell \in S^{(2)}(t_k)}} \varepsilon_\ell & \text{for } d > \tau_k^{(2)} \\
0 & \text{for } d = \tau_k^{(2)} \\
\left|\lambda_2(\tau_k^{(2)}, d)\right|\left(\left|\frac{\nu_k-\nu_{k-1}}{2}\right| - \operatorname{sgn}(\nu_k - \nu_{k-1})\hat{D}_k\right) - \operatorname{sgn}(\nu_k - \nu_{k-1}) \sum\limits_{\substack{d < \ell \leq \tau_k^{(2)} \\ \ell \in S^{(2)}(t_k)}} \varepsilon_\ell & \text{for } d < \tau_k^{(2)}
\end{cases} \qquad (120)
\end{aligned}$$

where

$$\hat{D}_k := \frac{(\nu_k - v_k) + (\nu_{k-1} - v_{k-1})}{2} \tag{121}$$

The maximal deviation of the signal estimates are at most $\rho_N \to 0$ away from the true signal estimates, and hence for all sufficiently large N, $\left|\hat{D}_k\right| \le |\rho_N| \le \frac{\underline{\Delta}}{2}$ (where $\underline{\Delta}$ bounds the minimum absolute jump from below). In this case

$$\left|\frac{\nu_{k+1} - \nu_k}{2} + \hat{D}_k\right| \ge \frac{\underline{\Delta}}{2} \quad \text{and} \quad \left|\frac{\nu_{k+1} - \nu_k}{2} - \hat{D}_k\right| \ge \frac{\underline{\Delta}}{2}. \tag{122}$$

Therefore the mean of the $\mathbb{M}_k(d)$'s satisfy

$$\mathbb{E}[\mathbb{M}_k(d)] = \left|\lambda_2(\tau_k^{(2)}, d)\right| \left(\left|\frac{\nu_{k+1} - \nu_k}{2}\right| \pm \operatorname{sgn}(\nu_k - \nu_{k-1})\hat{D}_k\right) \ge \left|\lambda_2(\tau_k^{(2)}, d)\right| \frac{\underline{\Delta}}{2}.$$

At the same time, $\mathbb{M}_k(d) - \mathbb{E}[\mathbb{M}_k(d)]$ is a zero mean subGaussian random variable with variance parameter $\sigma^2_{\max}\left|\lambda_2(\tau_k^{(2)}, d)\right|$, hence

$$\begin{aligned}
& \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) \le 0\right] \\
= \;& \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) - \mathbb{E}[\mathbb{M}_k(d)] \le -\mathbb{E}[\mathbb{M}_k(d)]\right] \\
\le \;& \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) - \mathbb{E}[\mathbb{M}_k(d)] \le -\left|\lambda_2(\tau_k^{(2)}, d)\right| \frac{\underline{\Delta}}{2}\right] \\
\le \;& \frac{1}{2}\exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}\left|\lambda_2(\tau_k^{(2)}, d)\right|\right) \qquad (123)
\end{aligned}$$

Going back to (119), this gives:

$$\begin{aligned}
& \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\hat{\tau}_k^{(2)} - \tau_k^{(2)}\right| > a_N\right] \\
\le \;& \sum_{\substack{d \in S^{(2)}(t_k) \\ \left|\lambda_2(\tau^{(2)}, d)\right| > a_N/2}} \mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\mathbb{M}_k(d) \le 0\right] \\
\le \;& \frac{1}{2} \sum_{\substack{d \in S^{(2)}(t_k) \\ \left|\lambda_2(\tau^{(2)}, d)\right| > a_N/2}} \exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}\left|\lambda_2(\tau_k^{(2)}, d)\right|\right) \\
\le \;& \frac{1}{2}\left[\sum_{k=\frac{a_N}{2}+1}^{\infty} \exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}k\right) + \sum_{k=-\frac{a_N}{2}-1}^{-\infty} \exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}|k|\right)\right] \\
= \;& \left(1 - \exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}\right)\right)^{-1} \exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}\left(\frac{a_N}{2} + 1\right)\right) \qquad (124)
\end{aligned}$$

Therefore in order to bound $\mathbb{P}_{J,\boldsymbol{v},\boldsymbol{t}}\left[\left|\hat{\tau}_k^{(2)} - \tau_k^{(2)}\right| > a_N\right]$ by C_ε/J as we stated back in (114), a legitimate choice of a_N which will satisfy this bound can be found by solving

$$\frac{\exp\left(-\frac{\underline{\Delta}^2}{8\bar{\sigma}^2_{\max}}\right)}{1-\exp\left(-\frac{\underline{\Delta}^2}{8\bar{\sigma}^2_{\max}}\right)}\exp\left(-\frac{\underline{\Delta}^2}{8\sigma^2_{\max}}\cdot\frac{a_N}{2}\right) = \frac{C_\epsilon}{J}. \tag{125}$$

The solution to this will be of the form $a_N = C_1\log(J) + C_2$, where C_1 and C_2 are constants not dependent on N, and because $\log(J) = O(\log(N))$, this is a valid solution. □

9.6. *Proof sketch of Theorem 5.*

PROOF. Let $\mathcal{R}_N$ be the event

$$\mathcal{R}_N := \left\{\hat{J} = J;\ \max_{j=1,\ldots,J}\left|\hat{\tau}_j^{(1)} - \tau_j\right| \le w(N);\ \max_{j=0,\ldots,J}\left|\hat{\nu}_j^{(1)} - \nu_j\right| \le \rho_N\right\} \tag{126}$$

Define $G_N()$ as the joint distribution of the first stage estimates $(\hat{J}, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}) = (\hat{J}, \hat{\tau}_1^{(1)}, \ldots, \hat{\tau}_{\hat{J}}^{(1)}, \hat{\nu}_0^{(1)}, \ldots, \hat{\nu}_{\hat{J}}^{(1)})$. Because $\mathbb{P}[\mathcal{R}_N] \to 1$, it is possible to show that the difference between

$$\mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \text{ for } j = 1,\ldots,J\right] \tag{127}$$

and $\prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})$, equals the expression

$$\int_{(k,\boldsymbol{t},\boldsymbol{\alpha})\in\mathcal{R}_N}\left(\prod_{j=1}^J \mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J} = k;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right]\right.$$
$$\left. - \prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) dG_N(k,\boldsymbol{t},\boldsymbol{v}) + o(1) \tag{128}$$

It is therefore sufficient to show that the difference inside the integral is, for all $(k,\boldsymbol{t},\boldsymbol{\alpha}) \in \mathcal{R}_N$, uniformly bounded in absolute value by a $o(1)$ term. Henceforth, consider only such admissible k's, $\boldsymbol{t}$'s, $\boldsymbol{v}$'s (which, of course, restricts k to J), and additionally that N is large enough so that $\rho_N \le |\underline{\Delta}|/8$ and distance between consecutive points in $S^{(1)}$ is more than 2 (i.e. $\min_{i,j\in S^{(1)}, i\ne j}|i-j| > 2$). We will proceed to show (23) by obtaining an upper bound for the following absolute difference, for every $j = 1,\ldots,J$:

$$\left|\mathbb{P}\left[\left|\hat{\tau}_j^{(2)} - \tau_j\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J} = J;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right]\right.$$
$$\left. - P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right|. \tag{129}$$

This upper bound will be derived in several components.

First Component: A more explicit expression for the change point estimates is, for every $j = 1, \ldots, J$,

$$\begin{aligned} \hat{\tau}_j^{(2)} &= \underset{t \in S^{(2)}(t_i)}{\arg\min} (\operatorname{sgn}(v_j - v_{j-1})) \sum_{i \in S^{(2)}(t_i)} \left(Y_i - \frac{v_j + v_{j-1}}{2} \right) (1(i \leq t) - 1(i \leq \tau_j^{(2)})) \\ &=: \underset{t \in S^{(2)}(t_i)}{\arg\min} \hat{X}_j^{(2)}(t) \end{aligned} \tag{130}$$

$\hat{X}^{(2)}(t)$ is a random walk over the set $t \in S^{(2)}(t_j)$, which is a set of integers skipping over the points of $S^{(1)}$. To express this as a random process over a set of integers without skipped points (a requirement to apply some probability bounds), define the one-to-one function $\pi_{2,j}(s) := \lambda_2(\tau_j^{(2)}, s)$ for $s \in \{1, \ldots, N\} - S^{(1)}$. This allows us to consider the random walk $\hat{X}_j^{(2)}(\pi_{2,j}^{-1}(t))$ over the set $t \in \pi_{2,j}(S^{(2)}(t_j))$. This is a set of integers containing no skipped points, and furthermore it can be shown that

$$\left[\pm Q_{(|\Delta_j - 2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \subset \pi_{2,j}(S^{(2)}(t_j)) \tag{131}$$

(could be done by showing $[\pm((K-1)/2)w(N)] \subset \pi_{2,j}(S^{(2)}(t_j))$ and using Lemma 11 to argue that $Q_{(|\Delta_j - 2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) = O(\log(N)) = o(w(N)))$.

From here, $\hat{X}_j^{(2)}(\pi_{2,j}^{-1}(t))$ is a random walk with non-symmetrical linear drifts on either side of $t = 0$. We wish to approximate $\hat{X}^{(2)}(t)$ with a random walk with symmetrical drift, so consider $X'_{|\Delta_j| - 2|\hat{D}_j|}(t)$, defined as

$$X'_{|\Delta_j| - 2|\hat{D}_j|}(t) := \begin{cases} t\frac{|\Delta_j| - 2|\hat{D}_j|}{2} + \operatorname{sgn}(\Delta_j) \sum_{i=1}^{t} \varepsilon_{\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + i\right)} & \text{for } t > 0 \\ 0 & t = 0 \\ |t|\frac{|\Delta_j| - 2|\hat{D}_j|}{2} - \operatorname{sgn}(\Delta_j) \sum_{i=t}^{-1} \varepsilon_{\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + i + 1\right)} & \text{for } t < 0 \end{cases} . \tag{132}$$

where

$$\hat{D}_j = \frac{\nu_j - v_j}{2} + \frac{\nu_{j-1} - v_{j-1}}{2}. \tag{133}$$

This random walk is very close to $\hat{X}^{(2)}(\pi_{2,j}^{-1}(t))$, since it equals either $X'_{|\Delta_j| - 2|\hat{D}_j|}(t) + 2|\hat{D}_j t| 1(t > 0)$ or $X'_{|\Delta_j| - 2|\hat{D}_j|}(t) + 2|\hat{D}_j t| 1(t < 0)$ for $t \in \pi_{2,j}^{-1}(S^{(2)}(t_j)$, and $|\hat{D}_j| < \rho_N \to 0$. In either case, this begs the use of Lemma 13 and Lemma 16, two results which allows us to compare the argmin of such similar random walks. The only important condition they require is satisfied by (131). Their

application gives

$$\begin{aligned}
&\left|\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)},\hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\,\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right.\\
&\left.-\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\,\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right|\\
=\;&\left|\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X_j^{(2)}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\,\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right.\\
&\left.-\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\,\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right|\\
(134)\le\;& C_1^*\rho_N
\end{aligned}$$

For some constant $C_1^* > 0$, not dependent on j or N.

Second Component: Now $X'_{|\Delta|-2|\hat{D}_j|}(t)/\sigma$ has the same exact distribution as $X_{(|\Delta|-2|\hat{D}_j|)/\sigma}(t)$, for all integers $t \in \pi_{2,j}^{-1}(S^{(2)}(t_j)$. It was also shown in the previous section that the set $\pi_{2,j}^{-1}(S^{(2)}(t_j)$ contains the interval of integers $\left[\pm Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]$ for all large N. Therefore apply Lemma 14 to obtain

$$\begin{aligned}
&\left|\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\,\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right.\\
(135)\quad&\left.-\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\right| \le C_2^*\rho_N
\end{aligned}$$

for some $C_2^* > 0$ not dependent on j or N.

Third Component: The set $\pi_{2,j}^{-1}(S^{(2)}(t_j))$ contains the set $\left[\pm\left(\frac{(K-1)w(N)}{2}-1\right)\right]$. This allows an application of Lemma 9 to obtain

$$\begin{aligned}
&\left|\mathbb{P}\left[\left|\underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\right.\\
&\left.-\mathbb{P}\left[\left|\underset{|t|\le\frac{(K-1)w(N)}{2}-1}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\right|\\
(136)\qquad\le\;& A_1\exp\left(-B_1\left(\frac{(K-1)w(N)}{2}-1\right)\right)
\end{aligned}$$

for some constant $A_1 > 0$, $B_1 > 0$ not dependent on j or N. The same application of the lemma can also yield

$$\begin{aligned} & \left| \mathbb{P}\left[\left| \underset{|t| \leq \frac{(K-1)w(N)}{2}-1}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right. \\ & \left. -\mathbb{P}\left[\left| \underset{t\in\mathbb{Z}}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right| \\ \leq \; & A_2 \exp\left(-B_2 \left(\frac{(K-1)w(N)}{2} - 1 \right) \right) \end{aligned} \tag{137}$$

for some $A_2 > 0$ and $B_2 > 0$ not dependent on j or N. Adding up these two upper bounds imply

$$\begin{aligned} & \left| \mathbb{P}\left[\left| \underset{t\in\pi_{2,j}^{-1}(S^{(2)}(t_j))}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right. \\ & \left. -\mathbb{P}\left[\left| \underset{t\in\mathbb{Z}}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right| \\ \leq \; & (A_1 \vee A_2) \exp\left(-(B_1 \wedge B_2) \left(\frac{(K-1)w(N)}{2} - 1 \right) \right) \end{aligned} \tag{138}$$

Sum of the Components: Adding up the differences in (169), (173), and (178):

$$\begin{aligned} & \left| \mathbb{P}\left[\left| \lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)} \right) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J} = J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v} \right] \right. \\ & \left. -\mathbb{P}\left[\left| \underset{t\in\mathbb{Z}}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right| \\ \leq \; & C_4\rho_N + C_5 \exp\left[-C_6(K-1)w(N) \right] \end{aligned} \tag{139}$$

for some constants C_4, C_5, and C_6. This allows us to bound

$$\begin{aligned} & \prod_{j=1}^{J} \mathbb{P}\left[\left| \lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)} \right) \right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J} = J; \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t}; \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v} \right] \\ & - \prod_{j=1}^{J} P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \end{aligned} \tag{140}$$

with

$$
\begin{aligned}
& J\left|\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right. \\
& \left.-\mathbb{P}_{|\Delta_j|-2\rho_N}(\sqrt[J]{1-\alpha})\right|\left(\mathbb{P}_{|\Delta_j|-2\rho_N}(\sqrt[J]{1-\alpha})\right)^{-1} \\
\leq\ & J\left(C_4\rho_N + C_5\exp\left[-C_6(K-1)w(N)\right]\right)(1-\alpha)^{-1/J} \\
\leq\ & C_4(1-\alpha)^{-1}J\rho_N + C_5(1-\alpha)^{-1}N\exp\left[-C_6(K-1)w(N)\right], \qquad (141)
\end{aligned}
$$

which goes to 0 since $J\rho_N \to 0$ and $w(N) \geq CN^{1-\gamma}$ for some constant C. □

9.7. *Proof of Theorem 5.* In this section we perform the proof in the special situation where all the error terms of the data sequence are iid $N(0,\sigma^2)$ for some positive σ. In the more general case where the error terms are independent Gaussian with variances bounded above and below by $\sigma^2_{\max}$ and $\sigma^2_{\min}$, with identical distributions between change points, similar results from Section 10 and methods from the following proof will yield the statement of Theorem 4. Because we are dealing with iid error terms, we could simplify the notations presented with the theorem in the main draft. First, we could simplify the random walks

$$
X_\Delta(t) := X_{\Delta,1,1}(t) = \begin{cases} t\frac{|\Delta|}{2} + \sum_{i=1}^{t}\varepsilon_i^* & t>0 \\ 0 & t=0 \\ \frac{|t\Delta|}{2} + \sum_{i=0}^{t-1}\varepsilon_{-i}^* & t<0 \end{cases} \qquad (142)
$$

$$
\text{where } \varepsilon_i^* \overset{\text{iid}}{\sim} N(0,1) \text{ for all } i, \qquad (143)
$$

denote the argmin of this random walk as $L_\Delta := \underset{t\in\mathbb{Z}}{\arg\min}\, X_\Delta(t)$, simplify the quantiles

$$
Q_\Delta(1-\alpha) := Q_{\Delta,1,1}(1-\alpha) = \min\ \{t\geq 0 : \mathbb{P}\left[|L_\Delta| \leq t\right] \geq 1-\alpha\} \qquad (144)
$$

for any $\alpha \in [0,1]$, and their associated probabilities

$$
P_\Delta(1-\alpha) = P_{\Delta,1,1}(1-\alpha)\left[|L_\Delta| \leq Q_\Delta(1-\alpha)\right]. \qquad (145)
$$

We will use these notations for the remainder of this section and the entirety of Section 10.

Using this notation and working within this simplified framework of iid errors, the results of the theorem translates to:

$$
\begin{aligned}
& \mathbb{P}\left[\hat{J}=J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \text{ for } j=1,\ldots,J\right] \\
=\ & \left(\prod_{j=1}^{J} P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) + o(1), \qquad (146)
\end{aligned}
$$

and the second inequality

$$\begin{aligned}
&\mathbb{P}\left[\hat{J}=J;\ \left|\lambda_2\left(\tau_j^{(2)},\hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \text{ for } j=1,\ldots,J\right] \\
&= \left(\prod_{j=1}^{J} P_{(|\Delta_j|+2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right)+o(1), \qquad (147)
\end{aligned}$$

Both these inequalities can be shown utilizing various probability bounds regarding the argmins of random walks of the form $X_\Delta(\cdot)$, and other similar random walks. To utilize such result, the estimators $\hat{\tau}_j^{(2)}$ must be expressed as the argmin of random walks with positive linear drifts. We thus introduce some new notation and new insight to facilitate this task.

First we define the indexing function $\pi_2 = \pi_{2,N}$, which maps the set of integers $\{1,\ldots,N\} - S^{(1)}$ to the set of integers $\{1,\ldots,N-|S^{(1)}|\}$:

$$\pi_2(k) = \sum_{j=1}^{k} 1(j \notin S^{(1)}). \qquad (148)$$

For every N, π_2 is a strictly increasing bijection, and it also has the property such that for any two values i and j in the domain of π_2, $\lambda_2(i,j) = \pi_2(j) - \pi_2(i)$. Additionally, consider the following subset of the full data:

$$Y_{\pi_2^{-1}(1)}, Y_{\pi_2^{-1}(2)}, \ldots, Y_{\pi_2^{-1}(N-|S^{(1)}|)}. \qquad (149)$$

These all the datapoints which were not used in the first stage subsample. This subset of the full dataset is also a change point model following conditions (M1) to (M4) with the same signal upper bound $\bar{\theta}$, the signal jump lower bound $\underline{\Delta}$, and the same error distribution. The change points for this subset is $\pi_2(\tau_j^{(2)})$ for $j=1,\ldots,J$. In fact, (149) and the first stage subsample $\{Y_i\}_{i\in S^{(1)}}$ can be considered statistically independent change point models, resulting in the first stage estimates obtained using $\{Y_i\}_{i\in S^{(1)}}$ being independent of (149). This means that conditional on the first stage estimates, the distribution of (149) does not change from their marginal distributions.

A property of the π_2 function that will be used is a relationship between the π_2^{-1} function and the "normal" subtraction, namely, we want to be able to compare $\pi_2^{-1}(a,b)$ to $b-a$ for $a,b \in \{1,\ldots,N-|S^{(1)}|\}$. For all N large enough such that the distance between consecutive points in $S^{(1)}$ is more than 2 (i.e. $\min_{i,j\in S^{(1)}, i\neq j} |i-j| > 2$), we have the following property: for any integers a and b such that a and $a+b$ are in $\{1,\ldots,N-|S^{(1)}|\}$,

$$\left|\pi_2^{-1}(a+b) - \pi_2^{-1}(a)\right| \le 2|b|. \qquad (150)$$

One way to see this is the following: for any $c,d \in \{1,\ldots,N\} - S^{(1)}$, $|\lambda_2(c,d)|$ counts the number of points in either $(c,d]$ (if $c \le d$) or $(d,c]$ (if otherwise), and therefore $|\lambda_2(c,d)| \ge |d-c|/2$ due to the $S^{(1)}$ containing no two points that are less than 2 apart. Knowing this, the fact that

$$\left|\lambda_2\left(\pi_2^{-1}(a), \pi_2^{-1}(a+b)\right)\right| = \left|\pi_2\left(\pi_2^{-1}(a+b)\right) - \pi_2\left(\pi_2^{-1}(a)\right)\right| = |b|, \qquad (151)$$

means that

$$2|b| = 2\left|\lambda_2\left(\pi_2^{-1}(a), \pi_2^{-1}(a+b)\right)\right| \geq \left|\pi_2^{-1}(a+b) - \pi_2^{-1}(a)\right| \tag{152}$$

We will offer a proof of the inequality given in (146). The steps for verifying (147) would only require a few modifications.

PROOF. Let $\mathcal{R}_N$ be the event

$$\mathcal{R}_N := \left\{\hat{J} = J;\ \max_{j=1,\ldots,J}\left|\hat{\tau}_j^{(1)} - \tau_j\right| \leq w(N);\ \max_{j=0,\ldots,J}\left|\hat{\nu}_j^{(1)} - \nu_j\right| \leq \rho_N\right\} \tag{153}$$

Again define $G_N()$ joint distribution of the first stage estimates $(\hat{J}, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}) = (\hat{J}, \hat{\tau}_1^{(1)}, \ldots, \hat{\tau}_{\hat{J}}^{(1)}, \hat{\nu}_0^{(1)}, \ldots, \hat{\nu}_{\hat{J}}^{(1)})$.

$$\begin{aligned}
& \mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \text{ for } j = 1, \ldots, J\right] \\
= \ & \mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\, \forall j;\ \mathcal{R}_N\right] \\
& + \mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\, \forall j;\ \text{not } \mathcal{R}_N\right] \\
= \ & \int_{(k,\boldsymbol{t},\boldsymbol{\alpha})\in\mathcal{R}_N} \mathbb{P}\left[\max_{j=1,\ldots,J}\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J} = k;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right] dG_N(k, \boldsymbol{t}, \boldsymbol{v}) \\
& + \mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\, \forall j;\ \text{not } \mathcal{R}_N\right]
\end{aligned}$$

Because the probability of $\mathcal{R}_N$ goes to 1, the difference between this probability and $\prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})$ is

$$\begin{aligned}
& \int_{(k,\boldsymbol{t},\boldsymbol{\alpha})\in\mathcal{R}_N} \left(\mathbb{P}\left[\max_{j=1,\ldots,J}\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J} = k;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right]\right. \\
& \left. - \prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) dG_N(k, \boldsymbol{t}, \boldsymbol{v}) + o(1)
\end{aligned} \tag{154}$$

It is therefore sufficient to show that the difference inside the integral is, for all $(k, \boldsymbol{t}, \boldsymbol{\alpha}) \in \mathcal{R}_N$, uniformly bounded in absolute value by a $o(1)$ term. In other words, show there is a sequence $C_{N,\alpha} = o(1)$ such that

$$\begin{aligned}
& \left|\mathbb{P}\left[\max_{j=1,\ldots,J}\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J} = k;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right]\right. \\
& \left. - \prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right| \leq C_{N,\alpha}
\end{aligned} \tag{155}$$

for all admissible $(k, \boldsymbol{t}, \boldsymbol{v}) \in \mathcal{R}_N$.

Henceforth, consider only such admissible k (which restricts to $k = J$), $\boldsymbol{t}$'s and $\boldsymbol{v}$'s, and additionally suppose that N is at least large enough so that $\rho_N \leq |\underline{\Delta}|/16$ and distance between consecutive points in $S^{(1)}$ is more than 2 (i.e. $\min_{i,j \in S^{(1)}, i \neq j} |i-j| > 2$). We will proceed to show (155), by obtaining an upper bound for the following absolute difference, for every $j = 1, \ldots, J$:

$$\left| \mathbb{P}\left[\left| \hat{\tau}_j^{(2)} - \tau_j \right| \leq Q_{(|\Delta_j| - 2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J} = J;\, \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t};\, \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v} \right] \right.$$

$$\left. - P_{(|\Delta_j| - 2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right| \tag{156}$$

This upper bound will be derived in several components.

First Component: A more explicit expression for the change point estimates is

$$\begin{aligned} \hat{\tau}_j^{(2)} &= \operatorname*{arg\,min}_{t \in S^{(2)}(t_i)} (\operatorname{sgn}(v_j - v_{j-1})) \sum_{i \in S^{(2)}(t_i)} \left(Y_i - \frac{v_j + v_{j-1}}{2} \right) (1(i \leq t) - 1(i \leq \tau_j^{(2)})) \\ &=: \operatorname*{arg\,min}_{t \in S^{(2)}(t_i)} \hat{X}_j^{(2)}(t) \end{aligned} \tag{157}$$

Since N was assumed to be large enough so that $\rho_N < |\underline{\Delta}|/8 \leq (\nu_j - \nu_{j-1})/8$, the sign of $v_j - v_{j-1}$ is the same as the sign of $\Delta_j := \nu_j - \nu_{j-1}$, making the optimized expression above equal

$$\begin{aligned} \hat{X}_j^{(2)}(t) &= \operatorname{sgn}(\Delta_j) \sum_{i \in S^{(2)}(t_i)} \left(Y_i - \frac{v_j + v_{j-1}}{2} \right) (1(i \leq t) - 1(i \leq \tau_j^{(2)})) \\ &= \begin{cases} \left| (\tau_j^{(2)}, t] \cap S^{(2)}(t_j) \right| \left[\frac{|\Delta_j|}{2} + \operatorname{sgn}(\Delta_j) \hat{D}_j \right] + \operatorname{sgn}(\Delta_j) \sum_{i \in (\tau_j^{(2)}, t] \cap S^{(2)}(t_j)} \varepsilon_i & \text{for } t > \tau_j \\ 0 & t = \tau_j \\ \left| (t, \tau_j^{(2)}] \cap S^{(2)}(t_j) \right| \left[\frac{|\Delta_j|}{2} - \operatorname{sgn}(\Delta_j) \hat{D}_j \right] - \operatorname{sgn}(\Delta_j) \sum_{i \in (t, \tau_j^{(2)}] \cap S^{(2)}(t_j)} \varepsilon_i & \text{for } t > \tau_j \end{cases} \\ &= \begin{cases} \left(\pi_2(t) - \pi_2(\tau_j^{(2)}) \right) \cdot \left[\frac{|\Delta_j|}{2} + \operatorname{sgn}(\Delta_j) \hat{D}_j \right] + \operatorname{sgn}(\Delta_j) \sum_{i = \pi_2(\tau_j^{(2)}) + 1}^{\pi_2(t)} \varepsilon_{\pi_2^{-1}(i)} & \text{for } t > \tau_j \\ 0 & t = \tau_j \\ \left(\pi_2(\tau_j^{(2)}) - \pi_2(t) \right) \cdot \left[\frac{|\Delta_j|}{2} - \operatorname{sgn}(\Delta_j) \hat{D}_j \right] - \operatorname{sgn}(\Delta_j) \sum_{i = \pi_2(t) + 1}^{\pi_2(\tau_j^{(2)})} \varepsilon_{\pi_2^{-1}(i)} & \text{for } t < \tau_j \end{cases} \\ &\text{since } (a,b] \cap S^{(2)}(t_j) = \pi_2^{-1}\left((\pi_2(a), \pi_2(b)] \right) \text{ for any } (a,b] \subset S^{(2)}(t_j) \end{aligned} \tag{158}$$

where

$$\hat{D}_j = \frac{\nu_j - v_j}{2} + \frac{\nu_{j-1} - v_{j-1}}{2} \tag{159}$$

which is less than ρ_N in absolute value. From the equalities written above, we can deduce that for any integer t such that $\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t\right) \in S^{(2)}(t_j)$,

$$\begin{aligned} &\hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t\right)\right) \\ &= \begin{cases} t \cdot \left[\frac{|\Delta_j|}{2} + \mathrm{sgn}(\Delta_j)\hat{D}_j\right] + \mathrm{sgn}(\Delta_j) \sum_{i=\pi_2(\tau_j^{(2)})+1}^{\pi_2(\tau_j^{(2)})+t} \varepsilon_{\pi_2^{-1}(i)} & \text{for } t > \tau_j \\ 0 & t = \tau_j \\ |t| \cdot \left[\frac{|\Delta_j|}{2} - \mathrm{sgn}(\Delta_j)\hat{D}_j\right] - \mathrm{sgn}(\Delta_j) \sum_{i=\pi_2(\tau_j^{(2)})+t+1}^{\pi_2(\tau_j^{(2)})} \varepsilon_{\pi_2^{-1}(i)} & \text{for } t < \tau_j \end{cases} \end{aligned} \tag{160}$$

This allows us to compare $\hat{X}_j^{(2)}$ with the random walk $X'_{|\Delta_j|-2|\hat{D}_j|}$:

$$X'_{|\Delta_j|-2|\hat{D}_j|}(t) := \begin{cases} t\frac{|\Delta_j|-2|\hat{D}_j|}{2} + \mathrm{sgn}(\Delta_j) \sum_{i=1}^{t} \varepsilon_{\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+i\right)} & \text{for } t > 0 \\ 0 & t = 0 \\ |t|\frac{|\Delta_j|-2|\hat{D}_j|}{2} - \mathrm{sgn}(\Delta_j) \sum_{i=t}^{-1} \varepsilon_{\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+i+1\right)} & \text{for } t < 0 \end{cases} . \tag{161}$$

Specifically, the random process $\hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t\right)\right)$ either equals the sum $X'_{|\Delta_j|-2|\hat{D}_j|}(t) + 2|\hat{D}_j t|1(t > 0)$ or $X'_{|\Delta_j|-2|\hat{D}_j|}(t) + 2|\hat{D}_j t|1(t < 0)$. In either case, this begs the use of Lemma 13 and Lemma 16, but first we must verify some the conditions of those results, namely that $|\Delta_j| > 2|\hat{D}_j|$ (automatically true since N was assumed to be large enough so that $|\Delta_j| \geq \underline{\Delta} \geq 8\rho_N$), and secondly, $\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t\right) \in S^{(2)}(t_j)$ for $|t| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})$. To see that this is true, first use (150) to arrive at

$$\begin{aligned} &\left|\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) \pm Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) - \tau_j^{(2)}\right| \\ &= \left|\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) \pm Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) - \pi_2^{-1}\left(\pi_2\left(\tau_j^{(2)}\right)\right)\right| \\ &\leq 2Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \end{aligned} \tag{162}$$

Therefore, given any $|t| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})$, $\pi_2^{-1}\left(\pi_2\left(\tau_j^{(2)}\right) + t\right) \in \left[\tau_j^{(2)} \pm 2Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] - S^{(1)}$. Next, due to Lemma 11, there are positive expressions

$C_1(\cdot)$, $C_2(\cdot)$, both decreasing, such that

$$\begin{aligned} Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) &\leq C_1(|\Delta_j|-2\rho_N)\log\left(\frac{C_2(|\Delta_j|-2\rho_N)J}{\alpha}\right) \\ &\leq C_1(\underline{\Delta}/2)\log\left(\frac{C_2(\underline{\Delta}/2)J}{\alpha}\right) \\ &\leq C_1(\underline{\Delta}/2)\log\left(\frac{C_2(\underline{\Delta}/2)N}{\alpha}\right) \end{aligned} \tag{163}$$

Since $w(N)$ is greater than order of $N^{1-\gamma}$, which in turn is greater in order than $\log(N)$, we see that for all large N:

$$\begin{aligned} & t_j - Kw(N) \leq \tau_j^{(2)} - (K-1)w(N) + 1 < \tau_j^{(2)} - 2Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \\ \leq\ & \pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) - Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) < \pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right) \\ \leq\ & \tau_j^{(2)} + 2Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \leq \tau_j^{(2)} + (K-1)w(N) - 1 \leq t_j + Kw(N) \end{aligned} \tag{164}$$

Therefore given any $|t| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})$,

$$\begin{aligned} \pi_2^{-1}\left(\pi_2\left(\tau_j^{(2)}\right)+t\right) &\in \left[\tau_j^{(2)} \pm 2Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] - S^{(1)} \\ &\subset [t_j \pm Kw(N)] - S^{(1)} \\ &= S^{(2)}(t_j) \end{aligned} \tag{165}$$

showing that the conditions of Lemma 13 are satisfied. Before continuing, we make a small point before continuing. Note that for any integer $t^* \in S^{(2)}(t_j)$, $\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right) = t^*$ is an equivalent event to $\underset{t:\, \pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min} \hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\right) = t^*$. This is because

$$\begin{aligned} & \underset{t:\, \pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min} \hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\right) = t^* \\ \longleftrightarrow\ & \underset{t\in S^{(2)}(t_j)}{\arg\min}\, \hat{X}_j^{(2)}(t) = \pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t^*\right) \\ \longleftrightarrow\ & \pi_2\left(\underset{t\in S^{(2)}(t_j)}{\arg\min}\, \hat{X}_j^{(2)}(t)\right) - \pi_2(\tau_j^{(2)}) = t^* \\ \longleftrightarrow\ & \lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right) = t^* \end{aligned} \tag{166}$$

We are now ready to apply Lemma 13:

$$\begin{aligned}
&\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)},\hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&-\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
=\;&\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min}\sigma^{-1}\hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|(\hat{J},\hat{\boldsymbol{\tau}}^{(\mathbf{1})},\hat{\boldsymbol{\nu}}^{(\mathbf{1})})=(J,\boldsymbol{t},\boldsymbol{v})\right]\\
&-\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min}\sigma^{-1}X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&\text{indexing change by (166); and division by constant } \sigma \text{ does not change argmin value}\\
\le\;&2\hat{D}_j\sigma^{-1}\Big[A_1^+\left((|\Delta_j|-2\hat{D}_j)/\sigma\right)\left(Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right)^{3/2}\\
&+B_1^+\left((|\Delta_j|-2\hat{D}_j)/\sigma\right)\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\Big]\\
&\times\exp\left[-C_1^+\left((|\Delta_j|-2\hat{D}_j)/\sigma\right)Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]
\end{aligned}$$

Because, as stated previously, $\hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\right)$ either equals $X'_{|\Delta_j|-2|\hat{D}_j|}(t)+2|\hat{D}_j t|1(t>0)$ or $X'_{|\Delta_j|-2|\hat{D}_j|}(t)+2|\hat{D}_j t|1(t<0)$, Lemma 13 leads to this inequality for some positive monotone expressions $A_1^+(),B_1^+()$, and $C_1^+()$, which are decreasing, decreasing, and increasing. This expression could be further bounded by

$$\begin{aligned}
&\le 2\sigma^{-1}\hat{D}_j\left[A_1^+\left(\underline{\Delta}/2\sigma\right)\left(Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right)^{3/2}+B_1^+\left(\underline{\Delta}/2\sigma\right)\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\right]\\
&\times\exp\left[-C_1^+\left(\underline{\Delta}/2\sigma\right)Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\\
&\quad\le 2C^*\left(\underline{\Delta}/2\sigma\right)\rho_N
\end{aligned}$$

(167)

where $C_+^*(\cdot)=\sup_{x\in\mathbb{R}^+}\left([A(\cdot)x^{3/2}+B(\cdot)\sqrt{x}]\exp(-C(\cdot)x)\right)$, guaranteed to be finite

In a similar manner, apply Lemma 16 to obtain

$$\begin{aligned}
&\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)},\hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&-\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&=\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min}\frac{1}{\sigma}\hat{X}_j^{(2)}\left(\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&-\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}}{\arg\min}\frac{1}{\sigma}X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\\
&\ge -2A_1^-\left(\left(|\Delta_j|-6|\hat{D}_j|\right)/\sigma\right)\rho_N\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\\
&\times\exp\left[-B_1^-\left(\left(|\Delta_j|-6|\hat{D}_j|\right)/\sigma\right)Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\\
&\text{for some positive decreasing expression } A_1^- \text{ and some positive increasing expression } B_1^-\\
&\ge -2A_1^-\left(\underline{\Delta}/2\sigma\right)\rho_N\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\exp\left[-B_1^-\left(\underline{\Delta}/2\sigma\right)Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\\
&\ge -2C_1^-(\underline{\Delta}/2\sigma)\rho_N\quad\text{ where } C_1^-(\cdot)=\sup_{x\in\mathbb{R}^+}A_1^-(\cdot)\sqrt{x}\exp\left[B_1^-(\cdot)x\right]
\end{aligned}$$

(168)

Altogether, both (167) and (168) together imply

$$\begin{aligned}
&\left|\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)},\hat{\tau}_j^{(2)}\right)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right.\\
&\left.-\mathbb{P}\left[\left|\underset{t\in S^{(2)}(t_j)-\tau_j}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J,\hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t},\hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right|\\
&\le 2\left(C_1^-(\underline{\Delta}/2\sigma)\vee C_1^+(\underline{\Delta}/2\sigma)\right)\rho_N
\end{aligned}\tag{169}$$

Second Component: Now $X'_{|\Delta|-2|\hat{D}_j|}(t)/\sigma$ has the same exact distribution as $X_{(|\Delta|-2|\hat{D}_j|)/\sigma}(t)$, for all integers t such that $\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)$. It was also shown in the previous section that the set $\left\{t:\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)\right\}$ contains the interval of integers

$\left[\pm Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]$. Therefore apply Lemma 14 to first obtain

$$\begin{aligned}
&\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right] \\
= \;& \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} \frac{1}{\sigma}X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2|\hat{D}_j|)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] \\
&\ge \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] \qquad (170)
\end{aligned}$$

and apply Lemma 14 to obtain another equality in the other direction

$$\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\middle|\hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]$$

$$\begin{aligned}
&= \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2|\hat{D}_j|)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] \\
&\le \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] + \qquad (171)
\end{aligned}$$

$$
\begin{aligned}
& (2\rho_N - 2|\hat{D}_j|)\left[A_2\left((|\Delta_j| - 2\rho_N)/\sigma\right)\left(Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right)^{3/2}\right. \\
& \left. + B_2\left((|\Delta_j| - 2\rho_N)/\sigma\right)\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\right] \\
& \times \exp\left[-C_2\left(|\Delta_j| - 2\rho_N\right) Q_{|\Delta_j|-2\rho_N}(\sqrt[J]{1-\alpha})\right] \\
\leq\ & \mathbb{P}\left[\left|\underset{t\in S^{(2)}(t_j)-\tau_j}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] + \\
& 2\rho_N\left[A_2\left(\underline{\Delta}/2\sigma\right)\left(Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right)^{3/2} + B_2\left(\underline{\Delta}/2\sigma\right)\sqrt{Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})}\right] \\
& \times \exp\left[-C_2\left(\underline{\Delta}/2\sigma\right) Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] \\
& \text{since } N \text{ is large enough such that } 2\rho_N < \underline{\Delta}/2, \text{ and by the monotonicity of } A_2, B_2, \text{ and } C_2 \\
\leq\ & \mathbb{P}\left[\left|\underset{t\in S^{(2)}(t_j)-\tau_j}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] + 2C_2^*\left(\frac{\underline{\Delta}}{2\sigma}\right)\rho_N
\end{aligned}
$$

$$
\text{where } C_2^*(\cdot) = \sup_{x\in\mathbb{R}^+}\left([A_2(\cdot)x^{3/2} + B_2(\cdot)\sqrt{x}]\exp(-C_2(\cdot)x)\right) \tag{172}
$$

These two inequalities together imply a bound on the absolute difference:

$$
\begin{aligned}
& \left|\mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X'_{|\Delta_j|-2|\hat{D}_j|}(t)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right| \hat{J} = J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})} = \boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})} = \boldsymbol{v}\right] \\
& - \mathbb{P}\left[\left|\underset{t:\,\pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right)\in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right]\Bigg| \\
& \leq 2C_2^*\left(\frac{\underline{\Delta}}{2\sigma}\right)\rho_N
\end{aligned} \tag{173}
$$

Third Component: We note that the set $\left\{t : \pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) + t\right) \in S^{(2)}(t_j)\right\}$ contains the set $\left[\pm\left(\frac{(K-1)w(N)}{2} - 1\right)\right]$. This is because by (150),

$$
\left|\pi_2^{-1}\left(\pi_2(\tau_j^{(2)}) \pm \left(\frac{(K-1)w(N)}{2} - 1\right)\right) - \tau_j^{(2)}\right| \leq 2\left(\frac{(K-1)w(N)}{2} - 1\right) \tag{174}
$$

secondly because

$$t_j - Kw(N) \le \tau_j^{(2)} - (K-1)w(N) + 1 < \tau_j^{(2)} - 2\left(\frac{(K-1)w(N)}{2} - 1\right)$$

$$< \tau_j^{(2)} + 2\left(\frac{(K-1)w(N)}{2} - 1\right) \le \tau_j^{(2)} + (K-1)w(N) \le t_j + Kw(N) \tag{175}$$

Therefore, for any t such that $|t| \le \left(\frac{(K-1)w(N)}{2} - 1\right)$, we have $\pi_2^{-1}(\pi_2(\tau_j^{(2)}) + t) \in [t_j \pm Kw(N)] - S^{(1)} = S^{(2)}(t_j)$. This allows an application of Lemma 9 to obtain

$$\left| \mathbb{P}\left[\left| \underset{t:\, \pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right) \in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right.$$

$$\left. - \mathbb{P}\left[\left| \underset{|t| \le \frac{(K-1)W(N)}{2}-1}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right|$$

$$\le A_3((|\Delta_j| - 2\rho_N)/\sigma) \exp\left(-B_3((|\Delta_j| - 2\rho_N)/\sigma) \left(\frac{(K-1)w(N)}{2} - 1 \right) \right)$$

for some decreasing expression $A_3()$ and increasing expression $B_3()$

$$\le A_3(\underline{\Delta}/2\sigma) \exp\left(-B_3(\underline{\Delta}/2\sigma) \left(\frac{(K-1)w(N)}{2} - 1 \right) \right) \tag{176}$$

The same application of the lemma can also yield

$$\left| \mathbb{P}\left[\left| \underset{|t| \le \frac{(K-1)W(N)}{2}-1}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right.$$

$$\left. - \mathbb{P}\left[\left| \underset{t \in \mathbb{Z}}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right|$$

$$\le A_3(\underline{\Delta}/2\sigma) \exp\left(-B'(\underline{\Delta}/2\sigma) \left(\frac{(K-1)w(N)}{2} - 1 \right) \right) \tag{177}$$

Adding up these two upper bounds imply

$$\left| \mathbb{P}\left[\left| \underset{t:\, \pi_2^{-1}\left(\pi_2(\tau_j^{(2)})+t\right) \in S^{(2)}(t_j)}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right.$$

$$\left. - \mathbb{P}\left[\left| \underset{t \in \mathbb{Z}}{\arg\min} X_{(|\Delta_j|-2\rho_N)/\sigma}(t) \right| \le Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \right] \right|$$

$$\le 2A_3(\underline{\Delta}/2\sigma) \exp\left(-B'(\underline{\Delta}/2\sigma) \left(\frac{(K-1)w(N)}{2} - 1 \right) \right) \tag{178}$$

Sum of the Components: Adding up the differences in (169), (173), and (178):

$$\begin{aligned}
&\left| \mathbb{P}\left[\left| \lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right] \right. \\
&\left. -\mathbb{P}\left[\left| \arg\min_{t\in\mathbb{Z}} X_{(|\Delta_j|-2\rho_N)/\sigma}(t)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right] \right| \\
\leq\ & C_4\rho_N + C_5\exp\left[-C_6(K-1)w(N)\right]
\end{aligned} \tag{179}$$

for some constants C_4, C_5, and C_6.

Finally, in order to bound (155), we note that given the two real valued triangular arrays $a_{N,1},\ldots,a_{N,J}$ and $b_{N,1},\ldots,b_{N,J}$, all of which contained in the continuous interval $[0,1]$, such that $\left|\frac{a_{N,i}-b_{N,i}}{b_{N,i}}\right| \leq C_N$ for $1\leq i\leq J$, where $JC_N\to 0$ as $N\to\infty$, then $|\prod a_{N,j}-\prod b_{N,j}|\to 0$. This is because

$$\begin{aligned}
\left|\prod_{j=1}^J a_{N,j} - \prod_{j=1}^J b_{N,j}\right| &= \left|\prod_{j=1}^J b_{N,j}\right| \left|\prod_{j=1}^J\left(1-\frac{a_{N,j}-b_{N,j}}{b_{N,j}}\right)-1\right|
\end{aligned} \tag{180}$$

Since $|\prod b_{N,j}|\in[0,1]$, the above converges to 0 if $\prod\left(1-\frac{a_{N,j}-b_{N,j}}{b_{N,j}}\right)\to 1$, which is true since

$$\begin{aligned}
&\prod_{j=1}^J\left(1-\frac{a_{N,j}-b_{N,j}}{b_{N,j}}\right) \geq (1-C_N)^J \geq 1-C_NJ \to 1 \\
&\prod_{j=1}^J\left(1-\frac{a_{N,j}-b_{N,j}}{b_{N,j}}\right) \leq (1+C_N)^J \leq (e^{C_N})^J \to e^0 = 1
\end{aligned} \tag{181}$$

This result is useful because for all N large enough so that $2Kw(N)\leq\delta_N$, the stage two sets $S^{(2)}(t_j)$'s for $j=1,\ldots,J$ are mutually exclusive sets, and hence by conditional independence, (155) equals

$$\begin{aligned}
&\left| \prod_{j=1}^J \mathbb{P}\left[\left| \lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J}=J;\ \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t};\ \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right] \right. \\
&\left. -\prod_{j=1}^J P_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha})\right|
\end{aligned} \tag{182}$$

More-ever, using (179),

$$
\begin{aligned}
& J\left|\mathbb{P}\left[\left|\lambda_2\left(\tau_j^{(2)}, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{(|\Delta_j|-2\rho_N)/\sigma}(\sqrt[J]{1-\alpha}) \middle| \hat{J}=J, \hat{\boldsymbol{\tau}}^{(\mathbf{1})}=\boldsymbol{t}, \hat{\boldsymbol{\nu}}^{(\mathbf{1})}=\boldsymbol{v}\right]\right. \\
& \left.-\mathbb{P}_{|\Delta_j|-2\rho_N}(\sqrt[J]{1-\alpha})\right|\left(\mathbb{P}_{|\Delta_j|-2\rho_N}(\sqrt[J]{1-\alpha})\right)^{-1} \\
\leq\ & J\left(C_4\rho_N + C_5\exp\left[-C_6(K-1)w(N)\right]\right)(1-\alpha)^{-1/J} \\
\leq\ & C_4(1-\alpha)^{-1}J\rho_N + C_5(1-\alpha)^{-1}N\exp\left[-C_6(K-1)w(N)\right], \qquad (183)
\end{aligned}
$$

which goes to 0 since $J\rho_N \to 0$ and $w(N) \geq CN^{1-\gamma}$ for some constant C. This lets us conclude that (182) converges to 0. □

9.8. *Proof of Theorem 4.* Notation-wise, we again utilize the π_2 function defined in (148), which is a bijection from the set $\{1,\ldots,N\} - S^{(1)}$ to the set $\{1,\ldots,N-|S^{(1)}|\}$.

PROOF. Let $S^{(2)}\left(\hat{\tau}_k^{(1)}\right)$ for $k = 1, ..., \hat{J}$ be the second stage subsamples. As in previous sections, define

$$
\mathcal{R}_N := \left\{\hat{J}=J;\ \max_{j=1,\ldots,J}\left|\hat{\tau}_j^{(1)}-\tau_j\right| \leq w(N);\ \max_{j=0,\ldots,J}\left|\hat{\nu}_j^{(1)}-\nu_j\right| \leq \rho_N\right\} \qquad (184)
$$

We also define the following random functions on integers: for $k = 1, ..., J$, on the event $\mathcal{R}_N$ let

$$
\begin{aligned}
\hat{X}_k^{(2)}(d) := \operatorname{sgn}\left(\hat{\nu}_k^{(1)}-\hat{\nu}_{k-1}^{(1)}\right) & \sum_{j\in S^{(2)}(\hat{\tau}_k^{(1)})}\left(Y_j - \frac{\hat{\nu}_k^{(1)}+\hat{\nu}_{k-1}^{(1)}}{2}\right)\left(1\left(j \leq \pi_2^{-1}(\pi_2(\tau_k^{(2)})+d)\right) - 1\left(j\leq\tau_k^{(2)}\right)\right) \\
& \text{for } \pi_2^{-1}(\pi_2(\tau_k^{(2)})+d) \in S_k^{(2)} \\
\hat{X}_k^{(2)}(d) := \infty \qquad & \text{otherwise}
\end{aligned} \qquad (185)
$$

and on the event $\mathcal{R}_N^C$ let

$$
\hat{X}_k^{(2)}(d) := d. \qquad (186)
$$

so that the arg min of $\hat{X}_k^{(2)}(d)$ is $d = -\infty$ on the event $\mathcal{R}_N^C$. Using this definition, for all sufficiently large N, the event $\{\hat{J}=J,\ \lambda_2(\tau_k^{(2)}, \hat{\tau}_k^{(2)}) = j_k \text{ for } k = 1, ..., J\} \cap \mathcal{R}_N$ is equivalent to the event

$$
\left\{\arg\min_{d\in\mathbb{Z}} \hat{X}_k^{(2)}(d) = j_k \text{ for } k = 1, ..., J\right\}.
$$

Further, for any integer $M > 0$, we can easily obtain convergence properties by restricting the function $\hat{X}_k^{(2)}$ to the set $\{-M, -(M-1), ..., M\}$. For sufficiently large N, when event $\mathcal{R}_N$ occurs, we have for any $d \in \{-M, ..., M\}$, and for all $k = 1, ..., J$,

$$\hat{X}_k^{(2)}(d) = \operatorname{sgn}(\Delta_k) \sum_{j \in S^{(2)}(\hat{\tau}_k^{(1)})} \left(Y_j - \frac{\hat{\nu}_k^{(1)} + \hat{\nu}_{k-1}^{(1)}}{2} \right) \left(1\left(j \le \pi_2^{-1}(\pi_2(\tau_k^{(2)}) + d) \right) - 1\left(j \le \tau_k^{(2)} \right) \right)$$

$$= \begin{cases} \operatorname{sgn}(\Delta_k) \sum_{j=\pi_2(\tau_k^{(2)})+1}^{\pi_2(\tau_k^{(2)})+d} \left(Y_{\pi^{-1}(j)} - \nu_k + \frac{\nu_k - \nu_{k-1}}{2} + \frac{1}{2}\left(\nu_k - \hat{\nu}_k^{(1)} + \nu_{k-1} - \hat{\nu}_{k-1}^{(1)} \right) \right) & \text{for } d > 0 \\ 0 & \text{for } d = 0 \\ -\operatorname{sgn}(\Delta_k) \sum_{j=\pi_2(\tau_k^{(2)})+d+1}^{\pi_2(\tau_k^{(2)})} \left(Y_{\pi_2^{-1}(j)} - \nu_{k-1} + \frac{\nu_{k-1} - \nu_k}{2} + \frac{1}{2}\left(\nu_k - \hat{\nu}_k^{(1)} + \nu_{k-1} - \hat{\nu}_{k-1}^{(1)} \right) \right) & \text{for } d < 0 \end{cases}$$

$$= \begin{cases} \frac{d|\Delta_k|}{2} + \operatorname{sgn}(\Delta_k) \left(\sum_{j=\pi_2(\tau_k^{(2)})+1}^{\pi_2(\tau_k^{(2)})+d} \varepsilon_{\pi_2^{-1}(j)} + \frac{d}{2}\left(\nu_k - \hat{\nu}_k^{(1)} + \nu_{k-1} - \hat{\nu}_{k-1}^{(1)} \right) \right) & \text{for } d > 0 \\ 0 & \text{for } d = 0 \\ -d\frac{|\Delta_k|}{2} - \operatorname{sgn}(\Delta_k) \left(\sum_{j=\pi_2(\tau_k^{(2)})+d+1}^{\pi_2(\tau_k^{(2)})} \varepsilon_{\pi_2^{-1}(j)} + \frac{d}{2}\left(\nu_k - \hat{\nu}_k^{(1)} + \nu_{k-1} - \hat{\nu}_{k-1}^{(1)} \right) \right) & \text{for } d < 0 \end{cases}$$

(187)

Because $\max_{i=1,...,J} |\hat{\nu}_i^{(1)} - \nu_o| \le \rho_N$ under $\mathcal{R}_N$, this gives the uniform bound

$$\left| \frac{d}{2\sigma} \left(\nu_k - \hat{\nu}_k^{(1)} + \nu_{k-1} - \hat{\nu}_{k-1}^{(1)} \right) \right| \le \frac{1}{2} M \rho_N. \tag{188}$$

for all k and d. The right side of the above inequality converges to 0 since $\rho_N \to 0$, and because all of this occurs with probability $\mathbb{P}[\mathcal{R}_N] \to 1$, this shows that the $\hat{X}_k^{(2)}(d)$'s all jointly converge. Specifically, let $\varepsilon_{j,k}^*$ for $j = 0, .., J$ and $k \in \mathbb{Z}$ be random variables such that $\{\varepsilon_{j,k}\}_{k \in \mathbb{Z}}$ are iid $\mathcal{E}_j$ variables for $j = 0, \ldots, J$. Next, we denote the random walks

$$X_j^*(d) = \begin{cases} d\frac{|\Delta_j|}{2} + \operatorname{sgn}(\Delta_j) \sum_{l=1}^{d} \varepsilon_{j,l}^* & \text{for } d > 0 \\ 0 & \text{for } d = 0 \\ -d\frac{|\Delta_j|}{2} - \operatorname{sgn}(\Delta_j) \sum_{l=d+1}^{0} \varepsilon_{j,l}^* & \text{for } d < 0 \end{cases} \tag{189}$$

(note that if the error terms are iid $N(0, \sigma^2)$, the random walk $X_{\Delta_k}^*(d)$ has precisely the same

distribution as the random walk $\sigma X_{\Delta_k/\sigma}(d)$). We have the joint weak convergence

$$\begin{pmatrix} \hat{X}_1^{(2)}(-M), & \hat{X}_1^{(2)}(-M+1), & \ldots, & \hat{X}_1^{(2)}(M), \\ \hat{X}_2^{(2)}(-M), & \hat{X}_2^{(2)}(-M+1), & \ldots, & \hat{X}_2^{(2)}(M), \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, \\ \hat{X}_J^{(2)}(-M), & \hat{X}_J^{(2)}(-M+1), & \ldots, & \hat{X}_J^{(2)}(M) \end{pmatrix} \quad \Rightarrow$$

$$\begin{pmatrix} X_1^*(-M), & X_1^*(-M+1), & \ldots, & X_1^*(M), \\ X_2^*(-M), & X_2^*(-M+1), & \ldots, & X_2^*(M), \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, \\ X_J^*(-M), & X_J^*(-M+1), & \ldots, & X_J^*(M) \end{pmatrix} \tag{190}$$

Define $L_j^* := \arg\min_{k\in\mathbb{Z}} X_j^*(k)$, and $L_j^{*(M)} := \arg\min_{|k|\le M} X_j(k)$ for $j = 1, \ldots, J$ (note that if the error terms of the data sequence is $N(0,1)$, $L_{\Delta_k}^*$ has the same distribution as L_{Δ_k}). We have the joint weak convergence

$$\left(\arg\max_{|j|\le M} \hat{X}_1^{(2)}(d), \ldots, \arg\max_{|j|\le M} \hat{X}_J^{(2)}(d)\right) \quad \Rightarrow \quad \left(L_{\Delta_1}^{*(M)}, \ldots, L_{\Delta_J}^{*(M)}\right) \tag{191}$$

by the continuous mapping theorem, because arg max is a continuous function on $\mathbb{R}^{2M+1}$ (except when at least two of the coordinates are equal, which has probability 0 if the error terms have densities).

Next, we establish that $\mathbb{P}\left[\hat{J} = J,\, \lambda_2(\tau_k^{(2)}, \hat{\tau}_k^{(2)}) = j_k \text{ for } k = 1, ..., J\right]$ converges to the product of $\mathbb{P}[L_k^* = j_k]$ for $k = 1, ..., J$. We will do this by showing for any fixed $\epsilon > 0$, the absolute difference between the two is smaller than ϵ for all large N. As in the proof of the single change point problem, this is accomplished through three main inequalities.

First Inequality: From the result of Theorem 3, and by the fact that $\mathbb{P}[\mathcal{R}_N] \to 1$, we can find an integer K_0 greater than $\max_k |j_k|$, such that for any $K_1 \ge K_0$, we have for sufficiently large N

$$\mathbb{P}\left[\hat{J} = J,\, \max_{k=1,\ldots,J} \left|\hat{\tau}_k^{(2)} - \tau_k\right| \le (2K_1 + 2),\, \mathcal{R}_N\right] \ge 1 - \frac{\epsilon}{4} \tag{192}$$

For all sufficiently large N, $\left\{\hat{J} = J,\, \max_{k=1,\ldots,J} \left|\hat{\tau}_k^{(2)} - \tau_k\right| \le (2K_1 + 2)\right\} \cap \mathcal{R}_N$ would mean

$$\left\{\hat{J} = J,\, \max_{k=1,..,J} \left|\lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right)\right| \le K_1\right\} \cap \mathcal{R}_N$$

and hence

$$1-\frac{\epsilon}{3} \leq \mathbb{P}\left[\hat{J}=J,\ \min_{k=1,..,J}\left|\lambda_2\left(\tau_k^{(2)},\hat{\tau}_k^{(2)}\right)\right|\leq K_1,\ \mathcal{R}_N\right]$$

$$= \mathbb{P}\left[\hat{J}=J;\ \max_{k=1,...,J}\left|\arg\min_{d\in\mathbb{Z}}\hat{X}_k^{(2)}(d)\right|\leq K_1\right] \tag{193}$$

Now

$$\arg\min_{d\in\mathbb{Z}}\hat{X}_k^{(2)}(d)=j_k \text{ for } k=1,...,J$$

$$\longleftrightarrow \arg\min_{|d|\leq K_1}\hat{X}_k^{(2)}(d)=j_k \text{ for } k=1,...,J, \text{ and } \max_{k=1,...,J}\left|\arg\min_{d\in\mathbb{Z}}\hat{X}_k^{(2)}(d)\right|\leq K_1 \tag{194}$$

With steps very similar to those used in (80), it can be shown that

$$\left|\mathbb{P}\left[\arg\min_{d\in\mathbb{Z}}\hat{X}_k^{(2)}(d)=j_k \text{ for } k=1,...,J\right]-\mathbb{P}\left[\arg\min_{|d|\leq K_1}\hat{X}_k^{(2)}(d)=j_k \text{ for } k=1,...,J\right]\right|$$

$$\leq \mathbb{P}\left[\max_{k=1,...,J}\left|\arg\min_{d\in\mathbb{Z}}\hat{X}_k^{(2)}(d)\right|>K_1\right]$$

$$\leq \epsilon/4 \tag{195}$$

Second Inequality We can find some integer $K_2>K_0$ such that

$$\mathbb{P}\left[\max_{k=1,..,J}|L_k^*|\leq K_2\right]\geq 1-\frac{\epsilon}{4} \tag{196}$$

Now $L_k^*=j_k$ for $k=1,...,J$ if and only if both $L_k^{*(K_2)}=j_k$ for $k=1,...,J$ and $\max_k|L_k|\leq K_2$. With steps very similar to those in (76), we have

$$\left|\mathbb{P}\left[L_k^*=j_k \text{ for } k=1,...,J\right]-\mathbb{P}\left[L_k^{*(K_2)}=j_k \text{ for } k=1,...,J\right]\right| \tag{197}$$

$$\leq \mathbb{P}\left[\max_{k=1,..,J}|L_k^*|>K_2\right]$$

$$\leq \epsilon/4 \tag{198}$$

Third Inequality By weak convergence, we have

$$\left|\mathbb{P}\left[L_k^{*(K_2)}=j_k \text{ for } k=1,...,J\right]-\mathbb{P}\left[\arg\min_{|d|\leq K_2}\hat{X}_k^{(2)}(d)=j_k \text{ for } k=1,...,J\right]\right|\leq\frac{\epsilon}{4} \tag{199}$$

for all sufficiently large N.

Combining the inequalities in (195), (197), and (199) will give

$$\begin{aligned}
& \left|\mathbb{P}\left[L_k^* = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[\hat{J}=J,\, \lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right) = j_k \text{ for } k=1,..,J,\, \mathcal{R}_N\right]\right| \\
= & \left|\mathbb{P}\left[L_k^* = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[\arg\min_{d\in\mathbb{Z}} \hat{X}_k^{(2)}(d) = j_k \text{ for } k=1,...,J\right]\right| \\
\leq & \left|\mathbb{P}\left[\arg\min_{d\in\mathbb{Z}} \hat{X}_k^{(2)}(d) = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[\arg\min_{|d|\leq K_2} \hat{X}_k^{(2)}(d) = j_k \text{ for } k=1,...,J\right]\right| \\
+ & \left|\mathbb{P}\left[L_k^{*(K_2)} = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[\arg\min_{|d|\leq K_2} \hat{X}_k^{(2)}(d) = j_k \text{ for } k=1,...,J\right]\right| \\
+ & \left|\mathbb{P}\left[L_k^* = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[L_k^{*(K_2)} = j_k \text{ for } k=1,...,J\right]\right| \\
\leq & \, 3\epsilon/4 \qquad (200)
\end{aligned}$$

Additionally, for sufficiently large N we have $\mathbb{P}[\text{not } \mathcal{R}_N] < \epsilon/4$ and hence

$$\begin{aligned}
& \left|\mathbb{P}\left[\hat{J}=J,\, \lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right) = j_k \text{ for } k=1,..,J,\, \mathcal{R}_N\right]\right. \\
- & \left.\mathbb{P}\left[\hat{J}=J,\, \lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right) = j_k \text{ for } k=1,..,J\right]\right| \\
< & \, \epsilon/4 \qquad (201)
\end{aligned}$$

and hence

$$\left|\mathbb{P}\left[L_k^* = j_k \text{ for } k=1,...,J\right] - \mathbb{P}\left[\hat{J}=J,\, \lambda_2\left(\tau_k^{(2)}, \hat{\tau}_k^{(2)}\right) = j_k \text{ for } k=1,..,J\right]\right| < \epsilon \qquad (202)$$

for all sufficiently large N. □

9.9. *Proof of Theorem 8.* We will show that

$$\mathbb{P}\left[\hat{J}=J;\, \left|\hat{\tau}_j^{re} - \tau_j^{**}\right| \leq Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\ \forall j\right] \geq 1+\alpha_N+o(1) \qquad (203)$$

Proof. Letting $\mathcal{R}_N$ be the event that

$$\left\{\hat{J}=J;\, \max_{j=1,...,J}|\hat{\tau}_j^* - \tau_j^{**}| \leq w^*(N^*);\, \max_{j=0,...,J}|\hat{\nu}_j^{(1)} - \nu_j| \leq \rho_N\right\}. \qquad (204)$$

In a similar way to the proof for Theorem 5, to prove

$$\begin{aligned}
& \mathbb{P}\left[\hat{J}=J;\, \max_{j=1,...,J}|\hat{\tau}_j^{re} - \tau_j^{**}| \leq Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right) \text{ for all } j=1\ldots,J\right] \\
\geq & \, 1-\alpha_N+o(1) \qquad (205)
\end{aligned}$$

it is sufficient to demonstrate that

$$\begin{aligned}
&\mathbb{P}\left[\max_{j=1,\ldots,J}|\hat{\tau}_j^{re}-\tau_j^{**}|\leq Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\ \forall j\middle|J=\hat{J},\hat{\tau}_j^*=t_j,\hat{\nu}_j^{(1)}=v_j\ \forall j\right]\\
&= 1-P_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)+o(1) \qquad (206)
\end{aligned}$$

for all t_j's and v_j's permissible within $\mathcal{R}_N$, which we will assume when we write t_j's and v_j's from here on.

We now try to bound the difference between

$$\mathbb{P}\left[|\hat{\tau}_j^{re}-\tau_j^{**}|\leq Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\middle|J=\hat{J},\hat{\tau}_j^*=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for all } j\right]$$

and $P_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)$ for all j. Each estimator equals the argmin of a random walk:

$$\begin{aligned}
\hat{\tau}_j^{re} &:= \operatorname*{arg\,min}_{t\in[t_j\pm\hat{d}_j]}(\operatorname{sgn}(v_j-v_{j-1}))\sum_{i\in[t_j\pm\hat{d}_j]}\left(Y_i-\frac{v_j+v_{j-1}}{2}\right)\left[1(i\leq t)-1(i\leq\tau_j^{**})\right]\\
&=: \operatorname*{arg\,min}_{t\in[t_j\pm\hat{d}_j]}\hat{X}_j(t). \qquad (207)
\end{aligned}$$

In comparison with the random walk

$$X_j'(t):=\begin{cases}t\left(\frac{|\Delta_j|}{2}+\operatorname{sgn}(\Delta_j)\hat{D}_j\right)+\operatorname{sgn}(\Delta_j)\sum_{i=1}^t\varepsilon_{\tau_j^{**}+i} & t>0\\ 0 & t=0\\ |t|\left(\frac{|\Delta_j|}{2}-\operatorname{sgn}(\Delta_j)\hat{D}_j\right)-\operatorname{sgn}(\Delta_j)\sum_{i=1}^t\varepsilon_{\tau_j^{**}-i+1} & \end{cases} \qquad (208)$$

where $\hat{D}_j=\frac{v_j-\nu_j+v_{j-1}-\nu_{j-1}}{2}$. We have, for all sufficiently large N, $\hat{X}_j(t+\tau_j^{**})$ equaling either $X_j'(t)-2|t|\hat{D}_j1(t>0)$ or $X'(t)-2|t|\hat{D}_j1(t<0)$ for all integers $t\in[t_j\pm\hat{d}_j]-\tau_j^{**}$. With regards to the set $[t_j\pm\hat{d}_j]-\tau_j^{**}$, for all large N (such that $3w^*(N^*)\leq\delta_{N^*}^*/2$) this set contains the interval $[-\delta_{N^*}^*/2,\delta_{N^*}^*/2]$. This can be seen by the series of inequalities

$$\begin{aligned}
&t_j-\hat{d}_j\leq\tau_j^{**}+w^*(N^*)-(\delta_{N^*}^*-2w^*(N^*))\leq\tau_j^{**}-\delta_{N^*}^*/2\\
&< \tau_j^{**}-\delta_{N^*}^*/2\leq\tau_j^{**}-w^*(N^*)+(\delta_{N^*}^*-2w^*(N^*))\leq t_j+\hat{d}_j. \qquad (209)
\end{aligned}$$

Furthermore, since $\log(J/\alpha_N)=o(\delta_{N^*}^*)$, we can use Lemma 8 to obtain, for all large N,

$$\begin{aligned}
&\tau_j^{**}-\delta_{N^*}^*/2\leq\tau_j^{**}-\frac{1}{B}\log\left(\frac{AJ}{\alpha_N}\right)\leq\tau_j^{**}-Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\\
&< \tau_j^{**}+Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\leq\tau_j^{**}+\frac{1}{B}\log\left(\frac{AJ}{\alpha_N}\right)\leq\tau_j^{**}+\delta_{N^*}^*/2 \qquad (210)
\end{aligned}$$

where A and B are some constants. Therefore, we can apply Lemma 13 to obtain the inequality

$$\begin{aligned}
&\mathbb{P}\left[|\hat{\tau}_j^{re} - \tau_j^{**}| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\middle| J=\hat{J}, \hat{\tau}_j^* = t_j, \hat{\nu}_j^{(1)} = v_j \text{ for all } j\right] \\
= \;& \mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ \hat{X}_j(t+\tau_j^{**})\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\middle| J=\hat{J}, \hat{\tau}_j^* = t_j, \hat{\nu}_j^{(1)} = v_j \text{ for all } j\right] \\
\ge \;& \mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ \frac{1}{\sigma}X_j'(t)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\right] - C_1(|\underline{\Delta}|/2)|\hat{D}_j| \\
\ge \;& \mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ \frac{1}{\sigma}X_j'(t)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\right] - C_1(|\underline{\Delta}|/2)\rho_N \qquad (211)
\end{aligned}$$

for all large N, where $C_1(|\underline{\Delta}|/2)$ is some expression independent of N and j.

Next, use Lemma 14 to obtain

$$\begin{aligned}
&\mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ \frac{1}{\sigma}X_j'(t)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\right] \\
\ge \;& \mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ X_{(|\Delta_j|+2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\right] \qquad (212)
\end{aligned}$$

We could also use Lemma 9 twice to obtain

$$\begin{aligned}
&\mathbb{P}\left[\left|\underset{t\in[t_j\pm\hat{d}_j]-\tau_j^{**}}{\arg\min}\ X_{(|\Delta_j|+2\rho_N)/\sigma}(t)\right| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\right] \\
\ge \;& P_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right) - C_2(\underline{\Delta})\exp(-C_3(\underline{\Delta})\Delta_{N^*}^*) \qquad (213)
\end{aligned}$$

for some positive expressions $C_2(\underline{\Delta})$ and $C_3(\underline{\Delta})$ independent of N and j. By combining these inequalities, we come to the conclusion that

$$\begin{aligned}
&\mathbb{P}\left[|\hat{\tau}_j^{re} - \tau_j^{**}| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\middle| J=\hat{J}, \hat{\tau}_j^* = t_j, \hat{\nu}_j^{(1)} = v_j \text{ for all } j\right] \\
\ge \;& P_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right) - C_1(|\underline{\Delta}|/2)\rho_N - C_2(\underline{\Delta})\exp(-C_3(\underline{\Delta})\delta_{N^*}^*) \qquad (214)
\end{aligned}$$

Therefore

$$\begin{aligned}
& \mathbb{P}\left[\max_{j=1,\ldots,J}|\hat{\tau}_j^{re}-\tau_j^{**}| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\ \forall j \middle| J=\hat{J},\hat{\tau}_j^*=t_j,\hat{\nu}_j^{(1)}=v_j\ \forall j\right] \\
= \quad & \mathbb{P}\left[\bigcap_{j=1,\ldots,J}|\hat{\tau}_j^{re}-\tau_j^{**}| \le Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\ \forall j \middle| J=\hat{J},\hat{\tau}_j^*=t_j,\hat{\nu}_j^{(1)}=v_j\ \forall j\right] \\
\ge \quad & 1-\sum_{j=1}^{J}\mathbb{P}\left[|\hat{\tau}_j^{re}-\tau_j^{**}| > Q_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)\middle| J=\hat{J},\hat{\tau}_j^*=t_j,\hat{\nu}_j^{(1)}=v_j \text{ for all } j\right] \\
\ge \quad & 1-J\left[1-P_{(|\Delta_j|+2\rho_N)/\sigma}\left(1-\frac{\alpha_N}{J}\right)+(C_1(|\underline{\Delta}|/2)\rho_N+C_2(\underline{\Delta})\exp(-C_3(\underline{\Delta})\delta_{N^*}^*))\right] \\
\ge \quad & 1-\alpha_N-J\left(C_1(|\underline{\Delta}|/2)\rho_N+C_2(\underline{\Delta})\exp(-C_3(\underline{\Delta})\delta_{N^*}^*)\right) \qquad (215)
\end{aligned}$$

which converges to 1 since $\alpha_N \to 0$, $J\rho_N \to 0$, and $\delta_{N^*}^* > C(N^*)^\eta$ for some positive constant C and η. □

9.10. *Proof of Lemma 2.* In order to prove Lemma 2, we will rely on the following result:

LEMMA 6. *(i) Suppose that we have an estimation scheme which when applied onto $Z_1,\ldots,Z_{N^*}$ gives estimates $\hat{J}$ for J and $\left(\hat{\tau}_1^*,\cdots,\hat{\tau}_{\hat{J}}^*\right)$ for $(\tau_1^*,\cdots,\tau_J^*)$ such that*

$$\mathbb{P}\left[\hat{J}=J;\quad \max_{i=1,\cdots,J}|\hat{\tau}_j^*-\tau_j^*| \le w^*(N^*)\right] \ge 1-B_{N^*}^* \qquad (216)$$

for some sequences $w^*(N^*)$ and B_{N^*}, with $w^*(N^*)=o(\delta_{N^*}^*)$ and $B_{N^*}\to 0$. Then, for any positive sequence $\{\rho_{N^*}^*\}$ such that $\frac{w^*(N^*)}{\delta_{N^*}^*}=o(\rho_{N^*}^*)$, there exist constants C_1 and C_2, where

$$\mathbb{P}\left[\hat{J}=J;\quad |\hat{\nu}_i^*-\nu_i^*| \ge \rho_{N^*}^*\right] \le B_{N^*}+C_1 w^*(N^*)\frac{\exp\left[-C_2\delta_{N^*}^*\rho_{N^*}^{*2}\right]}{\sqrt{\delta_{N^*}^*}\rho_{N^*}^*} \qquad (217)$$

for all $i=1,\cdots,J$, when N^* is sufficiently large.
(ii) Moreover, as a consequence of part (i),

$$\begin{aligned}
& \mathbb{P}\left[\hat{J}=J;\quad \max_{i=0,\cdots,J}|\hat{\nu}_i^*-\nu_i^*|<\rho_{N^*}^*\right] \ge 1-\left(\frac{N^*}{\delta_{N^*}^*}+2\right)B_{N^*} \\
& -C_1\left(\frac{N^*}{\delta_{N^*}^*}+1\right)w^*(N^*)\frac{\exp[-C_2\delta_{N^*}^*\rho_{N^*}^{*2}]}{\sqrt{\delta_{N^*}^*}\rho_{N^*}^*}. \qquad (218)
\end{aligned}$$

It follows that in addition to the conditions in (i), if, furthermore, $N^*B_{N^*}/\delta_{N^*}^* \to 0$ and $(N^*w^*(N^*)/\delta_{N^*}^{*3/2}\rho_{N^*}^*)=o(\exp[C_2\delta_{N^*}^*\rho_{N^*}^{*2}])$, then the probability in (218) goes to 1. The $\hat{\nu}_i^*$'s are simultaneously consistent if $\rho_{N^*}^*$ also converges to 0.

PROOF. See Section 9.11 □

We will show a more general version of Lemma 2 by finding a sequence $\rho^*_{N^*} \to 0$ such that $\rho^*_{N^*}$ satisfies all the conditions of Lemma 6.

LEMMA 7. *Suppose conditions (M1) to (M3) and (M4 (SBS)) are satisfied, then there exist constants C_1 and C_2, and sequence $\rho^*_{N^*} \to 0$, such that if SBS is performed at step ISM3 on the subsample $Z_1, \ldots, Z_{N^*}$ with $\zeta_{N^*} = C_1\sqrt{\log(N^*)}$, and signal estimates are obtained as in (31),*

$$\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\ldots,J} |\hat{\tau}^*_i - \tau^*_i| \le C_2 \log(N^*); \quad \max_{i=0,\ldots,J} |\hat{\nu}^*_i - \nu^*_i| \le \rho^*_{N^*}\right] \to 1 \tag{219}$$

as $N \to \infty$. If we impose the conditional condition of (M7 (SBS)) from Section 9.1, then this also holds true for some sequence $\rho^*_{N^*}$ where $J\rho^*_{N^*} \to 0$.

PROOF. We start off by defining some notations. Since $\delta_N \ge KN^{1-\Xi}$ for some $K > 0$ by (M3), we must have $J \le K_1 N^{\Lambda}$ for some $K_1 > 0$ and $\Lambda \in [0, \Xi]$. This has the implication that $\delta^*_{N^*} \ge K_2 (N^*)^{1-\frac{\Xi}{\gamma}}$ for some $K_2 > 0$. We also set $\rho^*_{N^*} = (N^*)^{-\theta}$ where θ is some positive value that will be specified.

By the results of Lemma 1, we know that there exists a positive C_1 such that $\zeta_{N^*} = C_1\sqrt{\log(N^*)}$ would imply

$$\mathbb{P}\Big[\hat{J} = J; \quad \max_{i=1,\ldots,J} |\hat{\tau}^*_i - \tau^*_i| \le C_2 \log(N^*)\Big] \ge 1 - C_3/N^*\,. \tag{220}$$

for some positive constants C_2 and C_3. Using this with Lemma 6, we have $w^*(N^*) = C_2 \log(N^*)$ and $B_{N^*} = C_3/N^*$. If we set θ to be any value that satisfies

$$0 < \theta < \frac{1}{2}\left(1 - \frac{\Xi}{\gamma}\right), \tag{221}$$

and now we can check that the other conditions of Lemma 6 are satisfied:

- Since $\log(N^*) = o((N^*)^{1-\Xi/\gamma})$, we have $w^*(N^*) = o(\delta^*_{N^*})$.
- $B_{N^*} = C_3/N^* \to 0$
- $N^* B_{N^*}/\delta^*_{N^*} = C_3/\delta^*_{N^*} \to 0$
- We have

  $$\frac{w^*(N^*)}{\delta^*_{N^*}\rho^*_{N^*}} \le \frac{C_2}{K_2} \cdot \frac{\log N^*}{(N^*)^{1-\frac{\Xi}{\gamma}-\theta}}, \tag{222}$$

  and because $1 - \frac{\Xi}{\gamma} - \theta > 0$, we have $\frac{w^*(N^*)}{\delta^*_{N^*}} = o(\rho^*_{N^*})$.
- We have

  $$\frac{N^* w^*(N^*)}{(\delta^*_{N^*})^{3/2}\rho^*_{N^*}} \le \frac{1}{K_2^{3/2}} (N^*)^{\frac{1}{2}\left(3\frac{\Xi}{\gamma}-1+\theta\right)} \log N^*$$

  $$\delta^*_{N^*}(\rho^*_{N^*})^2 \ge K_2 (N^*)^{1-\frac{\Xi}{\gamma}-2\theta}. \tag{223}$$

Because $\frac{N^* w^*(N^*)}{(\delta^*_{N^*})^{3/2}\rho^*_{N^*}}$ is a power of N^* times a log term and $\exp\left(\delta^*_{N^*}(\rho^*_{N^*})^2\right)$ is an exponent of a positive power of N^* (since $\theta < (1-\Xi/\gamma)/2$), we have $\frac{N^* w^*(N^*)}{(\delta^*_{N^*})^{3/2}\rho^*_{N^*}} = o\left(\exp\left(C^*\delta^*_{N^*}(\rho^*_{N^*})^2\right)\right)$ for any positive constant C^*.

Therefore, all conditions of Lemma 6 are satisfied. We could combine the two limit results

$$\begin{aligned}\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\dots,J} |\hat{\tau}^*_i - \tau^*_i| \le C_2 \log N^*\right] \to 1\\ \mathbb{P}\left[\hat{J} = J; \quad \max_{i=0,\dots,J} |\hat{\nu}^*_i - \nu^*_i| \le \rho^*_{N^*}\right] \to 1\end{aligned} \tag{224}$$

to get the following through the Bonferroni inequality:

$$\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\dots,J} |\hat{\tau}^*_i - \tau^*_i| \le C_2 \log N^*; \quad \max_{i=0,\dots,J} |\hat{\nu}^*_i - \nu^*_i| \le \rho^*_{N^*}\right] \to 1 \tag{225}$$

as $N^* \to \infty$ with increasing N.

If condition (M7 (SBS)) is satisfied, then we have $(1-\Xi/\gamma)/2 > \Xi/\gamma$, so we can choose a $\theta \in \left(\frac{\Xi}{\gamma}, \frac{1}{2}\left(1-\frac{\Xi}{\gamma}\right)\right)$. This way, θ is still within the interval $\left(0, \frac{1}{2}\left(1-\frac{\Xi}{\gamma}\right)\right)$, meaning that all conditions of Lemma 6 are satisfied. We also have

$$J\rho^*_{N^*} \le K_1 N^{\Lambda}(N^*)^{-\theta} \le K^* N^{\Xi} N^{-\theta\gamma} \tag{226}$$

for some constant $K^* > 0$, and because $\Xi - \theta\gamma < 0$, this means $J\rho_{N^*} \to 0$ as $N \to \infty$. □

9.11. *Proof For Lemma 6.*

Proof. First we focus on part (i). Consider the separate cases of τ_i for $i = 0$ or $i = J$ (case 1), and $1 \le i < J$ (case 2). We know

$$\mathbb{P}\Big[\hat{J} = J; \quad \max_{j=1,\dots,J} |\hat{\tau}^*_j - \tau^*_j| \le w^*(N^*)\Big] \ge 1 - B_{N^*} \tag{227}$$

for some sequences $w^*(N^*)$ and B_{N^*} which are $o(\delta^*_{N^*})$, and $o(1)$, respectively. Also as in the statement of the theorem, assume there is some sequence $\rho^*_{N^*}$ such that $w^*(N^*)/\delta^*_{N^*} = o(\rho_{N^*})$. For the rest of the proof assume N^* is large enough so that

- $1 < w^*(N^*) < \frac{\delta^*_{N^*}}{6}$
- $6\frac{w^*(N^*)}{\delta^*_{N^*}}\bar{\theta} < \frac{\rho^*_{N^*}}{2}$
- $\delta^*_{N^*} > 3$

Case 1: For $i = 0$, $\hat{\nu}_0^*$ is the average of all Z_t's where t lies between 1 and $\hat{\tau}_1^*$, inclusive. We have

$$\begin{aligned}
& \mathbb{P}\left[\hat{J} = J; \quad |\hat{\nu}_0^* - \nu_0^*| \geq \rho_{N^*}^*\right] \\
\leq \quad & \mathbb{P}\left[\hat{J} = J; \quad |\hat{\tau}_1^* - \tau_1^*| > w^*(N^*)\right] + \mathbb{P}\left[\hat{J} = J; \quad |\hat{\nu}_0^* - \nu_0^*| \geq \rho_{N^*}^*; \quad |\hat{\tau}_1^* - \tau_1^*| \leq w^*(N^*)\right] \\
\leq \quad & B_{N^*} + \sum_{\tau:\, |\tau - \tau_1^*| \leq w^*(N^*)} \mathbb{P}\left[\hat{J} = J; \quad |\hat{\nu}_0^* - \nu_0^*| \geq \rho_{N^*}^*; \quad \hat{\tau}_1^* = \tau\right] \\
\leq \quad & B_{N^*} + \sum_{\tau:\, |\tau - \tau_1^*| \leq w^*(N^*)} \mathbb{P}\left[\hat{J} = J; \quad \hat{\tau}_1^* = \tau; \quad \left|\frac{1}{\tau}\sum_{j=1}^{\tau} Z_j - \nu_0^*\right| \geq \rho_{N^*}^*\right] \\
\leq \quad & B_{N^*} + \sum_{\tau:\, |\tau - \tau_1^*| \leq w^*(N^*)} \mathbb{P}\left[\left|\frac{1}{\tau}\sum_{j=1}^{\tau} (Z_j - \nu_0^*)\right| \geq \rho_{N^*}^*\right] \qquad (228)
\end{aligned}$$

For all $\tau_1 - w^*(N^*) \leq \tau \leq \tau_1^*$, we have $\frac{1}{\tau}\sum_{j=1}^{\tau}(Z_j - \nu_0^*) \sim N(0, \sigma^2/\tau)$, and hence

$$\begin{aligned}
\mathbb{P}\left[\left|\frac{1}{\tau}\sum_{j=1}^{\tau}(Z_j - \tau_0)\right| \geq \rho_{N^*}^*\right] &= 2\left(1 - \Phi(\sqrt{\tau}\rho_{N^*}^*)\right) \\
&\leq 2\frac{\phi(\sqrt{\tau}\rho_{N^*}^*/\sigma)}{\sqrt{\tau}\rho_{N^*}^*/\sigma} \\
&\leq \sqrt{\frac{2}{\pi}} \cdot \frac{\exp\Big(-(\tau_1 - w^*(N^*))(\rho_{N^*}^*)^2/(2\sigma^2)\Big)}{\sqrt{\tau_1 - w^*(N^*)}\rho_{N^*}^*/\sigma} \\
&\leq \frac{2\sigma}{\sqrt{\pi}} \cdot \frac{\exp\Big(-\delta_{N^*}^*(\rho_{N^*}^*)^2/(4\sigma^2)\Big)}{\sqrt{\delta_{N^*}^*}\rho_{N^*}^*} \\
&\qquad \text{(by } \tau_1 w^*(N^*) > \delta_{N^*}^*/2) \qquad (229)
\end{aligned}$$

For all $\tau_1^* < \tau \leq \tau_1^* + w^*(N^*)$ we have $\frac{1}{\tau}\sum_{j=1}^{\tau}(Z_j - \nu_0^*) \sim N\left(\frac{\tau - \tau_1^*}{\tau}(\nu_1^* - \nu_0^*), \frac{\sigma^2}{\tau}\right)$. Because

$$\left|\frac{\tau - \tau_1^*}{\tau}(\nu_1^* - \nu_0^*)\right| \leq \frac{w^*(N^*)}{\delta_{N^*}^*}(2\bar{\theta}) \leq \frac{\rho_{N^*}^*}{2}, \qquad (230)$$

the magnitude of the z-score of both $\pm\rho_{N^*}^*$ for the $N\left(\frac{\tau - \tau_1^*}{\tau}(\nu_1^* - \nu_0^*), \frac{\sigma^2}{\tau}\right)$ distribution is at least

$\frac{\rho^*_{N^*}\sqrt{\tau}}{2\sigma}$, and hence

$$\begin{aligned}\mathbb{P}\left[\left|\frac{1}{\tau}\sum_{j=1}^{\tau}(Z_j-\tau_0^*)\right|\geq\rho^*_{N^*}\right] &\leq 2\left(1-\Phi\left(\frac{\rho^*_{N^*}\sqrt{\tau}}{2}\right)\right)\\ &\leq 2\frac{\phi\left(\frac{\rho^*_{N^*}\sqrt{\tau}}{2\sigma}\right)}{\frac{\rho^*_{N^*}}{2\sigma}\sqrt{\tau}/\sigma}\\ &\leq \frac{2\sigma\sqrt{2}}{\sqrt{\pi}}\cdot\frac{\exp\left(-\delta^*_{N^*}(\rho^*_{N^*})^2/(8\sigma^2)\right)}{\sqrt{\delta^*_{N^*}}\rho^*_{N^*}}\end{aligned} \tag{231}$$

Therefore, the expression (229) can be bounded from above by

$$\begin{aligned}&B_{N^*}+(w^*(N^*)+1)\frac{2\sigma}{\sqrt{\pi}}\cdot\frac{\exp\left(-\delta^*_{N^*}(\rho^*_{N^*})^2/(4\sigma^2)\right)}{\sqrt{\delta^*_{N^*}}\gamma_{N^*}}+w^*(N^*)\frac{2\sigma\sqrt{2}}{\sqrt{\pi}}\cdot\frac{\exp\left(-\delta^*_{N^*}(\rho^*_{N^*})^2/(8\sigma^2)\right)}{\sqrt{\delta^*_{N^*}}\rho^*_{N^*}}\\ &\leq B_{N^*}+\frac{6\sigma\sqrt{2}}{\sqrt{\pi}}\cdot\frac{w^*(N^*)\exp\left(-\delta^*_{N^*}(\rho^*_{N^*})^2/(8\sigma^2)\right)}{\sqrt{\delta^*_{N^*}}\rho^*_{N^*}}.\end{aligned} \tag{232}$$

For $i=J$, a very similar argument will bound $\mathbb{P}\left[\hat{J}=J;\quad|\hat{\nu}^*_J-\nu^*_J|\geq\rho^*_{N^*}\right]$ by the same expression in (232).

Case 2: The procedure for this case will be similar to the steps for Case 1, but there are a few modifications. For $0<i<J$, $\hat{\nu}^*_i$ is the average of all Z_t's for $\hat{\tau}^*_i<t\leq\hat{\tau}^*_{i+1}$. For the following part we re-write this average by considering the midpoint $\tau_i^{*(m)}:=\frac{\lceil\tau^*_i+\tau^*_{i+1}\rceil}{2}$ where $1<i<J$.

In the case where $\hat{\tau}^*_i$ and $\hat{\tau}^*_{i+1}$ are within $\delta^*_{N^*}/3$ (which is less than $|\tau^*_{i+1}-\tau^*_i|/3$) of τ^*_i

and τ_{i+1}^* respectively, we have $\hat{\tau}_i^* < \tau_i^{*(m)} < \hat{\tau}_{i+1}^*$, and hence we can bound $|\hat{\nu}_i^* - \nu_i^*|$ by

$$\begin{aligned}
&\left| \frac{1}{\hat{\tau}_{i+1}^* - \hat{\tau}_i^*} \sum_{j=\hat{\tau}_i^*+1}^{\hat{\tau}_{i+1}^*} (Z_j - \nu_i^*) \right| \\
&= \left| \frac{\tau_i^{*(m)} - \hat{\tau}_i^*}{\hat{\tau}_{i+1}^* - \hat{\tau}_i^*} \left(\frac{1}{\tau_i^{*(m)} - \hat{\tau}_i^*} \sum_{j=\hat{\tau}_i^*+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \right) + \frac{\hat{\tau}_{i+1}^* - \tau_i^{*(m)}}{\hat{\tau}_{i+1}^* - \hat{\tau}_i^*} \left(\frac{1}{\hat{\tau}_{i+1}^* - \tau_i^{*(m)}} \sum_{j=\tau_i^{*(m)}+1}^{\hat{\tau}_{i+1}^*} (Z_j - \nu_i^*) \right) \right| \\
&\leq \frac{\tau_i^{*(m)} - \hat{\tau}_i^*}{\hat{\tau}_{i+1}^* - \hat{\tau}_i^*} \left| \frac{1}{\tau_i^{*(m)} - \hat{\tau}_i^*} \sum_{j=\hat{\tau}_i^*+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \right| + \frac{\hat{\tau}_{i+1}^* - \tau_i^{*(m)}}{\hat{\tau}_{i+1}^* - \hat{\tau}_i^*} \left| \frac{1}{\hat{\tau}_{i+1}^* - \tau_i^{(m)}} \sum_{j=\tau_i^{(m)}+1}^{\hat{\tau}_{i+1}^*} (Z_j - \nu_i^*) \right|
\end{aligned} \tag{233}$$

In order for $|\hat{\nu}_i^* - \nu_i^*|$ to exceed $\rho_{N^*}^*$, at least one of $\left| \frac{1}{\tau_i^{*(m)} - \hat{\tau}_i^*} \sum_{j=\hat{\tau}_i^*+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \right|$ or $\left| \frac{1}{\hat{\tau}_{i+1}^* - \tau_i^{*(m)}} \sum_{j=\tau_i^{*(m)}+1}^{\hat{\tau}_{i+1}^*} (Z_j - \nu_i^*) \right|$ must exceed $\rho_{N^*}^*$, or in other words,

$$
\begin{aligned}
&\mathbb{P}\left[\hat{J}=J;\quad |\hat{\nu}_i^*-\nu_i^*|\geq \rho_{N^*}^*\right]\\
&\leq \mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_i^*-\tau_i^*|>w^*(N^*)\quad \text{or}\quad |\hat{\tau}_{i+1}^*-\tau_{i+1}^*|>w(N)\right]+\\
&\mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_i^*-\tau_i^*|\leq w^*(N^*);\quad |\hat{\tau}_{i+1}^*-\tau_{i+1}^*|\leq w^*(N^*);\quad |\hat{\nu}_i^*-\nu_i^*|\geq \rho_{N^*}^*\right]\\
&\leq B_{N^*}+\mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_i^*-\tau_i^*|\leq w^*(N^*);\quad |\hat{\tau}_{i+1}^*-\tau_{i+1}^*|\leq w^*(N^*);\quad \left|\frac{1}{\tau_i^{*(m)}-\hat{\tau}_i^*}\sum_{j=\hat{\tau}_i^*+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]\\
&+\mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_i^*-\tau_i^*|\leq w^*(N^*);\quad |\hat{\tau}_{i+1}^*-\tau_{i+1}^*|\leq w^*(N^*);\quad \left|\frac{1}{\hat{\tau}_{i+1}^*-\tau_i^{*(m)}}\sum_{j=\tau_i^{*(m)}+1}^{\hat{\tau}_{i+1}^*}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]\\
&\leq B_{N^*}+\mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_i^*-\tau_i^*|\leq w^*(N^*);\quad \left|\frac{1}{\tau_i^{*(m)}-\hat{\tau}_i^*}\sum_{j=\hat{\tau}_i^*+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]\\
&+\mathbb{P}\left[\hat{J}=J;\quad |\hat{\tau}_{i+1}^*-\tau_{i+1}^*|\leq w^*(N^*);\quad \left|\frac{1}{\hat{\tau}_{i+1}^*-\tau_i^{*(m)}}\sum_{j=\tau_i^{*(m)}+1}^{\hat{\tau}_{i+1}^*}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]\\
&\leq B_{N^*}+\sum_{\tau:\,|\tau-\tau_i^*|\leq w^*(N^*)}\mathbb{P}\left[\left|\frac{1}{\tau_i^{*(m)}-\tau}\sum_{j=\tau+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]\\
&+\sum_{\tau:\,|\tau-\tau_{i+1}^*|\leq w^*(N^*)}\mathbb{P}\left[\left|\frac{1}{\tau-\tau_i^{*(m)}}\sum_{j=\tau_i^{*(m)}+1}^{\tau}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]
\end{aligned}
\tag{234}
$$

Next, we will bound $\mathbb{P}\left[\left|\frac{1}{\tau_i^{*(m)}-\tau}\sum_{j=\tau+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right]$ for each τ such that $|\tau-\tau_i^*|\leq w^*(N^*)$. For $\tau_i^*\leq\tau\leq\tau_i^*+w^*(N^*)$ we have $\frac{1}{\tau_i^{*(m)}-\tau}\sum_{j=\tau+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\sim N\left(0,\frac{\sigma^2}{\tau_i^{*(m)}-\tau}\right)$, and hence

$$
\begin{aligned}
\mathbb{P}\left[\left|\frac{1}{\tau_i^{*(m)}-\tau}\sum_{j=\tau+1}^{\tau_i^{*(m)}}(Z_j-\nu_i^*)\right|\geq \rho_{N^*}^*\right] &\leq 2\left(1-\Phi\left(\rho_{N^*}^*\sigma^{-1}\sqrt{\tau_i^{*(m)}-\tau}\right)\right)\\
&\leq \frac{2\sigma\sqrt{3}}{\sqrt{\pi}}\cdot\frac{\exp(-\delta_{N^*}^*(\rho_{N^*}^*)^2/(12\sigma^2))}{\rho_{N^*}^*\sqrt{\delta_{N^*}^*}},
\end{aligned}
\tag{235}
$$

where we used the fact that $\tau_i^{*(m)} - \tau > \tau_i^{*(m)} - \tau_i^* - w^*(N^*) > \delta_{N^*}^*/3 - \delta_{N^*}^*/6$.

For $\tau_i^* - w^*(N^*) \le \tau < \tau_i^*$, we have $\frac{1}{\tau_i^{*(m)} - \tau} \sum_{j=\tau+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \sim N\left(\frac{\tau_i^* - \tau}{\tau_i^{*(m)} - \tau}(\nu_i^* - \nu_{i-1}^*), \frac{\sigma^2}{\tau_i^{*(m)} - \tau} \right)$. The z-scores of $\pm\rho_{N^*}^*$ would have magnitudes greater than

$$\begin{aligned} & \sigma^{-1}\sqrt{\tau_i^{*(m)} - \tau}\left(\rho_{N^*}^* - \left| \frac{\tau_i^* - \tau}{\tau_i^{*(m)} - \tau}(\nu_i^* - \nu_{i-1}^*) \right| \right) \\ & \ge \sigma^{-1}\sqrt{\frac{\delta_{N^*}^*}{3}}\left(\rho_{N^*}^* - \frac{w^*(N^*)}{\delta_{N^*}^*/3}(2\bar{\theta}) \right) \\ & \ge \sqrt{\frac{\delta_{N^*}^*}{3}} \cdot \frac{\rho_{N^*}^*}{2\sigma} \end{aligned} \tag{236}$$

Hence, this gives the probability bound

$$\begin{aligned} \mathbb{P}\left[\left| \frac{1}{\tau_i^{*(m)} - \tau} \sum_{j=\tau+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \right| \ge \rho_{N^*}^* \right] & \le 2\left(1 - \Phi\left(\sqrt{\frac{\delta_{N^*}^*}{3}} \cdot \frac{\rho_{N^*}^*}{2\sigma} \right) \right) \\ & \le \frac{2\sigma\sqrt{6}}{\sqrt{\pi}} \cdot \frac{\exp\left(-\delta(\gamma_{N^*}^*)^2/(24\sigma^2)\right)}{\rho_{N^*}^*\sqrt{\delta_{N^*}^*}}. \end{aligned} \tag{237}$$

Putting together the bounds in (235) and (237) will give

$$\begin{aligned} & \sum_{\tau:\, |\tau - \tau_i| \le w^*(N^*)} \mathbb{P}\left[\left| \frac{1}{\tau_i^{*(m)} - \tau} \sum_{j=\tau+1}^{\tau_i^{*(m)}} (Z_j - \nu_i^*) \right| \ge \rho_{N^*}^* \right] \\ & \le 3w^*(N^*) \cdot \frac{2\sigma\sqrt{6}}{\sqrt{\pi}} \cdot \frac{\exp\left(-\delta_{N^*}^*(\rho_{N^*}^*)^2/(24\sigma^2)\right)}{\rho_{N^*}^*\sqrt{\delta_{N^*}^*}} \end{aligned} \tag{238}$$

In an extremely similar manner, it can be argued that

$$\begin{aligned} & \sum_{\tau:\, |\tau - \tau_{i+1}^*| \le w^*(N^*)} \mathbb{P}\left[\left| \frac{1}{\tau - \tau_i^{*(m)}} \sum_{j=\tau_i^{*(m)}+1}^{\tau} (Z_j - \nu_i^*) \right| \ge \rho_{N^*}^* \right] \\ & \le 3w^*(N^*) \cdot \frac{2\sigma\sqrt{6}}{\sqrt{\pi}} \cdot \frac{\exp\left(-\delta_{N^*}^*(\rho_{N^*}^*)^2/(24\sigma^2)\right)}{\rho_{N^*}^*\sqrt{\delta_{N^*}^*}} \end{aligned} \tag{239}$$

Therefore, (234) can be bounded by

$$B_{N^*} + \frac{12\sigma\sqrt{6}}{\sqrt{\pi}} \cdot \frac{w^*(N^*)\exp\left(-\delta_{N^*}^*(\rho_{N^*}^*)^2/(24\sigma^2)\right)}{\rho_{N^*}^*\sqrt{\delta_{N^*}^*}} \tag{240}$$

By taking constants C_1 and C_2 to be the "worse" of the coefficients in (232) and (240), which are $\frac{12\sigma\sqrt{6}}{\sqrt{\pi}}$ and $1/(24\sigma^2)$ respectively, we can combine the result of both cases and establish

$$\mathbb{P}\left[\hat{J} = J;\ |\hat{\nu}_i^* - \nu_i^*| \geq \rho_{N^*}^*\right] \leq B_{N^*} + C_1 w^*(N^*) \frac{\exp\left(-C_2 \delta_{N^*}^* (\rho_{N^*}^*)^2\right)}{\rho_{N^*}^* \sqrt{\delta_{N^*}^*}} \tag{241}$$

for all $i = 1, \ldots, J$ □

Using part (i), previously shown, it is straightforward to show part (ii):

PROOF. The complement of the event $\{\hat{J} = J; \quad \max_{i=0,\ldots,J} |\hat{\nu}_i - \nu_i| < \rho_N\}$ is the event where either $\hat{J} \neq J$ or $\hat{J} = J$ and $|\hat{\nu}_i - \nu_i| \geq \rho_N$ for some i. For all sufficiently large N and some positive constants C_1 and C_2 we have

$$\begin{aligned}
& 1 - \mathbb{P}\left[\hat{J} = J; \quad \max_{i=0,\ldots,J} |\hat{\nu}_i^* - \nu_i^*| < \rho_{N^*}^*\right] \\
\leq\ & \mathbb{P}[\hat{J} \neq J] + \sum_{i=0}^{J} \mathbb{P}\left[\hat{J} = J; \quad |\hat{\nu}_i^* - \nu_i^*| \geq \rho_{N^*}^*\right] \\
\leq\ & B_{N^*} + (J+1)\left(B_{N^*} + C_1 w^*(N^*) \frac{\exp\left[-C_2 \delta_{N^*}^* (\rho_{N^*}^*)^2\right]}{\sqrt{\delta_{N^*}^*}\rho_{N^*}^*}\right) \\
\leq\ & B_{N^*} + \left(\frac{N^*}{\delta_{N^*}^*} + 1\right)\left(B_{N^*} + C_1 w^*(N^*) \frac{\exp\left[-C_2 \delta_{N^*}^* (\rho_{N^*}^*)^2\right]}{\sqrt{\delta_{N^*}^*}\rho_{N^*}^*}\right) \\
\to\ & 0
\end{aligned} \tag{242}$$

□

10. Supplement Part C (Probability Bounds on Argmin of Random Walks Absolute Value Drifts).

10.1. *Probability bound for Argmin of Random Walk.* Here we will derive a probability bound for random walks of the form

$$X_\Delta(t) := \begin{cases} t\left|\frac{\Delta}{2}\right| + \sum_{i=1}^{t} \varepsilon_i & t > 0 \\ 0 & t = 0 \\ |t| \cdot \left|\frac{\Delta}{2}\right| - \sum_{i=-1}^{|t|} \varepsilon_i & t < 0 \end{cases} \tag{243}$$

Specifically, the following exponential bound applies:

LEMMA 8. *Suppose that S is a set of integers and m a positive integer such that $[-m,m] \subset S$, then*

$$\mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| > m\right] \leq A(\Delta)\exp(-B(\Delta)m) \tag{244}$$

where A and B are expressions dependent only on $|\Delta|$, with A decreasing and B increasing in $|\Delta|$.

PROOF.

$$\begin{aligned}
&\mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| > m\right] \\
= &\sum_{j>m,\, j\in S} \mathbb{P}\left[\arg\min_{t\in S} X_\Delta(t) = j\right] + \sum_{j<-m,\, j\in S} \mathbb{P}\left[\arg\min_{t\in S} X_\Delta(t) = j\right] \\
\leq &\sum_{t>m,\, t\in S} \mathbb{P}\left[X_\Delta(t) < 0\right] + \sum_{t<-m,\, t\in S} \mathbb{P}\left[X_\Delta(t) < 0\right] \\
\leq &\sum_{t>m,\, t\in S} \mathbb{P}\left[N\left(t\frac{|\Delta|}{2}, t\right) < 0\right] + \sum_{t<-m,\, t\in S} \mathbb{P}\left[N\left(|t|\frac{|\Delta|}{2}, |t|\right) < 0\right] \\
\leq &\sum_{t=m+1}^{\infty} \mathbb{P}\left[N(0,1) < -\frac{\sqrt{t}|\Delta|}{2}\right] + \sum_{t=-m-1}^{-\infty} \mathbb{P}\left[N(0,1) < -\frac{\sqrt{|t|}|\Delta|}{2}\right] \\
\leq &\sum_{t=m+1}^{\infty} \exp\left(-\frac{t\Delta^2}{8}\right) + \sum_{t=-m-1}^{-\infty} \exp\left(-\frac{|t|\Delta^2}{8}\right) \\
= &\ 2\left(\frac{\exp\left(-\frac{\Delta^2}{8}\right)}{1+\exp\left(-\frac{\Delta^2}{8}\right)}\right)\exp\left(-\frac{m\Delta^2}{8}\right) \qquad (245)
\end{aligned}$$

□

This result has another implication. With probability approaching to 1 at an exponential pace, the argmin of $X_\Delta(t)$ equals the argmin over a smaller set:

LEMMA 9. *For any set of integers S and positive integer m such that $[-m,m] \in S$,*

$$\mathbb{P}\left(\arg\min_{t\in[-m,m]} X_\Delta(t) = \arg\min_{t\in S} X_\Delta(t)\right) \geq 1 - A(\Delta)\exp(-B(\Delta)m) \tag{246}$$

where A, B are expressions in Δ that are, respectively, decreasing and increasing in $|\Delta|$ Lemma 8.

PROOF. The two argmins of $X_\Delta(t)$ (over S and over $[m,m]$) are different if and only if the argmin

over S is outisde of the interval $[-m,m]$. Therefore

$$\begin{aligned} & \mathbb{P}\left(\arg\min_{t\in[-m,m]} X_\Delta(t) \neq \arg\min_{t\in S} X_\Delta(t)\right) \\ = \; & \mathbb{P}\left(\left|\arg\min_{t\in S} X_\Delta(t)\right| > m\right) \\ \leq \; & A(\Delta)\exp(-B(\Delta)m) \end{aligned} \tag{247}$$

□

This leads to

LEMMA 10. *Suppose that S is an integer set, ℓ and m are positive integers, and $[-\ell,\ell] \subset [-m,m] \subset S$, then*

$$\left|\mathbb{P}\left[\left|\arg\min_{t\in[-m,m]} X_\Delta(t)\right| \leq \ell\right] - \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \leq \ell\right]\right| \leq A(\Delta)\exp(-B(\Delta)m) \tag{248}$$

for some expressions $A()$ and $B()$ that are respectively, decreasing and increasing with respect to $|\Delta|$.

PROOF.

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in[-m,m]} X_\Delta(t)\right| \leq \ell\right] - \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \leq \ell\right] \\ = \; & \mathbb{P}\left[\left|\arg\min_{t\in[-m,m]} X_\Delta(t)\right| \leq \ell \text{ and } \left|\arg\min_{t\in S} X_\Delta(t)\right| > \ell\right] \\ \leq \; & \mathbb{P}\left[\left|\arg\min_{t\in[-m,m]} X_\Delta(t)\right| \neq \left|\arg\min_{t\in S} X_\Delta(t)\right|\right] \end{aligned} \tag{249}$$

The last line is greater than 0, and by Lemma 9, less than $A(\Delta)\exp(-B(\Delta)m)$ for some appropriate expressions $A()$ and $B()$. □

10.2. *Quantiles.* Due to Lemma 8, the following statement can be made regarding the quantile:

LEMMA 11. *Using the A and B from Lemma 8,*

$$Q_\Delta(\sqrt[J]{1-\alpha}) \leq \frac{1}{B}\log\frac{AJ}{\alpha} \tag{250}$$

PROOF. Using the inequality from Lemma 8,

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in\mathbb{Z}} X_\Delta(t)\right| \le \frac{1}{B}\log\frac{AJ}{\alpha}\right] \\ \ge\ & 1 - A\exp\left(-B\frac{1}{B}\log\frac{AJ}{\alpha}\right) \\ =\ & 1-\frac{\alpha}{J} \\ \ge\ & \sqrt[J]{1-\alpha} \end{aligned} \tag{251}$$

□

10.3. *Comparison Between Random Walks, Part 1.* Here will show some probability inequalities between random walks with different drifts. These results are useful in proving Theorem 5.

LEMMA 12. *Suppose that S is a set of integers, m is a positive integer, and $[-m, m] \subset S$. Define, for any positive Δ_1, Δ_2, the random walks*

$$W_{\Delta_1,\Delta_2}(t) = \begin{cases} |t|\frac{|\Delta_1|}{2} + \sum_{i=-1}^{t}\varepsilon_i & t<0 \\ 0 & t=0 \\ t\frac{|\Delta_2|}{2} + \sum_{i=1}^{t}\varepsilon_i & t>0 \end{cases} \tag{252}$$

Then for any $\eta > 0$,

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2}(t)\right| \le m\right] \ge \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2+2\eta}(t)\right| \le m\right] \\ & -\left(A(\Delta_2)m^{3/2} + B(\Delta_2)\sqrt{m}\right)\eta\exp(-C(\Delta_2)m) \end{aligned} \tag{253}$$

for some expressions $A(\Delta_2)$, $B(\Delta_2)$ and $C(\Delta_2)$ that are, respectively, decreasing, decreasing, and increasing with respect to $|\Delta_2|$. Similarly, the following inequality holds:

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2}(t)\right| \le m\right] \ge \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1+2\eta,\Delta_2}(t)\right| \le m\right] \\ & -\left(A(\Delta_1)m^{3/2} + B(\Delta_1)\sqrt{m}\right)\eta\exp(-C(\Delta_1)m) \end{aligned} \tag{254}$$

where the form of the expressions $A()$, $B()$, and $C()$ has identical forms as expressions used in (253).

PROOF. It is only required to prove the inequality between W_{Δ_1,Δ_2} and $W_{\Delta_1,\Delta_2+2\eta}$. This is because

$W_{\Delta_1,\Delta_2}(t)$ has the same distribution as $W_{\Delta_2,\Delta_1}(-t)$ for all $t \in S$, and therefore

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2}(t)\right| \le m\right] \\ = \; & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_2,\Delta_1}(-t)\right| \le m\right] \\ = \; & \mathbb{P}\left[\left|\arg\min_{t\in -S} W_{\Delta_2,\Delta_1}(t)\right| \le m\right] \\ & \text{where } -S := \{-t : t \in S\} \end{aligned} \tag{255}$$

where the last inequality is due to the fact that the existence of an $|\ell| \le m$ such that $W_{\Delta_2,\Delta_1}(-\ell) < W_{\Delta_2,\Delta_1}(-t)$ for all $t \in S$, $|t| > m$ could be true if and only if there exists an $|\ell| \le m$ such that $W_{\Delta_2,\Delta_1}(\ell) < W_{\Delta_2,\Delta_1}(t)$ for all $t \in -S$, $|t| > m$. Similarly, we also have

$$\mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1+2\eta,\Delta_2}(t)\right| \le m\right] = \mathbb{P}\left[\left|\arg\min_{t\in -S} W_{\Delta_2,\Delta_1+2\eta}(t)\right| \le m\right], \tag{256}$$

and from here, an inequality can be derived by comparing

$$\mathbb{P}\left[\left|\arg\min_{t\in -S} W_{\Delta_2,\Delta_1}(t)\right| \le m\right]$$

and

$$\mathbb{P}\left[\left|\arg\min_{t\in -S} W_{\Delta_2,\Delta_1+2\eta}(t)\right| \le m\right]$$

using (253). Therefore, the rest of the proof will only concern the random walks $W_{\Delta_1,\Delta_2}(\cdot)$ and $W_{\Delta_1,\Delta_2+2\eta}(\cdot)$.

For the sake of brevity here, we will use the shorthand notations $W(t)$ for the random walk $W_{\Delta_1,\Delta_2}(t)$, and $W_+(t)$ for the random walk $W_{\Delta_1,\Delta_2+2\eta}(t)$. We are interested in the probability of the event when $\left|\arg\min_{t\in S} W(t)\right| > m$ and $\left|\arg\min_{t\in S} W_+(t)\right| \le m$, so for now, assume that for some integer $k \in S$, such that $|k| > m$, $W(k) < W(t)$ for all $t \in S$, $|t| \le m$.

- If $k < -m$, then $W_+(k) = W(k) < W(t) \le W(t) + t\eta 1(t > 0) = W_+(t)$ for all $t \in S - \{k\}$; in other words $\left|\arg\min_{t\in S} W_+(t)\right| > m$, a contradiction. Therefore it is not possible for $k < -m$.
- This leaves the possibility that $k > m$. Additionally:
  - as how 'k' was defined, $W(k) < \min_{|t|\le m} W(t)$
  - because $W_+(k) = W(k) + k\eta$ is not the minimum among the $W_+(t)$'s for $t \in S$, we have

$$\begin{aligned} W(k) + k\eta &= W_+(k) \\ &\ge \min_{|t|\le m} W_+(t) \\ &\ge \min_{|t|\le m} W(t) \end{aligned} \tag{257}$$

This breakdown of events shows that in order for the argmin of $W_+(t)$ to be within $[-m,m]$ but for the argmin of $W(t)$ to be outside this interval, there must be a $k>m$ where $\left(\min_{|t|\le m} W(t)\right)-\eta k < W(k) \le \min_{|t|\le m} W(t)$. Therefore

$$\begin{aligned}
&\mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| > m \text{ and } \left|\arg\min_{t\in S} W_+(t)\right| \le m\right] \\
\le\ &\mathbb{P}\left[\exists k:\ k>m \text{ and } \left(\min_{|t|\le m} W(t)\right) - \eta k < W(k) \le \min_{|t|\le m} W(t)\right] \\
\le\ &\sum_{k\in S\cap(m,\infty)} \mathbb{P}\left[\left(\min_{|t|\le m} W(t)\right) - \eta k < W(k) \le \min_{|t|\le m} W(t)\right]. \qquad (258)
\end{aligned}$$

The random variable $\min_{|t|\le m} W(t)$ can either equal $\min_{t\in[0,m]} W(t)$ or $\min_{t\in[-m,0]} W(t)$. Therefore, for any specific k, the event $\left(\min_{|t|\le m} W(t)\right) - \eta k < W(k) \le \min_{|t|\le m} W(t)$ implies that either $\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)$ or $\left(\min_{t\in[-m,0]} W(t)\right) - \eta k < W(k) \le \min_{t\in[-m,0]} W(t)$, yielding the inequality

$$\begin{aligned}
&\mathbb{P}\left[\left(\min_{|t|\le m} W(t)\right) - \eta k < W(k) \le \min_{|t|\le m} W(t)\right] \\
\le\ &\mathbb{P}\left[\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)\right] \\
&+\mathbb{P}\left[\left(\min_{t\in[-m,0]} W(t)\right) - \eta k < W(k) \le \min_{t\in[-m,0]} W(t)\right] \qquad (259)
\end{aligned}$$

Both of the two probabilities in the last part can be bounded. First, because $k>m$, $W(k)$ is independent of $W(-1),\ldots,W(-m)$, the distribution of $W(k)$ is still $N\left(k\frac{|\Delta_2|}{2}, k\right)$ even after conditioning on the value of $\min_{t\in[-m,0]} W(t)$, hence

$$\begin{aligned}
&\mathbb{P}\left[\left(\min_{t\in[-m,0]} W(t)\right) - \eta k < W(k) \le \min_{t\in[-m,0]} W(t)\right] \\
=\ &\mathbb{E}\left[\mathbb{P}\left[x-\eta k < W(k) \le x \middle| \min_{t\in[-m,0]} W(t) = x\right]\right] \\
=\ &\mathbb{E}\left[\mathbb{P}\left[\frac{x}{\sqrt{k}} - \sqrt{k}\left(\frac{|\Delta_2|}{2}+\eta\right) < N(0,1) \le \frac{x}{\sqrt{k}} - \sqrt{k}\frac{|\Delta_2|}{2} \middle| \min_{t\in[-m,0]} W(t) = x\right]\right] \\
&\text{where } x<0 \text{ since } \min_{t\in[-m,0]} W(t) \le W(0) = 0 \\
=\ &\mathbb{E}\left[\int_{\frac{x}{\sqrt{k}}-\sqrt{k}\left(\frac{|\Delta_2|}{2}+\eta\right)}^{\frac{x}{\sqrt{k}}-\sqrt{k}\frac{|\Delta_2|}{2}} \frac{\exp(-z^2/2)}{\sqrt{2\pi}}\,dz \middle| \min_{t\in[-m,0]} W(t) = x\right] \\
\le\ &\int_{-\sqrt{k}\left(\frac{|\Delta_2|}{2}+\eta\right)}^{-\sqrt{k}\frac{|\Delta_2|}{2}} \frac{\exp(-z^2/2)}{\sqrt{2\pi}}\,dz \\
\le\ &\frac{\eta\sqrt{k}}{\sqrt{2\pi}}\exp\left[-\frac{\Delta_2^2}{8}k\right] \qquad (260)
\end{aligned}$$

As for the other inequality, consider the event that $\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)$. Because $\min_{t\in[0,m]} W(t) \le W(0) = 0$, this event implies that for some $\ell \in [0,m]$, we have $W(\ell) - \eta k < W(k) < W(\ell) \le 0$ (namely, letting $\ell = \arg\min_{t\in[0,m]} W(t)$ would work). Therefore

$$\begin{aligned}
&\mathbb{P}\left[\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)\right] \\
\le\ & \sum_{\ell=0}^{m} \mathbb{P}\left[W(\ell) - \eta k < W(k) \le W(\ell) \le 0\right] \\
=\ & \sum_{\ell=0}^{m} \mathbb{P}\left[-\eta k < W(k) - W(\ell) \le 0 \text{ and } W(\ell) \le 0\right] \qquad (261)
\end{aligned}$$

Now $W(\ell) = \ell\frac{|\Delta_2|}{2} + \sum_{j=1}^{\ell}\varepsilon_j$ and $W(k) - W(\ell) = (k-\ell)\frac{|\Delta_2|}{2} + \sum_{j=\ell+1}^{k}\varepsilon_j$ are independent random variables, with distributions $N\left(\ell\frac{|\Delta_2|}{2}, \ell\right)$ and $N\left((k-\ell)\frac{|\Delta_2|}{2}, (k-\ell)\right)$. Hence

$$\begin{aligned}
&\mathbb{P}\left[-\eta k < W(k) - W(\ell) \le 0 \text{ and } W(\ell) \le 0\right] \\
=\ & \mathbb{P}\left[-\eta k < W(k) - W(\ell) \le 0\right] \cdot \mathbb{P}\left[W(\ell) \le 0\right] \\
=\ & \mathbb{P}\left[-\sqrt{k-\ell}\frac{|\Delta_2|}{2} - \frac{\eta k}{\sqrt{k-\ell}} < N(0,1) \le -\sqrt{k-\ell}\frac{|\Delta_2|}{2}\right] \cdot \mathbb{P}\left[N(0,1) \le -\sqrt{\ell}\frac{|\Delta_2|}{2}\right] \\
=\ & \int_{-\sqrt{k-\ell}\frac{|\Delta_2|}{2} - \frac{\eta k}{\sqrt{k-\ell}}}^{-\sqrt{k-\ell}\frac{|\Delta_2|}{2}} \frac{\exp(-z^2/2)}{\sqrt{2\pi}}\, dz \cdot \mathbb{P}\left[N(0,1) \le -\sqrt{\ell}\frac{|\Delta_2|}{2}\right] \\
\le\ & \frac{\eta k}{\sqrt{k-\ell}} \frac{\exp\left(-\frac{\Delta_2^2}{8}(k-\ell)\right)}{\sqrt{2\pi}} \cdot \frac{1}{2}\exp\left[-\frac{\Delta_2^2}{8}\ell\right] \\
=\ & \frac{1}{2\sqrt{2\pi}} \cdot \frac{\eta k}{\sqrt{k-\ell}} \exp\left[-\frac{\Delta_2^2}{8}k\right] \qquad (262)
\end{aligned}$$

Therefore, using (261),

$$\begin{aligned}
&\mathbb{P}\left[\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)\right] \\
\le\ & \sum_{\ell=0}^{m} \frac{1}{2\sqrt{2\pi}} \cdot \frac{\eta k}{\sqrt{k-\ell}} \exp\left[-\frac{\Delta_2^2}{8}k\right] \\
\le\ & \frac{\exp\left[-\frac{\Delta_2^2}{8}k\right]}{2\sqrt{2\pi}} \cdot \eta k \cdot \int_0^{m+1} \frac{1}{\sqrt{k-x}}\, dx \\
=\ & \frac{\exp\left[-\frac{\Delta_2^2}{8}k\right]}{\sqrt{2\pi}} \cdot \eta k \cdot (\sqrt{k} - \sqrt{k-m-1}) \\
\le\ & \frac{\exp\left[-\frac{\Delta_2^2}{8}k\right]}{\sqrt{2\pi}} \cdot \eta k \cdot \sqrt{m+1} \qquad (263)
\end{aligned}$$

Therefore,

$$
\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| > m \text{ and } \left|\arg\min_{t\in S} W_+(t)\right| \le m\right] \\
\le & \sum_{k\in S\cap(m,\infty)} \mathbb{P}\left[\left(\min_{t\in[0,m]} W(t)\right) - \eta k < W(k) \le \min_{t\in[0,m]} W(t)\right] \\
& + \sum_{k\in S\cap(m,\infty)} \mathbb{P}\left[\left(\min_{t\in[-m,0]} W(t)\right) - \eta k < W(k) \le \min_{t\in[-m,0]} W(t)\right] \\
& \text{according to (258) and (259)} \\
\le & \sum_{k=m+1}^{\infty} \left(\frac{\exp\left[-\frac{\Delta_2^2}{8}k\right]}{\sqrt{2\pi}} \cdot \eta k \cdot \sqrt{m+1}\right) + \sum_{k=m+1}^{\infty} \left(\frac{\eta\sqrt{k}}{\sqrt{2\pi}} \exp\left[-\frac{\Delta_2^2}{8}k\right]\right) \\
& \text{according to (260) and (263)} \\
\le & \frac{\eta(\sqrt{m+1}+1)}{\sqrt{2\pi}} \sum_{m+1}^{\infty} k \exp\left[-\frac{\Delta_2^2}{8}k\right] \\
\le & \frac{3\sqrt{m}}{\sqrt{2\pi}} \left(\frac{m\exp\left[-\frac{\Delta_2^2}{8}\right]}{1-\exp\left[-\frac{\Delta_2^2}{8}\right]} + \frac{\exp\left[-\frac{\Delta_2^2}{8}\right]}{\left(1-\exp\left[-\frac{\Delta_2^2}{8}\right]\right)^2}\right)\left(\eta\exp\left[-\frac{\Delta_2^2}{8}m\right]\right) \\
& \text{since } \sum_{k=a}^{\infty} kx^k = \left(\frac{a-1}{1-x} + \frac{1}{(1-x)^2}\right)x^a \text{ for any } |x|<1, a\in\mathbb{N} \qquad (264)
\end{aligned}
$$

From here,

$$
\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| \le m\right] \\
\ge & \mathbb{P}\left[\left|\arg\min_{t\in S} W_+(t)\right| \le m\right] - \mathbb{P}\left[\left|\arg\min_{t\in S} W_+(t)\right| \le m \text{ and } \left|\arg\min_{t\in S} W(t)\right| > m\right] \\
\ge & \mathbb{P}\left[\left|\arg\min_{t\in S} W_+(t)\right| \le m\right] - \left(A(\Delta_2)m^{3/2} + B(\Delta_2)\sqrt{m}\right)\eta\exp(-C(\Delta_2)m) \qquad (265)
\end{aligned}
$$

for some expressions $A(\Delta_2)$, $B(\Delta_2)$ and $C(\Delta_2)$ that are, respectively, decreasing, decreasing, and increasing with respect to $|\Delta_2|$. □

This result immediately leads to some results concerning the random walks $X_\Delta(\cdot)$, as they are a special case of the random walks $W_{\Delta_1,\Delta_2}(\cdot)$ where $\Delta_1 = \Delta_2$.

LEMMA 13. *Suppose that for the random walk $Y_+(t)$ for $t \in S$ equals*

$$
Y_+(t) = \begin{cases} X_\Delta(t) & \text{for } t \le 0 \\ X_\Delta(t) + \eta t & \text{for } t > 0 \end{cases} \qquad (266)
$$

for some constant η *such that* $0 < \eta < \frac{|\Delta|}{2}$. *Then for any* $[-m, m] \subset S \subset \mathbb{Z}$,

$$\mathbb{P}\left[\left|\arg\min_{t\in S} Y_+(t)\right| \le m\right] \le \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \le m\right] + \eta\left[A(\Delta)m^{3/2} + B(\Delta)\sqrt{m}\right]\exp\left[-C(\Delta)m\right] \tag{267}$$

for some expressions $A()$, $B()$, *and* $C()$ *which are, respectively, decreasing, decreasing, and increasing with respect to* $|\Delta|$. *The same probability inequality will hold if* $Y_+(t) = X_\Delta(t) + \eta|t|1(t < 0)$.

A similar set of inequalities hold for the random walk $Y_-(t)$ *for* $t \in S$, *defined as*

$$Y_-(t) = \begin{cases} X_\Delta(t) & \text{for } t \le 0 \\ X_\Delta(t) - \eta t & \text{for } t > 0 \end{cases} \tag{268}$$

For $\eta < \frac{|\Delta|}{2}$, *and* $[-m, m] \subset S$, *we have*

$$\mathbb{P}\left[\left|\arg\min_{t\in S} Y_-(t)\right| \le m\right] \ge \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \le m\right] - \eta\left[A\left(\frac{|\Delta|}{2} - \eta\right)m^{3/2} + B\left(\frac{|\Delta|}{2} - \eta\right)\sqrt{m}\right]\exp\left[-C\left(\frac{|\Delta|}{2} - \eta\right)m\right] \tag{269}$$

for some expressions $A()$, $B()$, *and* $C()$ *which are, respectively, decreasing, decreasing, and increasing with respect to* $|\Delta|/2 - \eta$. *The same probability inequality will hold if* $Y_-(t) = X_\Delta(t) - \eta|t|1(t < 0)$.

Proof. Apply Lemma 12 with $\Delta_1 = \Delta_2 = \Delta$ to prove (267), and with $\Delta_1 = \Delta$, $\Delta_2 = \Delta - 2\eta$ to prove (269). □

Additionally, we can make probabilistic statements regarding the argmin of $X_\Delta(t)$ for two different values of Δ:

Lemma 14. *For any* $\Delta \neq 0$, $\eta > 0$, *and a set* S *which contains the interval* $[-m, m]$,

$$\mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \le m\right] \le \mathbb{P}\left[\left|\arg\min_{t\in S} X_{|\Delta|+2\eta}(t)\right| \le m\right] \tag{270}$$

and

$$\mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \ge m\right] \ge \mathbb{P}\left[\left|\arg\min_{t\in S} X_{|\Delta|+2\eta}(t)\right| \le m\right] - 2\eta\left[A(\Delta)m^{3/2} + B(\Delta)\sqrt{m}\right]\exp\left[-C(\Delta)m\right] \tag{271}$$

for some expressions $A()$, $B()$, *and* $C()$ *which can take the same form and have the same monotonicity properties as the ones used in Lemma 12.*

PROOF. The first inequality can be shown by noticing that the event $\left|\arg\min_{t\in S} X_\Delta(t)\right| \leq m$ implies that for some $|\ell| \leq m$, $X_\Delta(\ell) \leq X_\Delta(t)$ for all $t \in S$, $|t| > m$. This in turn implies that $X_{|\Delta|+\eta}(\ell) = X_\Delta(\ell) + |\ell|\eta < X_\Delta(t) + |t|\eta = X_{|\Delta|+\eta}(t)$ for all $t \in S$, $|t| > m$, which means $\left|\arg\min_{t\in S} X_{|\Delta|+\eta}(t)\right| \leq m$.

The second inequality can be shown by applying Lemma 12 twice:

$$\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \leq m\right] \\
\geq\ & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{|\Delta|,|\Delta|+2\eta}(t)\right| \leq m\right] - \eta\left[A(\Delta)m^{3/2} + B(\Delta)\sqrt{m}\right]\exp\left[-C(\Delta)m\right] \\
\geq\ & \mathbb{P}\left[\left|\arg\min_{t\in S} W_{|\Delta|+2\eta,|\Delta|+2\eta}(t)\right| \leq m\right] - 2\eta\left[A(\Delta)m^{3/2} + B(\Delta)\sqrt{m}\right]\exp\left[-C(\Delta)m\right] \\
=\ & \mathbb{P}\left[\left|\arg\min_{t\in S} X_{|\Delta|+\eta}(t)\right| \leq m\right] - 2\eta\left[A(\Delta)m^{3/2} + B(\Delta)\sqrt{m}\right]\exp\left[-C(\Delta)m\right] \qquad (272)
\end{aligned}$$

□

10.4. *Comparison between Random Walks, Part 2.* Here we will prove inequalities similar to those presented in Lemma 12, but in the other direction. These results are also useful in proving Theorem 5.

LEMMA 15. *Let the random walks W_{Δ_1,Δ_2} be as they were defined in Lemma 12. Then given any positive η, positive integer m, and set S such that $|\eta| < |\Delta_1|/2$ and $[-m, m] \subsetneq S$,*

$$\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2+2\eta}(t)\right| \leq m\right] \geq \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2}(t)\right| \leq m\right] \\
& -A\left(\frac{\Delta_1}{2} - \eta\right)\eta\sqrt{m}\exp\left(-B\left(\frac{\Delta_1}{2} - \eta\right)m\right) \qquad (273)
\end{aligned}$$

for some positive expressions $A()$ and $B()$ which are, respectively, decreasing and increasing in $\frac{\Delta_1}{2} - \eta$. Similarly, between the random walks W_{Δ_1,Δ_2} and $W_{\Delta_1+2\eta,\Delta_2}$ for $0 < \eta < \frac{\Delta_2}{2}$ there is the inequality

$$\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1+2\eta,\Delta_2}(t)\right| \leq m\right] \geq \mathbb{P}\left[\left|\arg\min_{t\in S} W_{\Delta_1,\Delta_2}(t)\right| \leq m\right] \\
& -A\left(\frac{\Delta_2}{2} - \eta\right)\eta\sqrt{m}\exp\left(-B\left(\frac{\Delta_2}{2} - \eta\right)m\right) \qquad (274)
\end{aligned}$$

for some positive expressions $A()$ and $B()$ which are, respectively, decreasing and increasing in $\frac{\Delta_2}{2} - \eta$.

PROOF. We will show the inequality between $W_{\Delta_1,\Delta_2+2\eta}$ and W_{Δ_1,Δ_2}, and the result between $W_{\Delta_1+2\eta,\Delta_2}$ and W_{Δ_1,Δ_2} can be shown in a similar fashion or in an argument similar to what was found in the proof for Lemma 12. As in the proof of that lemma, we will use the shorthand notation W for W_{Δ_1,Δ_2} and W_+ for $W_{\Delta_1,\Delta_2+2\eta}$.

We are interested in how $\left|\arg\min_{t\in S} W(t)\right| \leq m$ and $\left|\arg\min_{t\in S} W_+(t)\right| > m$ can simultaneously occur, which we will do by considering the possible values of the argmin of $W(t)$. If these two events are true, then first note that for some integer $k \in [-m, m]$, $W(k) \leq W(t)$ for all $t \in S$, $t \neq k$.

- if $k \leq 0$, then $W_+(k) = W(k) \leq W(t) + |t|\eta 1(t > 0) = W_+(t)$ for all $t \in S$ and $|t| > m$; in other words $\left|\arg\min_{t\in S} W_+(t)\right| \leq m$,
- thus $k > 0$ and $W(k) = \arg\min_{t\in[0,m]} W(t)$. The only possible way for $\left|\arg\min_{t\in S} W_+(t)\right| > m$ is for the argmin to be less than $-m$: for any $t \in S$:
  - if $t > m$ then $W_+(t) = W(t) + t\eta > W(k) + k\eta = W_+(k)$, which means that the argmin of $W_+(t)$ cannot be greater than m in absolute value
  - therefore the only possible argmin for $W_+(t)$ is for some $\ell < -m$, and it must satisfy

    $$W(\ell) = W_+(\ell) < W_+(k) = W(k) + \eta k \leq W(k) + \eta m \tag{275}$$

    but at the same time since $W(\ell)$ was not the minimum among the $W(t)$'s, we have $W(\ell) \geq W(k)$

This breakdown of events shows that in order for the argmin of $W(t)$ to be within $[-m, m]$ but for the argmin of $W_+(t)$ to be outside this interval, there must be an $\ell < -m$ where $\min_{t\in[0,m]} W(t) < W(\ell) \leq \min_{t\in[0,m]} W(t) + \eta m$. Hence:

$$\begin{aligned}
& \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| \leq m \text{ and } \left|\arg\min_{t\in S} W_+(t)\right| > m\right] \\
\leq\ & \mathbb{P}\left[\exists \ell < -m \text{ where } \min_{t\in[0,m]} W(t) < W(\ell) \leq \min_{t\in[0,m]} W(t) + \eta m\right] \\
\leq\ & \sum_{\ell=-m-1}^{-\infty} \mathbb{P}\left[\min_{t\in[0,m]} W(t) < W(\ell) \leq \min_{t\in[0,m]} W(t) + \eta m\right] \\
=\ & \sum_{\ell=-m-1}^{-\infty} \mathbb{E}\left[\mathbb{P}\left[x < W(\ell) \leq x + \eta m \,\middle|\, \min_{t\in[0,m]} W(t) = x\right]\right]
\end{aligned}$$

$$\leq \sum_{\ell=-m-1}^{-\infty} \mathbb{E}\left[\mathbb{P}\left[\frac{x}{\sqrt{|\ell|}} - \frac{\Delta_1}{2}\sqrt{|\ell|} < N(0,1) \leq \frac{x}{\sqrt{|\ell|}} - \frac{\Delta_1}{2}\sqrt{|\ell|} + \frac{\eta m}{\sqrt{|\ell|}} \middle| \min_{t\in[0,m]} W(t) = x\right]\right]$$

$$= \sum_{\ell=-m-1}^{-\infty} \mathbb{E}\left[\int_{\frac{x}{\sqrt{|\ell|}} - \frac{\Delta_1}{2}\sqrt{|\ell|}}^{\frac{x}{\sqrt{|\ell|}} - \frac{\Delta_1}{2}\sqrt{|\ell|} + \frac{\eta m}{\sqrt{|\ell|}}} \frac{1}{\sqrt{2\pi}} \exp(-z^2/2)\, dz \middle| \min_{t\in[0,m]} W(t) = x\right]$$

$$\leq \sum_{\ell=-m-1}^{-\infty} \int_{-\frac{\Delta_1}{2}\sqrt{|\ell|}}^{-\frac{\Delta_1}{2}\sqrt{|\ell|} + \frac{\eta m}{\sqrt{|\ell|}}} \frac{1}{\sqrt{2\pi}} \exp(-z^2/2)\, dz$$

since all possible values of x are negative, and the integrated density is monotone

$$\leq \sum_{\ell=-m-1}^{-\infty} \int_{-\frac{\Delta_1}{2}\sqrt{|\ell|}}^{-\frac{\Delta_1}{2}\sqrt{|\ell|} + \eta\sqrt{m}} \frac{1}{\sqrt{2\pi}} \exp(-z^2/2)\, dz$$

$$\leq \sum_{\ell=-m-1}^{-\infty} \frac{\eta}{\sqrt{2\pi}}\sqrt{m} \exp\left[-\frac{1}{2}\left(\frac{\Delta_1}{2}\sqrt{|\ell|} - \eta\sqrt{m}\right)^2\right]$$

$$\leq \frac{\eta\sqrt{m}}{\sqrt{2\pi}} \sum_{\ell=-m-1}^{-\infty} \exp\left[-\frac{1}{2}\left(\frac{\Delta_1}{2} - \eta\right)^2 |\ell|\right]$$

$$\leq \frac{1}{\sqrt{2\pi}} \left(\frac{\exp\left[-\frac{1}{2}\left(\frac{\Delta_1}{2} - \eta\right)^2\right]}{1 - \exp\left[-\frac{1}{2}\left(\frac{\Delta_1}{2} - \eta\right)^2\right]}\right) \eta\sqrt{m} \exp\left(-\frac{1}{2}\left(\frac{\Delta_1}{2} - \eta\right)^2 m\right).$$

(276)

Therefore,

$$\begin{aligned} & \mathbb{P}\left[\left|\arg\min_{t\in S} W_+(t)\right| \leq m\right] \\ \geq\ & \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| \leq m\right] - \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| \leq m \text{ and } \left|\arg\min_{t\in S} Y_+(t)\right| > m\right] \\ \geq\ & \mathbb{P}\left[\left|\arg\min_{t\in S} W(t)\right| \leq m\right] - A'\eta\sqrt{m}\exp(-B'm) \end{aligned} \tag{277}$$

for some constants A' and B' depending only on $\frac{\Delta_1}{2} - \eta$.

□

We can immediately apply this result to random walks of the form X_Δ:

LEMMA 16. *Suppose that for the random walk $Y_+(t)$ for $t \in \mathbb{Z}$ equals*

$$Y_+(t) = \begin{cases} X_\Delta(t) & \text{for } t \leq 0 \\ X_\Delta(t) + \eta t & \text{for } t > 0 \end{cases} \tag{278}$$

for some constant η *such that* $0 < \eta < \frac{|\Delta|}{2}$. *Then for any* $[-m, m] \subset S \subset \mathbb{Z}$,

$$\mathbb{P}\left[\left|\arg\min_{t\in S} Y_+(t)\right| \le m\right] \ge \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \le m\right] - A'\left(\frac{|\Delta|}{2} - \eta\right)\eta\sqrt{m}\exp\left(-B'\left(\frac{|\Delta|}{2} - \eta\right)m\right) \tag{279}$$

for some expressions $A'()$ *and* $B'()$ *which are, respectively, decreasing and increasing with respect to* $\left(\frac{|\Delta|}{2} - \eta\right)$. *The same probability inequality will hold if* $Y_+(t) = X_\Delta(t) + \eta|t|1(t < 0)$.

In addition, if $Y_-(t)$ *is defined differently as*

$$Y_-(t) = \begin{cases} X_\Delta(t) & \text{for } t \le 0 \\ X_\Delta(t) - \eta t & \text{for } t > 0 \end{cases} \tag{280}$$

then we have the inequality

$$\mathbb{P}\left[\left|\arg\min_{t\in S} Y_-(t)\right| \le m\right] \le \mathbb{P}\left[\left|\arg\min_{t\in S} X_\Delta(t)\right| \le m\right] + A'\left(\frac{|\Delta|}{2} - \eta\right)\eta\sqrt{m}\exp\left(-B'\left(\frac{|\Delta|}{2} - \eta\right)m\right) \tag{281}$$

for some expressions $A'()$ *and* $B'()$ *which are, respectively, decreasing and increasing with respect to* $\left(\frac{|\Delta|}{2} - \eta\right)$.

11. Supplement Part D (Intelligent Sampling using Wild Binary Segmentation).

11.1. *Wild Binary Segmentation.* We next discuss the Wild Binary Segmentation (WBinSeg) algorithm, introduced in [1]. Similar to our treatment of the BinSeg procedure, we will explain the WBinSeg procedure in the context of applying it to the dataset $Z_1, \ldots, Z_{N^*}$, a size $\sim N^\gamma$ size subsample of a larger dataset $Y_1, \ldots, Y_N$ which satisfies the conditions (M1)-(M4). The steps of this algorithm are:

1. Fix a threshold value ζ_{N^*} and initialize the segment set $SS = \{(1, N)\}$, the change point estimate set $\hat{\underline{\tau}} = \emptyset$, and M_{N^*} intervals $[s_1, e_1], \ldots, [s_{M_{N^*}}, e_{M_{N^*}}]$, where each s_j and e_j are uniformly picked from $\{1, \ldots, N^*\}$.
2. Pick any ordered pair $(s, e) \in SS$, remove it from SS (update SS by $SS \leftarrow SS - \{(s, e)\}$). If $s \ge e$ then skip to step 6, otherwise continue to step 3.
3. Define $\mathcal{M}_{s,e} := \{[s_i, e_i] : [s_i, e_i] \subseteq [s, e]\}$.
   - As an optional step, also take $\mathcal{M}_{s,e} \leftarrow \mathcal{M}_{s,e} \cup \{(s, e)\}$.
4. Find a $[s^*, e^*] \in \mathcal{M}_{s,e}$ such that

$$\max_{b\in\{s^*,\ldots,e^*-1\}} |\bar{Y}^b_{s^*,e^*}| = \max_{[s',e']\in\mathcal{M}_{s,e}} \left(\max_{b\in\{s',\ldots,e'-1\}} |\bar{Y}^b_{s',e'}|\right)$$

and let $b_0 = \underset{b\in\{s^*,\ldots,e^*-1\}}{\arg\max} |\bar{Z}^b_{s^*,e^*}|$.

5. If $|\bar{Z}^{b_0}_{s^*,e^*}| \geq \zeta_{N^*}$, then add b_0 to the list of change point estimates (add b_0 to $\hat{\underline{\tau}}$), and add ordered pairs (s, b_0) and $(b_0 + 1, e)$ to SS, otherwise skip to step 5.
6. Repeat steps 2-4 until SS contains no elements.

Roughly speaking, WBinSeg performs very much like binary segmentation but with steps that maximize change point estimates over M_{N^*} randomly chosen intervals. The consistency results in [1] imply that in our setting, the following holds:

THEOREM 9. *Suppose conditions (M1) to (M4) are satisfied and the tuning parameter ζ_{N^*} is chosen appropriately such that there exists positive constants C_1 and C_2 with $C_1\sqrt{\log(N^*)} \leq \zeta_{N^*} \leq C_2\sqrt{\delta_{N^*}}$. Denote $\hat{J}$, $\hat{\tau}_1, \ldots, \hat{\tau}_{\hat{J}}$ as the estimates obtained from wild binary segmentation. Then, there exists positive constants C_3, C_4 where*

$$\mathbb{P}\Big[\hat{J} = J; \quad \max_{i=1,\ldots,J} |\hat{\tau}_i - \tau_i| \leq C_3 \log(N^*)\Big] \geq 1 - C_4(N^*)^{-1} - \left(\frac{N^*}{\delta^*_{N^*}}\right)\left(1 - \left(\frac{\delta^*_{N^*}}{3N^*}\right)^2\right)^{M_{N^*}} \tag{282}$$

We remark that the right side of (282) does not necessarily converge to 1 unless $M_{N^*} \to \infty$ fast enough. Using some simple algebra, it was shown in the original paper, that for sufficiently large N, this expression can be bounded from below by $1 - CN^{-1}$ for some $C > 0$ if $M_{N^*} \geq \left(\frac{3N^*}{\delta_{N^*}}\right)^2 \log(N^{*2}/\delta_{N^*})$, a condition on M_{N^*} which we we assume from here on in order to simplify some later analysis.

Compared with the consistency result for binary segmentation given in Lemma 1, $\max_{j=1,\ldots,J} |\hat{\tau}_j - \tau_j|$ can be bounded by some constant times $\log(N^*)$, which can grow much slower than $E_{N^*} = (N^*/\delta^*_{N^*})^2 \log(N^*)$ whenever $N^*/\delta^*_{N^*} \to \infty$. However, this comes at the cost of computational time. Suppose we perform wild binary segmentation with M_{N^*} random intervals $[s_1, e_1], \ldots, [s_{M_{N^*}}, e_{M_{N^*}}]$, then for large N^* the most time consuming part of the operation is to maximize the CUSUM statistic over each interval, with the other tasks in the WBinSeg procedure taking much less time. This takes an order of $\sum_{j=1}^{M_{N^*}}(e_j - s_j)$ time, and since the interval endpoints are drawn from $\{1, \ldots, N^*\}$ with equal probability, we have $\mathbb{E}[e_j - s_j] = \frac{N^*}{3}(1 + o(1))$ for all $j = 1, \ldots, M_{N^*}$. Hence the scaling of the average computational time for maximizing the CUSUM statistic in all random intervals, and WBinSeg as a whole, is $O(N^* M_{N^*}) = O\left(\frac{N^{*3}}{(\delta^*_{N^*})^2} \log((N^*)^2/\delta^*_{N^*})\right)$ time, which is greater than $O(N^* \log(N^*))$ time for binary segmentation whenever $N^*/\delta^*_{N^*} \to \infty$. The trade-off between the increased accuracy of the estimates and the bigger computational time, as they pertain to intelligent sampling, will be analyzed later.

As with the BinSeg method, we need to verify that the WBinSeg method also satisfies the consistency condition of (13) at step (ISM3) of intelligent sampling, so that the results of Theorem 5 continue to hold with the latter as the first stage procedure. To this end, we need to demonstrate

a set of signal estimators that satisfy the condition of Lemma 6 with a $\rho^*_{N^*}$ such that $J\rho^*_{N^*} \to 0$. We do this by using the estimator proposed in (31), and also by imposing two further conditions:

(M8 (WBinSeg)): Ξ (from condition (M3)) is further restricted by $\Xi \in [0, 1/3)$,
(M9 (WBinSeg)): N_1, from step (ISM2), is chosen so that $N_1 = K_1 N^\gamma$ for some $K_1 > 0$ and $\gamma > 3\Xi$.

LEMMA 17. *Under conditions (M1) through (M4), (M8 (WBinSeg)), and (M9 (WBinSeg)), we have*

$$\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\dots,J} |\hat{\tau}_i - \tau_i| \le w^*(N^*); \quad \max_{i=0,\dots,J} |\hat{\nu}_i - \nu_i| \le \rho^*_{N^*}\right] \to 1 \tag{283}$$

for some $\rho^*_{N^*}$ where $J\rho^*_{N^*} \to 0$.

PROOF. We need to show that wild binary segmentation does satisfy the requirements of Lemma 6. We have $\delta^*_{N^*} \gtrsim (N^*)^{1-\frac{\Xi}{\gamma}}$ and $J \lesssim (N^*)^{\frac{\Xi}{\gamma}}$. By the properties of the WBinSeg estimator shown in Theorem 9, $w^*(N^*) \sim \log(N^*)$ and $B_{N^*} \sim N^{*-1}$. We shall show that for $\rho = (N^*)^\theta$ where $\theta \in \left(\frac{\Xi}{2\gamma} - \frac{1}{2}, -\frac{\Xi}{\gamma}\right)$, the conditions of Lemma 6 are satisfied, and in addition, $J\rho^*_{N^*} \to 0$.

First, the set $\left(\frac{\Xi}{2\gamma} - \frac{1}{2}, -\frac{\Xi}{\gamma}\right)$ is a valid set since

$$\frac{3\Xi}{2\gamma} - \frac{1}{2} = \frac{1}{2}\left(3\frac{\Xi}{\gamma} - 1\right) < 0. \tag{284}$$

by condition (M9 (WBinSeg)). Hence $\frac{\Xi}{2\gamma} - \frac{1}{2} < -\frac{\Xi}{\gamma}$. Second, $J\rho^*_{N^*} \to 0$ since

$$J\rho^*_{N^*} \lesssim (N^*)^{\frac{\Xi}{\gamma}}(N^*)^\theta \tag{285}$$

But

$$\theta + \frac{\Xi}{\gamma} < 0.$$

Finally, as for the conditions of Lemma 6, we have

- $\frac{w^*(N^*)}{\delta^*_{N^*}} \lesssim (N^*)^{-(1-\frac{\Xi}{\gamma})} \log(N^*) \to 0$, hence $w^*(N^*) = o(\delta^*_{N^*})$
- $\frac{N^*}{\delta^*_{N^*}} B_{N^*} \sim \frac{1}{\delta^*_{N^*}} \to 0$
- because $\rho^*_{N^*} = (N^*)^\theta$ where $\theta \in \left(\frac{\Xi}{2\gamma} - \frac{1}{2}, -\frac{\Xi}{\gamma}\right)$, we have $\rho^*_{N^*} = o(1)$
- $\frac{N^* w^*(N^*)}{(\delta^*_{N^*})^{3/2}\rho^*_{N^*}} \lesssim (N^*)^{1-\theta} \log(N^*)$, and $\delta^*_{N^*}(\rho^*_{N^*})^2 \gtrsim (N^*)^{2\theta+1-\frac{\Xi}{\gamma}}$; since $2\theta + 1 - \frac{\Xi}{\gamma} > 0$, this means that $\frac{N^* w^*(N^*)}{(\delta^*_{N^*})^{3/2}\rho^*_{N^*}} = o(\exp(C_2 \delta^*_{N^*}(\rho^*_{N^*})^2))$ for any positive constant C_2

Since all conditions of Lemma 6 are satisfied, the signal estimators satisfy

$$\mathbb{P}\left[\hat{J} = J; \max_{j=0,\dots,J} |\hat{\nu}^*_j - \nu_j| \le \rho^*_{N^*}\right] \to 1 \tag{286}$$

which combines with the consistency of the change point estimators through a Bonferroni inequality to obtain the consistency result (13). □

TABLE 11
Table of γ_{min} and computational times for various values of Ξ, using WBinSeg at stage 1.

Ξ	$[0, 1/9)$	$[1/9, 1/7)$	$[1/7, 1/3)$
γ_{min}	$\frac{1-2\Xi+\Lambda}{2}$	$\max\left\{\frac{1-2\Xi+\Lambda}{2}, 3\Xi+\eta\right\}$	$\Xi+\eta$
Order of Time	$N^{(1+2\Xi+\Lambda)/2}\log(N)$	$N^{(1+2\Xi+\Lambda)/2}\log(N)$ or $N^{5\Xi+\eta}\log(N)$	$N^{5\Xi+\eta}\log(N)$
γ_{min} $(\Lambda=0)$	$\frac{1-2\Xi}{2}$	$3\Xi+\eta$	$3\Xi+\eta$
Time $(\Lambda=0)$	$N^{(1+2\Xi)/2}\log(N)$	$N^{5\Xi+\eta}\log(N)$	$N^{5\Xi+\eta}\log(N)$
γ_{min} $(\Lambda=\Xi)$	$\frac{1-\Xi}{2}$	$\frac{1-\Xi}{2}$	$3\Xi+\eta$
Time $(\Lambda=\Xi)$	$N^{(1+3\Xi)/2}\log(N)$	$N^{(1+3\Xi)/2}\log(N)$	$N^{5\Xi+\eta}\log(N)$

REMARK 16. *As with the BinSeg algorithm, the WBinSeg procedure is asymptotically consistent but faces the same issues as BinSeg in a practical setting where the goal is to obtain confidence bounds $[\hat{\tau}_i \pm C_3\log(N^*)]$ for the change point τ_i. Namely, there are unspecified constants associated with the tuning parameter ζ_{N^*} and the confidence interval width $C_3\log(N^*)$ in (282). The issue of choosing a confidence interval width will can be resolved by applying the procedure of Section 9.1.*

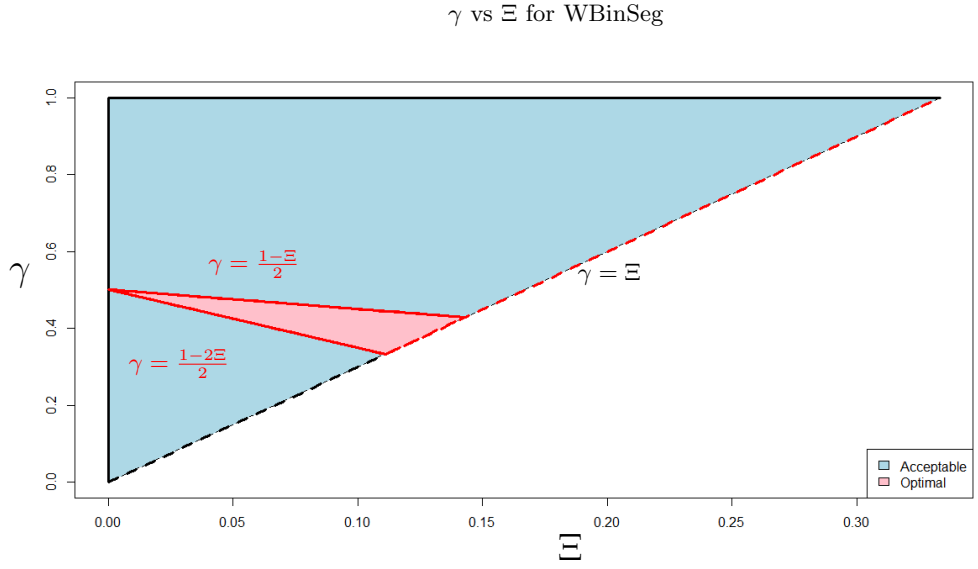


Fig 28: Blue triangle encompasses all valid values of γ vs Ξ as set by (M8 (WBinSeg)). Pink region, solid red lines, and dotted red lines denotes γ_{min} for each Ξ.

REMARK 17. *Although the γ_{min} values are, across the board, smaller than those given in Table 9 of the main paper (which records the γ_{min} values and computational time of Binseg at Stage 1), the order of the actual computational time is also greater across the board. There is some advantage in using WBinSeg since consistency condition (13) with $J\rho_N \to 0$ is satisfied in a greater regime of Ξ*

($\Xi < 1/3$ as opposed to $\Xi < 1/4$ for BinSeg), but in scenarios where the change points are placed far apart BinSeg would run in shorter time.

11.2. *Computational Time Order for Multiple Change Points.* To analyze the computational time when using WBinSeg at stage 1, we again assume that $\delta_N/N^{\Xi} \to K_1$ and $J(N)/N^{\Lambda} \to K_2$ for some constant $\Lambda \leq \Xi$ and positive constants K_1, K_2. To summarize the details, for $N_1 \sim N^{\gamma}$, the average time for the first stage is $O(N^{\gamma+2\Xi}\log(N))$, while the second stage takes, on average, $O(N^{1-\gamma+\Lambda}\log(N))$ time. Together with condition (M9 (WBinSeg)) and setting, $\gamma > 3\Xi$, the order of average time for both stages combined is minimized by setting $\gamma_{min} = \max\left\{\frac{1-2\Xi+\Lambda}{2}, \Xi+\eta\right\}$ for any small constant η, with the average total computational time being $O(N^{\gamma_{min}+2\Xi}\log(N))$.

Detailed Analysis: We have $\delta_N \sim N^{1-\Xi}$ for some $\Xi \in [0, 1/3)$ and $J \sim N^{\Lambda}$ for some $\Lambda \in [0, \Xi]$. Given n data points from (11) with minimum separation δ_n between change points, it takes an order of $\frac{n^3}{\delta_n^2}\log(n)$ time to perform the procedure, due to having to use $M_n \sim \left(\frac{n}{\delta_n}\right)^2 \log(n^2/\delta_n)$ random intervals, each requiring $O(n)$ time on average. The first stage works with a time series data of length order N^{γ} with minimal separation $\delta^*_{N^*}$, and hence has an average computational time that is of the same order as$(N^*/\delta^*_{N^*})^2 \cdot N^*\log(N^*)$, which is the same order as $N^{\gamma+2\Xi}\log(N)$. The second stage works with $\hat{J}$ intervals, each of width $CN^{1-\gamma}\log(N)$ for some constant C. Because we have

$$\mathbb{P}[\hat{J} = J] \geq 1 - C(N^*)^{-1} \qquad \text{for some } C > 0 \tag{287}$$

by our earlier condition on M_N, we arrive at $\mathbb{E}[\hat{J}] = O(J)$ because

$$\mathbb{E}[\hat{J}] \leq J(1 - C(N^*)^{-1}) + N(C(N^*)^{-1}) \leq (C+1)J.$$

This in turn shows the expected computational time of the second stage is $O(JN^{1-\gamma}\log(N))$ which simplifies to $O(N^{1-\gamma+\Lambda}\log(N))$.

Both stages combined are expected to take $O\left(N^{(\gamma+2\Xi)\vee(1-\gamma+\Lambda)}\log(N)\right)$ time. This fact combined with the requirement $\gamma > 3\Xi$ lead to an optimal way to choose γ to minimize the amount of computational time:

- On the region $\Xi < 1/9$ we can solve the equation $\gamma_{min} + 2\Xi = 1 - \gamma_{min} + \Lambda$ to get the minimizing γ as $\gamma_{min} = \frac{1-2\Xi+\Lambda}{2}$, which satisfies $\gamma_{min} > 3\Xi$. This results in $O(N^{\frac{1+2\Xi+\Lambda}{2}}\log(N))$ computational time.
- On the region $\Xi \in [1/9, 1/7)$:
  - If $\frac{1-2\Xi+\Lambda}{2} > 3\Xi$, set $\gamma_{min} = \frac{1-2\Xi+\Lambda}{2}$ resulting in $O(N^{\frac{1+2\Xi+\Lambda}{2}}\log(N))$ computational time.
  - Otherwise if $\frac{1-2\Xi+\Lambda}{2} \leq 3\Xi$, set $\gamma_{min} = 3\Xi+\eta$, where $\eta > 0$ is small, for $O(N^{5\Xi+\eta}\log(N))$ computational time.
- For $\Xi \in [1/7, 1/3)$ also set $\gamma_{min} = \Xi + \eta$, where $\eta > 0$ is small, for $O(N^{5\Xi+\eta}\log(N))$ computational time.

11.3. *Simulation Results for WBinSeg.* We next looked at how effective intelligent sampling with WBinSeg would work in practice, by running a set of simulations with the same set of model parameters as in Setup 2 of Section 5, and used the exact same method of estimation except with WBinSeg used in place of the Binseg algorithm. For the tuning parameters of WBinSeg, the same ζ_N was retained and the number of random intervals was taken as $M_n = 20000$. Although we could have used the theoretically prescribed value of $M_N = 9(N_1/\delta^*_{N_1})^2 \log(N_1^2/\delta^*_{N_1})$, this turns out to be over 400,000 and is excessive as setting $M_N = 20,000$ gave accurate estimates.

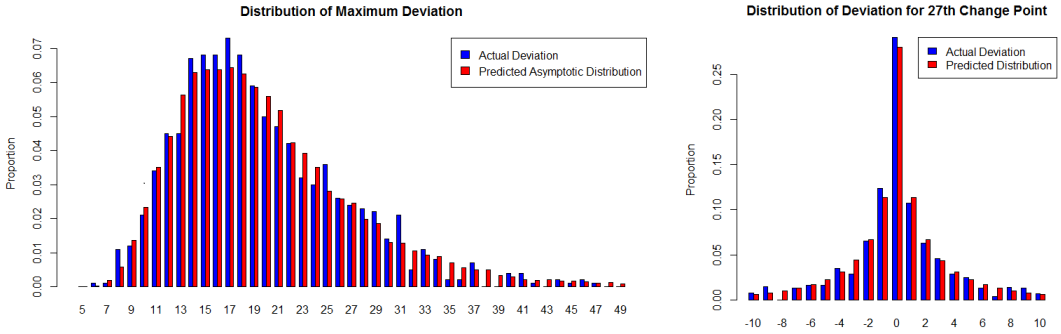


Fig 29: Distributions of $\max_{1\le j\le 55} \lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right)$ and $\lambda_2\left(\tau_{27}, \hat{\tau}_{27}^{(2)}\right)$ from 1000 trials using the same parameters as setup 2 but employing WBinSeg instead of BinSeg.

Using WBinSeg along with steps (D1) and (D2), the event $\{\hat{J} = J\}$ also occurred over 99% of the time during simulations. One can also see from Figure 29 that the distribution of the $\hat{\tau}_j^{(2)}$'s again match with Theorem 4. Because the performance of intelligent sampling is near identical for this setup, regardless of whether BinSeg or WBinSeg was used, the reader may wonder why the latter isn't used for the previous few simulations. The reason is the following: when the refitting method from Section 9.1 is implemented, it results in the second stage intervals being of width $Q_1(1-\alpha/J)N^{1-\gamma}$, irrespective of which of BinSeg or WBinSeg was used at stage one [where $Q_1(1-\alpha/J)$ is the $1-\alpha/J$ quantile of $|L_1|$]. Hence, WBinSeg loses any possible advantage from the tighter confidence bound of width $O(\log(N))$ rather than $O(E_N)$ for BinSeg from stage one. So, in a sparse change point setting and with stage 1 refitting, WBinSeg provides no accuracy advantages but adds to the computational time, e.g., the 1000 iterations used to create Figure 29 averaged ≈ 293 seconds, while the iterations used to create Figure 3 averaged ≈ 7 seconds.

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